Unemployment Insurance, Inequality of Opportunity, and Labor Market Conditions*

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Abstract

Social benefits are the fundamental reason why unemployment insurance exists, and are therefore central to understanding whether benefit generosity should vary with economic conditions. Yet, little is known about how the social benefits of unemployment insurance evolve over the business cycle, even though downturns may change both the magnitude of consumption losses upon unemployment and who experiences them. I propose and empirically evaluate a framework to assess the welfare effects of extending benefit duration, accounting for selection into unemployment induced by changing work opportunities. Welfare effects depend on social preferences for redistribution across unemployed types, consumption-smoothing gains, and efficiency costs. Exploiting regional and time variation in Spain, where the unemployment rate tripled, I examine the evolution of these components. A newly collected survey demonstrates stronger support for providing benefits to high-willingness-to-work unemployed, whose share increases in downturns. Consumption data reveal larger drops upon job loss when unemployment is high. Regression-discontinuity estimates using administrative records show efficiency costs fall in recessions. Taken together, results indicate that benefit duration should have been longer at the Great Recession's peak than before. Three quarters of the welfare gain from shifting benefit duration from booms to recessions comes from changes in social benefits, rather than from lower efficiency costs. More broadly, these findings imply that existing literature overlooking changes in social benefits understates the optimal duration cyclicality.

JEL Classification: D63, E21, J64, J65

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1 Introduction

Unemployment insurance (UI) is a crucial tool for policymakers seeking to support individuals when hit by adverse employment shocks. Its importance is even higher during economic downturns, when large numbers of individuals unexpectedly lose their jobs and rely on this income support to alleviate the hardship of unemployment. Despite this being a widespread issue, there is significant variation across countries in how the generosity of the program varies with economic conditions. While some, like the United States, usually provide more generous transfers during recessions, others, like many countries in Western Europe, keep transfers constant regardless of economic conditions.

Economists have long debated whether the unemployed should receive more generous transfers when economic conditions are bad. The debate centers around how the efficiency costs and social benefits associated with UI (Baily, 1978; Chetty, 2006) evolve with economic conditions. Existing evidence on social benefits over the cycle, which focuses on the consumption insurance value, is scant and remains inconclusive, since empirical studies usually exploit relatively small variations in the unemployment rate which may limit the power to detect fluctuations of these components. More fundamentally, the canonical framework fails to capture the inherent distributional feature of the system, since benefit reforms that differentiate according to labor market conditions effectively transfer resources across groups of unemployed individuals that may differ drastically. These individuals may not only differ in their ability to smooth consumption, but also in the source of their unemployment status: while a labor supply narrative may be better suited for the unemployed in good times, involuntary unemployment due to lacking work opportunities becomes increasingly significant as labor market conditions deteriorate (Summers, 2010).[1] Consequently, accounting for shifts in the composition of the unemployed becomes crucial, possibly affecting the welfare implications of unemployment benefit reforms over the cycle.

This paper proposes, and empirically evaluates, a framework to assess the welfare consequences of changing UI generosity with economic conditions. The framework characterizes the welfare impact of benefit duration extensions depending on three components: preferences for redistribution between different types of unemployed, consumption smoothing gains, and efficiency costs. It extends previous work by accounting for differential selection of individuals into unemployment, incorporating inequality of opportunity in job attainment (Roemer, 1998) due to variation in labor market conditions. In the model, the social benefits of UI reforms do not only arise from alleviating consumption inequality due to unemployment. The origin of this inequality also matters, becoming key to disentangle whether unemployment emerges from unequal economic opportunities (e.g., job offers) or disparities in individual effort (e.g., job search).

The framework delivers a set of high-level "sufficient statistics" that can be estimated empirically,

[1] Intuitively, the role of these two components changes substantially over the cycle, as conveyed by Summers: *"In an economy that is as demand constrained as ours, whatever small changes in search intensity may be associated with unemployment insurance are not the reason for the persistence of joblessness. With five unemployed Americans seeking work for every job opening available, there can be little doubt that the overwhelming cause of unemployment is not a lack of will among the jobless to find work, but a lack of work opportunities".*

providing a direct link between theory and evidence. I implement this framework in the context of Spain, which is particularly well-suited for this analysis since it features unusually large variation in the unemployment rate. Over the period of study, 2005-2017, the national unemployment rate increased from 7.9% to 26.9%, making this context remarkable relative to international standards (see Figure A1). The empirical analysis combines administrative data on unemployment spells, survey data on consumption, time use survey data on job search, and a representative online survey I conducted on preferences for redistribution. Along with policy variation and the aforementioned variation in economic conditions, it enables a comprehensive assessment of all the inputs needed to evaluate the welfare effects of differentiating UI duration across good and bad times. This assessment reveals a central result: changes in social benefits account for 73% of the welfare gains from benefit duration extensions.

The theoretical framework follows Baily (1978), Lentz and Tranaes (2005), Chetty (2008), Schmieder et al. (2012), and Kolsrud et al. (2018), where a population of heterogeneous unemployed individuals look for jobs by making search effort and savings decisions. Individuals are also subject to different economic conditions (good or bad), where search effort is more or less effective in terms of job finding (unequal economic opportunity). The government provides unemployed individuals with unemployment benefits that are paid over a limited number of periods, and this duration can be made dependent on economic conditions.

I extend the standard framework by allowing the unemployed to differ in their willingness to work: the extent to which they search for jobs conditional on the economic environment. The government maximizes the welfare-weighted utility of employed and unemployed individuals. In the spirit of the generalized social marginal welfare weights of Saez and Stantcheva (2016), the welfare weights can depend on the unemployed's willingness-to-work type. In this setting, the social benefits of UI reforms are allowed to reflect social preferences that depend on whether inequality emerges because of bad luck (not finding a job when labor demand is dampened) or lack of effort (not finding a job when plenty of opportunities are available). Focusing on this effort dimension is natural for several reasons. First, effort is the main choice variable determining unemployment duration in canonical job search models which are employed in the UI literature. Second, it is consistent with an influential body of work showing that effort constitutes a central driver of demand for redistribution across a broad range of settings (Alesina and Angeletos, 2005; Almås et al., 2010; Cappelen et al., 2007; 2013; Durante et al., 2014; Alesina et al., 2018; Almås et al., 2020). Third, given that the importance of effort as a determinant of unemployment varies greatly over the cycle (Summers, 2010), and thus large shifts in the pool of unemployed can potentially take place, it becomes particularly relevant to account for this dimension when assessing the welfare effects of UI reforms that transfer resources across individuals in different labor market conditions.

The optimal cycle-dependent benefit duration follows a Baily-Chetty intuition in this economy: social benefits and costs of unemployment insurance in good times must balance the benefits and costs in bad times. This condition provides a transparent link between theory and evidence that is robust to a wide range of specific assumptions about the underlying model. The benefits and costs of UI benefit extensions

are fully captured by three sufficient statistics that can be estimated in the data. First, the framework introduces a novel channel determining the social benefits provided by UI transfers over the cycle. In the presence of preferences for redistribution across the unemployed based on their willingness to work, the social benefits of UI reforms take into account that the composition of the unemployed may differ in booms and recessions. Although individual willingness to work is not directly observed, aggregate economic conditions seen through the lens of this framework reveal information about the type of the average unemployed. In good times, when job opportunities are available, individuals with a high willingness to work often find a job, and so the pool of unemployed is on average composed of low-willingness-to-work individuals. However, in bad times, even individuals with high willingness to work are likely to remain unemployed, and thus the pool of unemployed is composed of both low- but also high-willingness-to-work types. This implies that, all else constant, the willingness to work of the average unemployed increases with the unemployment rate.[2] Therefore, in the presence of asymmetric information, a social planner who prefers to provide income support for individuals whose unemployment status is not caused by limited job search effort can use aggregate conditions to target resources towards them. As a result, the larger the gap in society's relative valuations between the unemployed in recession and the unemployed in boom, the larger the welfare effects of having higher potential UI duration in recession relative to boom.

Second, benefit duration depends on the consumption smoothing gains provided by the transfers. This component captures differences in the marginal utility of consumption of a euro of unemployment benefits across unemployed individuals in different economic conditions. On the one hand, it captures the standard insurance value, where the transfers are more valuable when individuals are less able to self-insure themselves against job loss. On the other hand, it also captures the fact that the system does not only provide insurance, but also potentially redistributes by effectively transferring resources across groups of individuals, who may differ in their pre-unemployment consumption levels. Consequently, the larger the gap in marginal utilities of consumption between the unemployed in recessions and the unemployed in booms, the larger the welfare impact of UI reforms that differentiate benefit duration between these two economic conditions.

Third, the derived formula highlights that the welfare effects of duration extensions also depend on the evolution of the efficiency costs of the policy over the cycle. Higher potential benefit duration (PBD) reduces incentives to look for jobs, which may result in distortions of labor supply with individuals spending more time unemployed and thus taking longer to find a job. This imposes an additional cost on the government budget, since the longer unemployment duration translates into more resources spent on unemployment benefits and less resources collected through taxes when employed. In this context, a larger gap between the labor supply distortion caused by extending benefit duration in good times relative to bad times corresponds to a larger welfare effect of cycle-dependent reforms.

[2] In other words, aggregate unemployment rate can be seen as a shifter of the signal-to-noise ratio in the spirit of Alesina and Angeletos (2005), where the signal here corresponds to the voluntary part of unemployment and the noise to the involuntary one. The ratio in this context is decreasing with the unemployment rate, so that the moment of the cycle provides additional information to the government about the role of lack of effort vs bad luck in individuals' employment status.

The framework described above guides the empirical analysis, which is structured in three parts. The first part quantifies how the component capturing the social benefits of the transfers — due to both preferences for redistribution and changes in the unemployed's composition — evolves with economic conditions. I proxy willingness to work by the search effort level that the unemployed exert in a common economic environment. Individuals who search on a weekly basis are considered to have a high willingness to work, while those who do not are characterized by low willingness. I assess whether society has a preference for redistribution between these types of unemployed individuals. To do so, I explicitly design a survey and administer it to a sample of 1022 individuals who are representative of the Spanish active population. The survey design contains features such as order randomization and open-ended questions to make sure that results are not driven by order effects or specific characteristics of the implementation. In a first step towards eliciting preferences, respondents are asked to express preference towards resource allocation between two types of unemployed: one who searched for jobs last week and one who did not. Remarkably, around 75% conveys a desire to reward job search behavior. In a second step, with the aim of quantifying relative valuations between types, respondents are presented with multiple pairwise choices that differ in the amounts received by each type of unemployed. The design varies experimentally the cost of the transfer between individuals, allowing me to elicit the distribution of relative valuations by society between the two types. This approach yields estimates that imply strong preferences for redistribution, with society valuing a transfer to the low-willingness-to-work type at only 8% of the value placed on an equivalent transfer to the high-willingness-to-work type.

After documenting significant preferences for redistribution along the willingness-to-work dimension, I study changes in the composition of the unemployed over the cycle. By exploiting the Spanish Time Use data, I show that the share of unemployed who search for jobs weekly increases strongly with the unemployment rate. Nevertheless, this increase could reflect both a change in composition of the unemployed and a behavioral change. I follow Mukoyama et al. (2018) to gauge the part driven by composition, and find that there is a large change over the business cycle driven just by the shift in composition: the share of weekly searchers (high willingness to work) goes from 54% in boom to 70% in recession. I additionally provide a sensitivity analysis including a prediction exercise just using the Spanish data. The results confirm the main finding of a considerable shift in the pool of unemployed towards high-willingness-to-work types in bad times. Combining the change in the composition of types over the cycle with the relative valuations between types, I estimate a ratio of welfare weights of 1.27. That is, society values a €1 transfer to the type of individuals unemployed in recession as much as €1.27 to those unemployed during a boom.

The second part of the analysis explores how the component of social benefits due to consumption smoothing gains of the transfers varies with economic conditions. In doing so, I employ a difference-in-differences design similar to Gruber (1997) where I compare the consumption patterns of individuals becoming unemployed with those of individuals who remain working. I exploit rich survey panel data on household consumption that allow to distinguish between UI recipients and those who exhaust benefits,

and which provide very detailed information on different consumption categories. This allows me to study the effect on both food consumption — that has traditionally been the focus of the literature due to data limitations — and on overall consumption including housing, utilities, clothing and transportation among others. On the one hand, assessing self-insurance ability, I find that on average, households experience an overall 10.1% (s.e. 1.3%) reduction in consumption upon job loss while receiving UI benefits. This reduction is larger for those exhausting benefits, reaching 22.1% (s.e. 3.1%). Moreover, this drop increases in bad times for both groups of unemployed, with a 1 p.p. increase in unemployment rate resulting in an additional 0.6% - 0.8% significant reduction relative to pre-unemployment consumption levels. Similarly to Kroft and Notowidigdo (2016), I find limited variation in the drop in food consumption over the business cycle. Instead, the drop in overall consumption seems to be driven by both durable and non-durable categories.

On the other hand, I investigate whether in addition to consumption drops upon unemployment, the consumption levels of the unemployed also differ because of disparities in pre-layoff consumption. I find that individuals getting unemployed already had lower consumption prior to job loss relative to those who remained employed. However, this gap does not vary with economic conditions. This implies that, although the UI system redistributes between people of different pre-layoff consumption levels, it does not do so differentially across booms and recessions and so this component does not have empirical relevance for assessing the cyclicality of the policy. As a result, overall, extending UI duration in recessions provides larger consumption smoothing gains relative to extensions in boom periods, arising from its insurance role. Moreover, it underscores the importance of leveraging detailed consumption data and helps reconcile previous work which documented a constant drop in consumption over the cycle.

The third part employs administrative data on unemployment spells to analyze the evolution of the efficiency cost over the cycle. This cost depends on two components. First, a mechanical one, captured by the share of people who remain unemployed after benefit exhaustion, which I estimate nonparametrically. Second, a behavioral one, captured by the response of unemployment duration to changes in potential benefit duration, which I estimate through a regression discontinuity design. To estimate the latter, I leverage variation in the structure of the Spanish UI system, where the statutory duration of the benefits varies discretely at several work experience thresholds. I find that, on average, for every €1 transferred mechanically to the unemployed, the government needs to levy €0.59 on top of that because of the behavioral response. Consistent with results by Schmieder et al. (2012), I also find that the efficiency cost declines in recessions, and that this is driven by a higher share of individuals exhausting benefits in bad times.

Finally, the empirical findings are used to inform the statistics of interest in the derived framework, with the aim of assessing the welfare effects of a hypothetical reform making UI benefit duration dependent on the business cycle. Specifically, I estimate that a local budget-balanced reform of the UI system that increases the duration of benefits for the unemployed in recession, and reduces the duration for the unemployed in boom, would yield a net welfare gain of €1.06 per euro mechanically transferred to

individuals unemployed in recession. Around 73% of this welfare gain comes from an increase in the social benefits provided by UI transfers, with contributions of 43% from consumption smoothing and 30% from the novel preferences for redistribution component. The remaining 27% is due to the lower efficiency cost of increasing generosity in bad times.

The large welfare effects imply that the potential duration of benefits should increase with the unemployment rate, and therefore it should have been more generous at the peak of the Great Recession than before. Importantly, these findings highlight the quantitative relevance of the change in social benefits over the cycle that has been overlooked in related empirical literature, which mostly focuses on the change in efficiency costs. In contrast, the total welfare gains of extending UI in recessions estimated in this paper are almost four times as large relative to the gains considering only the efficiency cost. This implies that UI benefit duration should increase more with the unemployment rate than predicted by previous work.[3] Note that this finding is not driven by the specific welfare criterion emphasized here. Even abstracting from welfare weights and considering only the changes in consumption smoothing benefits over the cycle, which would correspond to the standard utilitarian framework, I estimate that the welfare gains are more than double those implied by changes in the efficiency cost alone.

My work relates to several strands of the literature. It first relates to a strand that employs a sufficient-statistics approach to characterize the welfare effects of UI reforms, building on seminal work by Baily (1978), Gruber (1997) and Chetty (2006).[4] Within this tradition, it adds to work focusing on changes of generosity over the business cycle, where one strand applies the Baily-Chetty insights (Schmieder et al., 2012; Kroft and Notowidigdo, 2016; Landais et al., 2018a;b), while another one relies on model-based calibrations (Mitman and Rabinovich, 2015; McKay and Reis, 2021; Kekre, 2023). Relative to most prior work following a stylized characterization, the proposed framework recognizes that benefit duration extensions over the cycle effectively transfer resources across groups of heterogeneous unemployed individuals. This parallels recent studies considering the distributional features of pension reforms (Kolsrud et al., 2024), and UI reforms in a static environment (Ferey, 2026). Taking this feature into consideration, it explores the welfare consequences of UI policy differentiation by incorporating social preferences for redistribution along the main choice variable in canonical job search models and a well-documented driver of society's fairness views and demand for redistribution (e.g., Cappelen et al., 2007; 2013; Durante et al., 2014; Almås et al., 2020). Considering this dimension — with potentially large variation with economic conditions — is not only relevant to understand society's actual tradeoffs and properly account for them when assessing welfare effects, but also to inform the set of socially acceptable reforms that policymakers can implement.

In doing so, it draws on a separate line of work regarding equality of opportunity (e.g., Roemer, 1998;

[3]For example, based on the survival rates presented in Kroft and Notowidigdo (2016), back-of-the-envelope calculations assuming a constant marginal response over the cycle suggest that, under the standard framework and considering only the efficiency cost, it is difficult to rationalize extending benefits from 26 to 99 weeks as implemented in the US. This is a motivation for further understanding of the changes of social benefits, focusing both on the consumption smoothing benefits and incorporating empirically-relevant social preferences.

[4]Later work enriches the sufficient statistics approach considering other dimensions of the generosity of unemployment benefits (e.g., Card et al., 2007; Shimer and Werning, 2008; Kolsrud et al., 2018; Lindner and Reizer, 2020; Campos et al., 2022; Wunsch and Zabrodina, 2024). For a recent review, see Le Barbanchon et al. (2024).

Roemer et al., 2003; Alesina and Angeletos, 2005; Alesina and La Ferrara, 2005; Alesina et al., 2018), and focuses on this aspect in the context of unemployment policy (Haber and Murray, 1966). Evaluating the social benefits of this type of UI reforms boils down to quantifying what is the value of a one euro transfer between the unemployed in boom and in recession. For the empirical application integrating these insights, it builds on the generalized social marginal welfare weights approach proposed by Saez and Stantcheva (2016), and is related to contemporaneous work by Capozza and Srinivasan (2024) in the context of US income taxation. I contribute by designing and administering a new survey which is explicitly constructed to estimate preferences for redistribution between types of unemployed individuals that differ in their willingness to work. By empirically incorporating the social preference for reducing the degree of inequality induced by luck, while rewarding individual effort, the framework evaluates reforms to the benefit system, capturing both the standard consumption smoothing motive and accounting for disparities in economic opportunity.[5]

The second part of the empirical analysis relates to work on the social benefits of UI transfers.[6] Specifically, the literature has focused on its consumption insurance value, starting with the seminal work on consumption behavior by Gruber (1997) and followed by Browning and Crossley (2001); Stephens Jr (2001); Hendren (2017); Kolsrud et al. (2018); Campos and Reggio (2020); Ganong and Noel (2019); Gerard and Naritomi (2021); Landais and Spinnewijn (2021).[7] However, evidence on how the value of UI changes with economic conditions is surprisingly scant. A few recent studies document consumption patterns over the period of the Covid pandemic (Chetty et al., 2024; Ganong et al., 2024), while, to the best of my knowledge, only one paper considers changes over the business cycle (Kroft and Notowidigdo, 2016). This study finds that the insurance value does not vary with economic conditions. Unfortunately, given the institutional setting and the available data, their results only consider food consumption, cannot distinguish between UI recipients and the rest of unemployed, and the variation in unemployment rate effectively exploited is relatively small, raising concerns about potential lack of power to detect fluctuations.[8] I revisit this aspect by providing evidence from a setting where rich consumption data beyond food consumption are available, together with information on UI status and large variation in unemployment rate. I find that the insurance value changes over the cycle, both for recipients and those exhausting benefits, and that the result is driven by non-food consumption categories, thus being able to

[5]It is also related to the work on "tagging" (Akerlof, 1978), where here economic conditions reveal additional information about the average unemployed's willingness-to-work type and are used as an imperfect tag.

[6]The evidence on the value of transfers has historically lagged behind the evidence on disincentive effects. This historical emphasis in the literature on the costs of government programs, as opposed to their benefits, is not only present in Unemployment Insurance, but extends to other programs including Medicaid, AFDC/TANF and SNAP in the US as documented by Aizer et al. (2022).

[7]More generally, it also relates to studies on consumption responses to income shocks (e.g., Blundell et al., 2008; Low and Pistaferri, 2015; Autor et al., 2019; Gross et al., 2020; Ganong et al., 2020).

[8]The smaller variation in unemployment rate relative to my setting arises from two factors. First, overall fluctuations in the US Great Recession are an order of magnitude smaller than in the case of Spain. Second, benefit generosity in the US tends to increase when the unemployment rate increases. Given that the interest is on understanding consumption insurance over the cycle, while keeping generosity *constant*, the relation between benefits and unemployment needs to be controlled for and so the effective variation used is smaller. The potential duration of benefits in Spain is constant over the cycle, which allows to exploit relatively more variation.

reconcile previous findings in the literature.

Finally, this paper also relates to studies analyzing labor supply disincentive effects induced by increases in unemployment benefit generosity. While a sizable literature quantifies labor supply distortions at a given moment of the cycle (e.g., Katz and Meyer, 1990; Card and Levine, 2000; Krueger and Meyer, 2002; Rothstein, 2011; Landais, 2015; Lalive et al., 2015; Le Barbanchon, 2016; Chodorow-Reich et al., 2019; Hagedorn et al., 2019; Dieterle et al., 2020; Boone et al., 2021; Gerard and Gonzaga, 2021; Khoury, 2021; Lichter and Schiprowski, 2021; Marinescu et al., 2021; Domenech-Arumi and Vannutelli, 2025; Acosta et al., 2025; Hernandez Martinez and Liu, 2025), few studies analyze how the disincentive effects evolve over the cycle (Schmieder et al., 2012; Farber et al., 2015; Kroft and Notowidigdo, 2016). My paper provides new insights on the evolution of labor supply responses induced by potential benefit duration extensions at different points of the cycle, in a setting with remarkably large variation in economic conditions.

The rest of the paper proceeds as follows. Section 2 provides the theoretical framework used to evaluate the marginal welfare gains from benefit duration extensions. Section 3 describes the institutional context and the data. Section 4 presents the empirical methodology and the results. Section 5 describes the welfare implications, and Section 6 concludes.

2 Framework

2.1 Setup

Here I present a stylized version of the main framework, which will guide the empirical analysis. It builds on Baily (1978), Lentz and Tranaes (2005), Chetty (2008), Schmieder et al. (2012), and Kolsrud et al. (2018), and characterizes the job search problem of the unemployed and the policy choice of the social planner in good and bad times. I study changes in benefit duration over the cycle, while keeping taxes fixed. This allows me to focus on the unemployed, considering transfers between individuals unemployed at different moments of the business cycle, and accounting for composition shifts in the pool of unemployed. A more detailed derivation is provided in Appendix B.1.

Environment. There is a continuum of agents with unit mass. They become unemployed at their own $t = 0$, where time corresponds to periods of the individual's spell measured since the event of job loss. Let $k_{i,t} \in \{B, R\}$ denote the aggregate economic condition faced by individual i at spell duration t, with B and R referring to boom and recession. The realization $k_{i,0}$ denotes economic conditions at unemployment entry, while current economic conditions $k_{i,t}$ may evolve over the spell and determine current and expected job-finding opportunities. In each period, agents can be employed or unemployed until the terminal period T when they retire. Labor market status is $s_{i,t} \in \{U, E\}$, where U and E denote unemployment and employment, respectively. Agents make saving decisions $a_{i,t}$ and, while unemployed, choose search effort $e_{i,t}$ facing a fixed wage that is high enough to ensure that all job offers will be accepted. Search is costly as given by $\psi_i(e_{i,t})$, which is assumed to be differentiable, increasing and convex. Effort increases the probability of finding a job next period, $h^{k_{i,t}}(e_{i,t})$, with a mapping that potentially depends on current

economic conditions. For instance, $h^B(e_{i,t}) > h^R(e_{i,t})$ (i.e. for a given effort level, job finding is higher in boom than in recession).[9]

Formally, the problem of an unemployed individual with assets $a_{i,t}$, current economic conditions $k_{i,t}$ and entry conditions $k_{i,0}$ is:

$$V^U_{i,t}(a_{i,t}, k_{i,t}; k_{i,0}) = \max_{e_{i,t}, a_{i,t+1}} u(c^U_{i,t}) - \psi_i(e_{i,t}) + \beta \mathbb{E}_t \Big[(1 - h^{k_{i,t}}(e_{i,t})) V^U_{i,t+1}(a_{i,t+1}, k_{i,t+1}; k_{i,0}) + h^{k_{i,t}}(e_{i,t}) V^E_{i,t+1}(a_{i,t+1}, k_{i,t+1}) \Big], \tag{1}$$

subject to $a_{i,t+1} = (1+r)a_{i,t} + b_t^{k_{i,0}} + y^U_t - c^U_{i,t}$ and $a_{i,t+1} \geq \underline{a}_i$. When re-employed, the agent solves the following problem:

$$V^E_{i,t}(a_{i,t}, k_{i,t}) = \max_{a_{i,t+1}} u(c^E_{i,t}) + \beta \mathbb{E}_t V^E_{i,t+1}(a_{i,t+1}, k_{i,t+1}), \tag{2}$$

where $a_{i,t+1} = (1+r)a_{i,t} + w_i - \tau - c^E_{i,t}$ and $a_{i,t+1} \geq \underline{a}_i$. Flow utilities for the unemployed and employed are $u(c^U_{i,t}) - \psi_i(e_{i,t})$ and $u(c^E_{i,t})$, with $u(\cdot)$ being increasing and concave. Unemployed's consumption is denoted by $c^U_{i,t}$, and employed's consumption by $c^E_{i,t}$. Each individual receives a wage w_i and pays a tax τ when employed, and receives benefits $b_t^{k_{i,0}}$ (which may depend on economic conditions) and non-labor income y^U_t when unemployed. Savings cannot be lower than a borrowing constraint $\underline{a}_i$.

Government policy. I consider an unemployment benefit policy $\mathbf{P} = (P^B, P^R)$ characterized by a benefit profile $\{b_t^{k_0}\}_{t=1}^T$. The government commits to the policy before the start of the model. The starting point is an environment with a fixed benefit level and a common potential benefit duration which does not depend on the cycle ($P^B = P^R$), such that agents receive benefit $b_t^{k_0} = b$ from $t = 1$ to $t = P$, and $b_t^{k_0} = 0$ afterwards. The choice of benefit generosity by the planner is the benefit duration P^{k_0}, considering a local reform where the duration can be made dependent on economic conditions at job loss. Thus, current conditions affect behavior and job finding but do not change the entitlement attached to an ongoing unemployment spell.

Let $\alpha^{k_0} = \Pr(k_{i,0} = k_0)$ be the share of unemployment entrants whose spells begin in economic state k_0. The survival probability $S_t^{k_0} = \Pr(s_{i,t} = U \mid k_{i,0} = k_0; \mathbf{P})$ is the fraction of that entry cohort remaining unemployed t periods after job loss under policy $\mathbf{P}$. Moreover, denote $D^{k_0} = \sum_{t=1}^T S_t^{k_0}$ as the expected nonemployment duration of individuals in that cohort, measured over the horizon T. For simplicity, the policy characterization and quantitative exercises abstract from time discounting, setting $r = 0$ and $\beta = 1$. Appendix B.1 considers the general problem and makes explicit these assumptions. The government's budget is given by:

$$G(\mathbf{P}) = \sum_{k_0 \in \{B,R\}} \alpha^{k_0} \left[(T - D^{k_0})\tau - \sum_{t=1}^T S_t^{k_0} b_t^{k_0} \right] \tag{3}$$

The budget aggregates income taxes and benefit spending over time and across economic conditions.

[9] Thus, in each period, agents can be in four mutually exclusive labor market statuses, which result from the combination of employment/unemployment and boom/recession. The stylized version of the model considers two aggregate economic conditions for simplicity, but can be easily extended to consider a continuum of them.

Social welfare for a given unemployment benefit policy is:

$$W(\mathbf{P}) = \int \omega(\overline{e}_i) V_{i,0}(\mathbf{P})\, di + \lambda[G(\mathbf{P}) - \overline{G}], \tag{4}$$

where $V_{i,0}(\mathbf{P})$ refers to indirect lifetime utility, and the social planner places welfare weights $\omega(\overline{e}_i)$, which depend on the willingness-to-work type, capturing specific preferences for redistribution towards individuals with specific attitudes towards job search $\overline{e}_i$ (Saez and Stantcheva, 2016). Specifically, individuals are characterized by a willingness-to-work type defined as the average effort level exerted over the unemployment spell when faced with benchmark economic conditions $\overline{k}$ and a fixed baseline benefit policy $\overline{\mathbf{P}}$, such that $\overline{e}_i = \mathbb{E}[e_{i,t} \mid \overline{k}, \overline{\mathbf{P}}]$. Accordingly, this characterization thus assigns a type to each individual regardless of her actual aggregate economic condition, evaluating individual effort conditional on everyone facing the same economic circumstances. Moreover, because the type is evaluated at the fixed baseline policy $\overline{\mathbf{P}}$, the welfare weights do not respond to changes in the actual benefit policy $\mathbf{P}$. The Lagrange multiplier on the budget constraint is denoted by λ, and $\overline{G}$ is an exogenous revenue constraint.

Reform of benefit generosity. The planner considers increasing the duration of benefits for the unemployed in recession and reducing the duration of benefits for the unemployed in boom $\left(\frac{db^R_{P+1}}{db^B_{P+1}} < 0\right)$, such that the budget remains balanced. Optimality of the differentiated benefit policy for the unemployed in recession and boom requires that the following expression holds:

$$\underbrace{\frac{\mathbb{E}^{U,R}_{P+1}[\omega(\overline{e}_i)u'(c^U_{i,P+1})]}{\mathbb{E}^{U,B}_{P+1}[\omega(\overline{e}_i)u'(c^U_{i,P+1})]}}_{\textbf{Marginal Benefit}} = \underbrace{\frac{1+FE^R}{1+FE^B}}_{\textbf{Marginal Cost}} \tag{5}$$

The expression highlights the key statistics needed to evaluate the welfare implications of changes in unemployment benefit duration over the cycle. The left hand side refers to the marginal benefit, which captures the social benefits of UI, and is shown to depend on two terms. First, it depends on the ratio of welfare weights between those unemployed in recession and in boom. Given the definition of the types, note that only the differences in mean effort of the unemployed across the cycle that are driven by composition enter the welfare weights, and so changes in behavior are not considered. Second, it also depends on the ratio of marginal utilities of consumption between the unemployed in recession and in boom who are at the benefit exhaustion point. The operator $\mathbb{E}^{U,k_0}_{P+1}[\,\cdot\,] = \mathbb{E}\,[\,\cdot \mid s_{i,P+1} = U,\ k_{i,0} = k_0; \mathbf{P}]$ indicates that the expectation averages over individuals who enter unemployment in economic condition k_0 and remain unemployed at benefit exhaustion, integrating over their current economic conditions at period $P+1$. The right hand side, which refers to the marginal cost and captures the efficiency costs of UI, depends on the relative extra costs — or fiscal externalities — due to labor supply distortions that are induced when increasing benefit duration in boom and recession.

This approach yields a characterization that is simple, yet remains valid under a wide range of assumptions about the underlying model. For example, the expression is robust to alternative dimensions of individual heterogeneity (e.g., job finding), and to different choice variables (e.g., reservation wages).

Invoking the envelope theorem, individuals already optimize with respect to other variables before the policy change, so changes in these alternative variables in response to the policy change do not have a first-order effect on welfare. As a result, the highlighted expression is general, remaining valid to assess the welfare consequences of benefit duration extensions.

In summary, the expression implies that differentiating the UI policy with respect to economic conditions is welfare improving if and only if the marginal benefit is greater than the marginal cost. This motivates the empirical part of the paper, which implements the expression in order to assess the welfare consequences of the reform.

2.2 Implementation: Marginal Benefit

In order to connect the conceptual framework to the data, I follow a consumption-based approach (Gruber, 1997; Chetty, 2006; Kolsrud et al., 2018; 2024). Under the implementation assumptions stated in Appendix B.2, the approximated empirical counterpart of the marginal benefit term is:

$$\frac{\mathbb{E}^{U,R}_{P+1}[\omega(\overline{e}_i)u'(c^U_{i,P+1})]}{\mathbb{E}^{U,B}_{P+1}[\omega(\overline{e}_i)u'(c^U_{i,P+1})]} \approx \underbrace{\frac{\omega^R}{\omega^B}}_{\substack{\textbf{Welfare}\\ \textbf{Weights}}} \underbrace{\frac{1+\gamma\frac{c^{E,R}-c^{U,R}}{c^{E,R}}}{1+\gamma\frac{c^{E,B}-c^{U,B}}{c^{E,B}}}\left[1+\gamma\frac{c^{E,B}-c^{E,R}}{c^{E,B}}\right]}_{\textbf{Consumption Smoothing}} \tag{6}$$

The equation shows the main components behind the social benefits associated with UI reforms: the value of these transfers depends on both the relative welfare weights and relative consumption differences between the individuals unemployed at each moment of the business cycle.

Welfare Weights. UI benefit duration reforms that differentiate with respect to labor market conditions effectively transfer resources across different individuals. Beyond consumption differences, which are captured by the consumption smoothing term, the type of individuals who are unemployed in boom and recession may be different along other dimensions, which the planner may wish to capture (Saez and Stantcheva, 2016) when assessing the welfare effects of the reform. Specifically, here the focus is on incorporating a specific dimension — willingness to work ($\overline{e}_i$) — which may particularly vary with economic conditions.

In the event that preferences for redistribution depended on this dimension, the first best would be to target benefits based on $\overline{e}_i$, but individual types are not observed. However, given the outlined framework, the moment of the cycle when people are unemployed reveals some information about their type. Intuitively, when the unemployment rate is low and there are many jobs available, search effort is very effective to find a job and so the low-effort individuals will be more likely to remain unemployed. In contrast, when the unemployment rate is high and there are fewer jobs available, labor supply becomes less relevant and even individuals who search hard are likely to remain unemployed, so the pool of unemployed shifts towards individuals who are, on average, of higher-effort types.

Given social preferences, and the selection mechanism described, the average welfare weight for unemployed individuals may vary over the cycle given the differential dynamic selection of the unemployed.

Thus, a planner that conditions benefits on aggregate conditions is imperfectly targeting individuals with different willingness to work (Akerlof, 1978).[10] The empirical part focuses on testing empirically these predictions and quantifying the different components of $\frac{\omega^R}{\omega^B}$, namely whether society has a preference for redistribution along the willingness-to-work dimension and whether the composition of the unemployed changes over the cycle. In addition, the main analysis is complemented with alternative quantification exercises where the welfare assessment only captures the effects under a standard (utilitarian) social planner that does not place different welfare weights across individuals.

Consumption Smoothing. The value provided by the transfers depends on the differences in marginal utility of consumption between the different groups of unemployed at each point of the cycle. The expression highlights two data moments which are relevant to assess the consumption component. First, the transfers provide insurance value, which is a function of a risk aversion parameter and the relative consumption changes upon job loss by the unemployed at different labor market conditions, as captured by the first term in the consumption smoothing component $(1+\gamma\frac{c^{E,R}-c^{U,R}}{c^{E,R}})/(1+\gamma\frac{c^{E,B}-c^{U,B}}{c^{E,B}})$. Intuitively, the larger the consumption drop individuals experience upon unemployment, the higher the value of transferring resources towards that state of the world.

Second, the marginal utility of consumption for the unemployed can differ not only because of different ability to insure consumption over the cycle, but also due to pre-unemployment differences in consumption levels. This means that a benefit reform of this type could effectively redistribute across pre-unemployment consumption levels if there is differential selection in terms of consumption, as captured by the second term $(1+\gamma\frac{c^{E,B}-c^{E,R}}{c^{E,B}})$.[11] The empirical analysis focuses on assessing whether the marginal utility of consumption for the unemployed varies with economic conditions, by estimating how both the consumption changes upon job loss and pre-unemployment levels evolve over the cycle.

2.3 Implementation: Marginal Cost

When changing benefit duration, affected unemployed individuals may take longer to find a job, which imposes an efficiency cost on the government budget. The ratio of total fiscal costs per euro transferred mechanically, induced by changing potential benefit duration across economic conditions is:

$$\frac{1+FE^R}{1+FE^B} \approx \frac{\frac{1}{S^R_{P+1}}\left[\frac{\partial D^R_b}{\partial b^R_{P+1}}b+\frac{\partial D^R}{\partial b^R_{P+1}}\tau\right]}{\frac{1}{S^B_{P+1}}\left[\frac{\partial D^B_b}{\partial b^B_{P+1}}b+\frac{\partial D^B}{\partial b^B_{P+1}}\tau\right]} \tag{7}$$

where $S^{k_0}_{P+1}$ is the share of entry-cohort-k_0 individuals who remain unemployed at the exhaustion point, $D^{k_0}_b=\sum_{t=1}^{T}(b^{k_0}_t/b)S^{k_0}_t$ measures expected unemployment duration in full-benefit days, and D^{k_0} denotes

[10] This idea applies more generally and could be implemented along other dimensions rather than national aggregate conditions, such as job availability in specific sectors or regions. Moreover, it is not necessary that all the selection along the willingness to work dimension occurs through dynamic selection over the unemployment spell, being also possible that composition shifts arise from differential layoff composition over the cycle.

[11] See Kolsrud et al. (2024) for a similar setting in the context of pension benefit reforms.

expected nonemployment duration.[12] Unemployment duration refers to time spent receiving benefits, while nonemployment duration refers to time between job loss and start of the new job. Intuitively, when the benefit at the exhaustion point is marginally increased, unemployed individuals may react by taking longer to find a job. This has a negative effect on the budget because people spend more time receiving benefits and also take longer to start paying taxes. The expression measures the total efficiency cost responses per euro transferred mechanically to the unemployed.

3 Institutional Background and Data

3.1 Great Recession and Unemployment Insurance System in Spain

Great Recession. The Spanish Great Recession was characterized by a strong boom period over the years up to 2007, followed by a slowdown in 2008 and then a substantial bust in the subsequent years. Internal demand had increased by around 20% in real terms since the early 2000s over the years leading up to the recession (Almunia et al., 2021). The rapid growth was in large part due to an increasingly important construction sector, which accounted for 12.4% of Spain's GDP at the peak in 2007. The growth, incentivized by the relatively cheap credit available in the Spanish economy, fueled a substantial housing boom (Martínez-Toledano, 2020). With a one year delay, after the unraveling of the subprime mortgage market in the US, the reduction in credit supply affected the real economy, which led to a collapse in internal demand together with a massive reduction in the importance of this sector, which was 5.4% of GDP by 2014. Overall, this resulted in large effects that extended across sectors, having a substantial impact in terms of labor market outcomes. The national unemployment rate skyrocketed from 7.9% to 26.9% over a period of 6 years. As mentioned before, this magnitude is remarkable relative to international standards as shown in Figure A1.

UI System. The Spanish UI system is a national program financed via payroll taxes. Benefit levels and durations are set by the central government and are the same for all regions. In order to qualify for unemployment insurance, a worker must become unemployed involuntarily and have worked for at least 1 year within the last 6 years after the last time of UI receipt. Individuals who quit voluntarily, receive full-time disability benefits or whose age is above 65 are not eligible to get UI. Potential benefit duration, which refers to the maximum time an individual can spend receiving UI benefits, is a function of work experience within the last 6 years before becoming unemployed. This duration is invariant over the business cycle. It is similar to systems in other countries in Western Europe such as Germany, and different from countries such as the US where the duration is usually extended when the economy is in a bad state. Specifically, potential benefit duration ranges from 4 to 24 months, increasing by two months every six months worked. In relation to benefit levels, they are determined as a function of pre-unemployment labor income and individual characteristics. The labor income considered is average gross earnings within the

[12] For a flat-benefit profile, we have $D_b^{k_0} = \sum_{t=1}^{P^{k_0}} S_t^{k_0}$.

last 6 months and the replacement rate is 70%, getting reduced to 60% after 6 months unemployed.[13] From July 2012 onwards, the latter replacement rate was set to 50%. After UI benefit exhaustion and depending on individual characteristics, individuals remaining unemployed may apply for unemployment assistance.

3.2 Data

Preferences for Redistribution. I design and administer an online survey to understand preferences for redistribution between the unemployed based on search effort types. To do so, I program a new survey using oTree and partner with the company Respondi/Bilendi to obtain respondents from Spain. This panel company has been previously used in the literature for other studies (e.g., Alesina et al., 2018; 2023). The survey was administered in August 2023, with a sample size of N=1022, and respondents being rewarded with €1.15 for an average completion time of 8 minutes. The data collection was performed to target 24 age-education-sex-employment cells based on the Spanish Labor Force Survey (*Encuesta de Población Activa*).[14] The sample is representative of the Spanish active population aged 25-64, as shown in Table A1. In order to maintain data quality, I further restrict the sample to individuals with a completion time between the 5th and 95th percentiles, and whose answers to the qualitative and quantitative questions explained below are consistent. This results in a final sample of 709 individuals. In order to address order effects, and understand reasons behind the responses, the survey has features such as order randomization and open-ended questions to make sure that results are not driven by specific characteristics of the implementation.[15]

The survey begins with a section collecting information on demographics and labor market status. It then elicits respondents' preferences for redistribution between the unemployed based on effort types, in both qualitative and quantitative ways. To do so, respondents are presented with scenarios where they are asked to allocate government funds between two individuals with different search effort behavior in the previous week, allowing to estimate relative valuations between them. Finally, the survey concludes with open-ended questions to understand the reasons for their choices, and multiple choice questions to know whether they would support a hypothetical UI reform of the type studied in the paper. Appendix C provides more detailed information about the survey structure including the full questionnaire.

Search Effort. The main measure of search effort is constructed from the Spanish Time Use Survey (*Encuesta de Empleo del Tiempo*). Specifically, in order to explore variation over the business cycle, I use the two available waves in years 2002-2003 and 2009-2010. The survey is harmonized at the European level by Eurostat as part of the Harmonized European Time Use Surveys, and is similar to the American Time Use Survey (ATUS). Its objective is quantifying the time spent by individuals on various activities

[13] In the stylized model, I focus on the duration of the benefits and make the simplifying assumption that the benefit level is constant up to the exhaustion point. See e.g., Kolsrud et al. (2018) for work considering variation in benefit profiles over the unemployment spell.

[14] In terms of implementation, the panel company sends invitations to their sample of individuals recently active in their platform who fulfill the criteria, and I only observe them conditional on accepting the invite. The company's reported response rate is approximately 50%.

[15] For a recent review on recommended survey practices in the literature, see Stantcheva (2023).

throughout the day. To that end, respondents fill questionnaires and time diaries where they record their daily activities in 10-minute slots over one day, so the survey structure is a cross section.
For the analysis, I focus on the unemployed aged 25-64 and use the information available on job-search-related activities to construct a measure capturing whether individuals report searching for jobs in a given day. More details on the construction of the search effort measure are presented in Appendix D.

Consumption. I use household expenditure survey data from the Spanish EPF (*Encuesta de Presupuestos Familiares*) over 2006-2015. This yearly survey provides detailed diary-based information on household consumption, labor market status, benefit income, and demographic characteristics. It is available over a long period of time, so proves useful to study dynamics over the business cycle far from the current period.[16] The survey has a panel dimension, with households being followed for two consecutive years, and contains 80 consumption categories which are classified following the COICOP/HBS classification. It accounts for 87% of aggregate consumption, which compares favorably to figures obtained in other countries (Heathcote et al., 2010; Campos and Reggio, 2015). I construct an expenditure measure that includes expenditure on a wide variety of goods and services: Food, Alcohol/tobacco, Clothing, Housing, Inventories, Health, Transport, Telecom, Recreation, Education, Hotels.[17] Throughout the analysis, I also provide results aggregated at the food/durables/non-durables level as well as at much more granular categorizations (see Table A3).
Another advantage relative to prior work is the information available regarding benefit income. I use reported household unemployment benefit or subsidy income to distinguish unemployment with benefits, denoted as UI recipients, from unemployment without such income after prior observed receipt, denoted as UI exhaustees. The constructed sample includes households with the following observed transitions: employment to employment, employment to unemployment with benefits, unemployment with benefits to unemployment with benefits, and unemployment with benefits to unemployment without benefits.[18] I exclude rent for homeowners (which is imputed), and deflate this measure using annual CPI. I restrict the sample to household heads aged 25-64 and who had less than a threefold change in consumption (Gruber, 1997; Kroft and Notowidigdo, 2016; Hendren, 2017).

Unemployment Spells. Information on unemployment spells and unemployment benefit transfers comes from the Social Security administrative records of the Continuous Sample of Employment Histories (*Muestra Continua de Vidas Laborales*). They are effectively matched employer-employee data with daily longitudinal information created by matching income tax data, Social Security records and census data. They cover a 4% non-stratified random sample of individuals that, for any year, have a relationship with the Social Security. The datasets comprise the years 2006-2017 and are used to build a sample containing the complete history of employment and unemployment spells since entrance in the labor market for

[16]The waves covered here correspond to an improved version of the survey where the sample size, duration of households' diaries completion and quality have increased substantially with respect to previous waves (INE, 2008).
[17]This corresponds to a more comprehensive version of the expenditure measure in Campos and Reggio (2020).
[18]Individuals with transitions from employment to unemployment without benefits are excluded because prior benefit receipt is not observed for those individuals, so they cannot be characterized as exhausting benefits.

individuals who had any relationship with Spain's Social Security during those years. An important variable in the analysis is potential benefit duration, which is not present in the dataset. For this reason, I compute it applying the UI eligibility rules to the working history of every individual. I measure both unemployment duration (days receiving UI, counted as days of nonemployment since job loss up to the potential benefit duration entitlement) and nonemployment duration (days between job loss and the beginning of a new employment spell). For the main analysis, given difficulties in the empirical computation of potential benefit duration, I focus on prime-age individuals with PBD between 8 and 22 months.[19] I cap my measure of nonemployment duration at 3 years (e.g., Schmieder et al., 2012; Lindner and Reizer, 2020).

Unemployment Rate and Demographics. The information on economic conditions is based on the microdata files of the Labor Force Survey (*Encuesta de Población Activa*) provided by the National Statistics Institute (INE). I compute unemployment rate measures at the region-year level over the period 2006-2017 for the analysis of the different components over the business cycle. Given differences in availability of geographical information across datasets, regions correspond to (52) provinces in the unemployment spell analysis, and (17) autonomous communities (CCAA) in the consumption and search effort analyses. I also use this dataset for two additional tasks: (i) obtaining shares of population for each demographic cell to achieve representativeness of the online survey; (ii) performing the prediction exercise based on search effort as described in Section 4.1.2.

4 Empirical Approach and Results

The framework described in Section 2 guides the empirical analysis, which aims to recover the required statistics for assessing the welfare effects of UI reforms that make benefit duration contingent on labor market conditions. I start by examining the evolution of the welfare weights and consumption moments, and continue analyzing the efficiency cost of the transfers.

4.1 Estimating Welfare Weights

A key statistic to evaluate how the social benefits provided by UI transfers vary with economic conditions is the ratio of welfare weights, $\frac{\omega^R}{\omega^B}$. Specifically, these weights capture society's relative valuations between the type of unemployed individuals receiving the transfers at different moments of the cycle, who may

[19] I focus on individuals with PBD from 8 to 22 months, discarding the cutoffs at the bottom and at the very top. Given these rules together with short contracts for individuals located around the first cutoffs, apparent bunching appears near these cutoffs. The last cutoff also presents bunching which is mechanical, given that the maximum work experience counted is 6 years, and those individuals tend to have been employed longer than that. In addition, I focus on individuals aged 26-54 given the longer work spells that enable better assignment into treatment, and the lower likelihood to be impacted by retirement and unemployment assistance policies. Given the statutory rules, an individual who gets unemployed but does not exhaust the PBD and works for a certain amount of time, can choose upon job loss what the new PBD will be between two alternatives: the remaining months from the old spell and the new ones. This information is missing in the data and thus makes the assignment of the running variable more difficult so I also restrict to the first spell of each individual in the sample. I also discard individuals whose previous job was less than part time, whose unemployment spell was not immediately preceded by an involuntary job loss, and whose unemployment spell was suspended. Finally, the most recent individuals included are the ones becoming unemployed in 2014, so that there are at least three years after the start of UI receipt for everyone in the sample.

differ in their underlying willingness to work. In the empirical implementation, I proxy $\overline{e}_i$ along the extensive margin of weekly job search, classifying the unemployed into two types: high willingness to work ($\overline{e}_A$, who search for jobs on a weekly basis under the given economic environment) and low willingness to work ($\overline{e}_B$, who do not search on a weekly basis).

To estimate the parameter of interest, I proceed in three steps. First, I examine preferences for redistribution between individuals of different willingness-to-work types. Second, I analyze whether the composition of the unemployed changes along the willingness-to-work dimension over the cycle. Third, I combine both results in order to estimate the ratio of welfare weights between the unemployed in recession and the unemployed in boom.

4.1.1 Preferences for Redistribution over Willingness-to-Work Types

I build on the insights from the generalized social marginal welfare weights approach of Saez and Stantcheva (2016) in order to understand whether society has a preference for redistribution over the willingness-to-work dimension. To operationalize this, I design and administer an online survey to estimate the statistics of interest highlighted by the framework. Here I describe the key questions, and refer the reader to Appendix C for more detailed information.

After responding to a set of demographic questions, individuals are shown the first question that is intended to elicit preferences for redistribution in a qualitative way. The question reads as follows:[20]

> *Imagine there are 2 unemployed individuals, A and B, who receive €950 while unemployed and are entitled to receive benefits for up to 14 months if remaining unemployed:*
>
> > *A looked for jobs last week*
> > *B did not look for jobs last week*
>
> *The government wants to transfer €1000 to individuals A and B in a one-time payment, and is thinking of how to allocate them. Please choose what allocation option you would prefer:*
>
> - *More money to A than to B*
> - *More money to B than to A*
> - *Same amount to A and B*

The answers to this question are shown in Panel (a) of Figure 1, which illustrates that around 75% of respondents prefer to allocate more resources to the individual who was looking for work in the previous week. Almost all of the other respondents would allocate the same resources to both individuals —meaning that they would not redistribute based on this dimension—, with virtually no respondents preferring to give more to the individual who did not search.

Although a stark qualitative picture emerges from the answers to this question, this is not enough to estimate the relative valuations between types. Therefore, in order to make progress on that front, after

[20] Note that the specific hypothetical benefit amount and potential duration correspond to the national average at the time of the survey as reported by the Spanish Employment Agency (SEPE).

the qualitative question, respondents are presented with 10 quantitative questions designed to obtain the statistic we are after. They are of the following form:

> *Imagine there are 2 unemployed individuals, A and B, who receive €950 while unemployed and are entitled to receive benefits for up to 14 months if remaining unemployed:*
>
> *A looked for jobs last week*
> *B did not look for jobs last week*
>
> *The government wants to transfer €1000 to individuals A and B in a one-time payment, and is thinking of how to allocate them. Now imagine that an unequal allocation may be costly. That is, it can allocate €500 each, or more to one than the other, but the latter option may result in the total amount of resources that can be distributed being reduced. Please choose what allocation option you would prefer:*
>
> – *500 for A and 500 for B*
> – *550 for A and 400 for B*

For each of these 10 questions, there are always two allocation options: one offering the same amount for both individuals, and the other one where the amounts differ between them. Under the equal allocation option, both individuals receive the same amount of €500 (i.e. $r_0^A = r_0^B = 500$). Under the alternative option, one individual gets €550 ($r_1^A = 550$) and the other individual an amount r_1^B obtained from the set $\{50, 100, 150, 200, 250, 300, 350, 400, 450\}$.[21] The identity of individuals A and B, the order of questions, as well as the order in which the equal allocation option appears within the question are all randomized. This intends to alleviate concerns about order effects.[22] Notice that because both hypothetical individuals face the same aggregate economic environment and the same baseline benefit policy, the difference in their weekly job search behavior isolates their willingness-to-work types, $\overline{e}_A$ and $\overline{e}_B$. In practice, the experimental variation created in the cost of the transfer, or Marginal Rate of Transformation (MRT), allows to identify the relative valuation over individuals, or Marginal Rate of Substitution (MRS), and induces individuals to accept the unequal allocation whenever:

$$\frac{\omega(\overline{e}_A)}{\omega(\overline{e}_B)} = MRS \geq MRT = \frac{r_0^B - r_1^B}{r_1^A - r_0^A} \tag{8}$$

Panel (b) of Figure 1 depicts the share of individuals who choose an unequal allocation for different values of the cost of the transfer. First, the figure shows that when the cost of the transfer is low, a large share of respondents choose the unequal allocation that gives more to individual A. Specifically, when there is no cost of transferring resources between individuals (MRT=1), the share is around 75%, which equals the figure obtained in the qualitative question.[23] Moreover, the share declines as the cost

[21] In practice, all individuals are presented with the same two options of 550 to A (B) and 450 to B (A). This allows me to assess data quality by comparing answers, for the same individual, to a question with no cost of transfer and asked in both qualitative and quantitative way. The other 8 questions are randomly drawn from the remaining set of transfer amounts.

[22] Note that the fact that each individual is presented only with 10 of these questions, and that there is order randomization, only has the objective of reducing fatigue and increasing data quality.

[23] As mentioned before in Section 3.2, the final sample excludes individuals that provide contradictory responses to the same question, asked in both a qualitative and a quantitative way. Table A1 shows this analysis sample is very similar to the broader sample in terms of observable characteristics.

increases, implying lower willingness to redistribute as more resources are lost to implement a given transfer. Overall, respondents exhibit a strong preference for redistribution from the individual who did not search last week and towards the one who searched. However, when asked to redistribute in the opposite direction, towards the individual who did not search, almost no one is in favor of that allocation. Leveraging the responses of individuals, I estimate the relative valuation (or average marginal rate of substitution) between the type of individual A and B as follows:

$$\frac{\omega(\overline{e}_A)}{\omega(\overline{e}_B)} = \sum_j (s_j - s_{j+1}) MRT_j \tag{9}$$

where s_j corresponds to the share of individuals accepting an unequal allocation when presented with an allocation option of cost MRT_j. Thus, I effectively compute a weighted average of the (binned) MRS distribution by summing over bins corresponding to all MRT values. Given the relatively high share still present at the largest cost considered in the design (MRT=9), my preferred specification assumes a linear extrapolation with a downward slope as estimated over the range before that point. Following this approach I obtain that $\frac{\omega(\overline{e}_A)}{\omega(\overline{e}_B)} = 12.8$, which reveals that respondents have a strong preference for targeting resources at individuals who exert effort.[24] Intuitively, this indicates a valuation of a €1 transfer to the individual who did not search for jobs at only €0.08 of the value placed on an equivalent €1 transfer to the individual who searched for jobs last week.

Preference heterogeneity. I examine whether the average preferences for redistribution elicited reflect homogeneity or are instead driven by specific groups. Results are shown in Figure A2. The findings reveal that preferences are remarkably homogeneous, even among groups that may be more likely to benefit from the policy. Unemployed individuals present slightly lower support for redistribution based on willingness to work relative to the employed, but still more than 50% wish to reward effort. While the baseline implementation elicits preferences for redistribution between types while fixing the economic environment, it could be the case that the welfare weight function also depends on economic conditions itself, on top of the unemployed's composition changes. To explore the importance of this issue, I examine whether preferences are heterogeneous by the unemployment rate of the respondents' region, and find that preferences are stable across economic conditions. I also consider whether previous experiences of respondents, during their impressionable years (18-25) (Krosnick and Alwin, 1989), affect their stated preferences. Specifically, I do not find that factors such as long-term unemployment of the parents or household's economic difficulties during this important period impact preferences along the willingness-to-work dimension.

Dimension of interest. It is plausible that respondents support redistribution based on effort, not due to a direct concern for effort itself, but rather because of some other underlying factor that correlates with

[24] Without extrapolating, assuming that all individuals at the upper tail of the distribution have exactly MRS=9, I obtain a lower bound of $\frac{\omega(\overline{e}_A)}{\omega(\overline{e}_B)} = 6.3$. As I show later, since the main statistic of interest is a ratio of welfare weights between the unemployed in boom and recession, the same assumption is made in both numerator and denominator and so the main results are not very sensitive to this issue.

this dimension. In order to understand what respondents have in mind when answering the questions, I ask them open-ended questions about the reasons for their allocations. A wordcloud with their answers is depicted in Panel (c) of Figure 1, where the most common expressions include "search for job", "effort" and "reward", implying that they directly consider as relevant the main dimension of interest.[25]

4.1.2 Unemployed's Composition

The second piece of information required is whether the composition of the unemployed shifts towards high-willingness-to-work types when economic conditions worsen. Intuitively, when labor demand falls during recessions, the offer arrival rate declines, search effort becomes less effective in terms of job finding, and a selection mechanism arises such that, all else equal, high-effort types are more likely to remain unemployed than when the overall unemployment rate is low. Here I provide empirical evidence that demonstrates the relevance of this mechanism.[26]

To do so, building on the survey evidence, I focus on the share of unemployed individuals who search for jobs in a given week.[27] Specifically, to match the theoretical definition of willingness-to-work types, we are interested in how the shares of high- and low-willingness-to-work types evolve over the cycle due to composition shifts, holding the benchmark environment fixed at boom economic conditions ($\overline{k} = B$). Because unemployed workers may also adjust their search behavior $e_{i,t}$ in response to lower job-finding opportunities during downturns, observed search effort in a recession reflects both changes in the composition of fixed types $\overline{e}_i$ and behavioral responses to changing economic conditions ($k = R$ relative to $\overline{k} = B$). The empirical analysis therefore estimates the effort levels that would be observed if all unemployed individuals faced a common boom environment, isolating how the composition of the unemployed pool shifts in terms of fixed types between booms and recessions.

For this analysis, I leverage the two available waves of Spanish Time Use data (2002-2003 and 2009-2010).[28] Exploiting the variation in unemployment rate across regions over the two time periods, I obtain

[25] Nevertheless, although respondents care about the effort dimension per se, there seems to be an understanding of confounders and heterogeneity in the population. For example, the analysis of the open-ended answers reveals that some individuals have in mind the case of single mothers with kids. They suggest that, despite this group potentially exerting less effort compared to others, they prefer not to support redistribution based on the effort dimension in this context, consistent with the results of the qualitative question. For this reason, the question used to elicit preferences intentionally does not fix other characteristics of the hypothetical individuals beyond the benefit policy, given that the unemployed's composition may change over the cycle along various dimensions in addition to effort. As a result, while understanding how preferences for redistribution vary depending on the source of effort differences is beyond the scope of this paper, the exercise below, which estimates welfare weights over the cycle, accounts for changes in the composition of the unemployed along dimensions that correlate with effort to the extent that respondents incorporate that heterogeneity/correlation. In relation to this, it is also important to note that the question fixes the potential unemployment benefit policy, and the transfers are lump-sum. This aims to obtain responses that are orthogonal to differences in consumption and to capture preferences based on effort.

[26] As mentioned before, since the statistic of interest is average willingness to work over the cycle, it is not necessary that the main mechanism driving composition changes is a decrease in job finding in bad times. This could also be explained by a different composition of layoffs.

[27] Given that according to the International Labour Organization (ILO) definition of unemployment, a requirement to be considered unemployed is to have searched for jobs in the last month, this explores the degree of search intensity within that period. It is also a relevant measure considered in other surveys like the Survey of Consumer Expectations (SCE) carried out by the Federal Reserve Bank of New York (Faberman et al., 2022).

[28] For more details on data construction, see Appendix D.

the results shown in Panel (e) of Figure 1.[29] The figure shows that the observed share of unemployed individuals who search in a given week increases strongly with the unemployment rate. Specifically, it goes from around 54% when the unemployment rate is 8% to more than 80% when the unemployment rate exceeds 20%.

Importantly, not all the fluctuations in search effort are necessarily due to a composition effect.[30] As mentioned before, individuals may also change their search behavior while unemployed in response to changes in the availability of jobs. Evidence from Mukoyama et al. (2018) using panel data from the US indicates that around 50% of the search effort increase observed over the cycle is driven by composition shifts, with both observable and unobservable characteristics playing an important role.[31] Applying that statistic to our context in order to correct for the behavioral response and isolate changes in type composition (i.e., evaluating all individuals at $\overline{k} = B$), I find that the share of high-willingness-to-work types among the unemployed increases from 54% in boom to 70% in recession.

For the estimation of the welfare weights below, I perform a sensitivity analysis where I consider different magnitudes for the change due to composition effects. In addition, I also perform a complementary prediction exercise using only Spanish data, where I predict search effort based on the set of demographics present in both the Time Use data and the Labor Force Survey. Specifically, I estimate the relationship between search effort and the demographic characteristics based on the Time Use data and then impute search effort in the Labor Force Survey using the same demographics, being able to condition on a sample of individuals no longer receiving UI benefits.[32] Note that the set of demographics in both datasets is limited and the cross-sectional nature of the dataset used from the Labor Force Survey for this analysis does not allow to account for unobservables, and so this exercise only captures composition changes associated with the observed demographics here in contrast to the richer data setup in Mukoyama et al. (2018). Nonetheless, Figure A4 reassuringly shows that predicted search effort changes over the business cycle, even when only considering this limited set of demographics present in the Spanish data.

4.1.3 Combining Information: Unemployed in Recession vs Boom

The previous sections showed that society exhibits strong preferences for transferring resources from low- to high-willingness-to-work individuals. Moreover, the composition of the unemployed shifts towards high-willingness-to-work types in bad times. Here I combine both pieces of information to obtain an

[29] Given data availability the analysis uses only two years, and residualizes the variables on region FEs to not consider variation from structural differences across regions that are constant over time. Note that the relationship documented is not only present in Spain, and does not depend on the specific search effort measure or the specification used. See Table A2 for evidence of overall countercyclicality of usage of different search methods over 15 years across 34 countries using data from Eurostat.

[30] For related literature that investigates the drivers of the cyclicality in the job finding rate instead of search effort, see e.g., Kroft et al. (2016) and Mueller and Spinnewijn (2025).

[31] The authors use market tightness as the measure of economic conditions and minutes of search as the measure of search effort. They exploit the panel dimension of the Current Population Survey (CPS) to account for both observables and unobservables. Findings from their working paper using unemployment rate instead present similar results.

[32] For this exercise, I estimate the model in the two Time Use waves and predict search effort in the LFS over 2005-2015. The demographics present in both datasets that are used in the prediction model are interactions between age categories and sex; between education categories and sex; and between married status and sex.

estimate of how society values transfers between the unemployed in recession and boom given the changes in composition along the effort dimension.

From Section 4.1.1, society's relative valuation across willingness-to-work types is $\frac{\omega(\overline{e}_A)}{\omega(\overline{e}_B)} = 12.8$. From the unemployed composition analysis in Section 4.1.2, the shares of high- and low-willingness-to-work types ($\overline{e}_A$ and $\overline{e}_B$, evaluated at the benchmark economic environment) among the unemployed in boom and recession are: $s_A^{Boom} = 0.54; s_B^{Boom} = 0.46; s_A^{Recess.} = 0.70; s_B^{Recess.} = 0.30$. Combining these two inputs, the average welfare weights in recession and boom are $\omega^R = (s_A^{Recess.})\omega(\overline{e}_A) + (s_B^{Recess.})\omega(\overline{e}_B)$ and $\omega^B = (s_A^{Boom})\omega(\overline{e}_A) + (s_B^{Boom})\omega(\overline{e}_B)$. This yields a ratio of welfare weights of $\frac{\omega^R}{\omega^B} = 1.27$, which indicates that society values a €1 transfer to the type of individuals unemployed in recession as much as €1.27 to those unemployed during a boom.

Composition changes along willingness to work. The baseline results documented in this section rely on an assumption about changes in the composition of the unemployed. Here I explore how sensitive estimates of the ratio of welfare weights are to implementation assumptions. On the one hand, under the strong assumption that the distribution of MRS does not have values larger than the ones randomized, we get a ratio of 1.23. On the other hand, using the baseline estimate, even if the part of the search effort increase in recession that is due to composition is 25% — half of the one found in the US —, the magnitude is 1.13. Alternatively, using the variation in effort due to composition effects obtained from the prediction exercise that only uses the Spanish Time Use data combined with data on unemployed without benefits from LFS, I obtain a similar magnitude: 1.12.[33] As explained before, this likely understates the composition shift given that the design does not account for unobservables and the predictors set present in the data is sparse. Yet, this approach has the advantage of focusing on UI exhaustees rather than the average unemployed, reassuringly showing that this group also exhibits substantial composition changes over the cycle. Finally, assuming all change in the baseline sample is due to composition effects, the magnitude is 1.53. Overall, the values remain close to the baseline estimate of 1.27.

This implies strong preferences for redistribution along the specific effort dimension, a result that is robust to different implementation assumptions. In summary, the findings demonstrate that there is an important shift in the composition of the unemployed over the cycle, and that society places a substantially higher value on transfers towards the type of individuals who tend to be unemployed in bad times.

4.2 Estimating Consumption Smoothing

In addition to the ratio of welfare weights, the second key element needed to assess the social benefits of UI transfers is how moments of consumption during unemployment vary with labor market conditions. As mentioned before, differences in consumption while unemployed for two groups of individuals (unemployed in recessions vs boom) depend on both consumption changes upon job loss and pre-unemployment consumption levels.

[33] This specification restricts the LFS sample to registered unemployed without benefits, and compares the change in composition in terms of observables between 2007 and 2013 (Panel (b) of Figure A4).

I start by evaluating how individuals change their consumption upon unemployment, and whether this change varies over the cycle. This sheds light on the relative insurance value of providing more generous UI transfers in bad times relative to good times. To quantify these patterns, I adopt a difference-in-differences strategy similar to Gruber (1997), where I compare the consumption behavior of household heads getting unemployed with those who remain employed over the business cycle. Importantly, these consumption moments are analyzed separately for both unemployed receiving benefits and those beyond the exhaustion point. Specifically, I estimate the following model:

$$\begin{aligned} logC_{it} = \beta_0 UIRecipient_{it} + \beta_1 UIExhausted_{it} + \beta_2 UIRecipient_{it} \times U.Rate_{r(i,t)t} \\ + \beta_3 UIExhausted_{it} \times U.Rate_{r(i,t)t} + \lambda_i + \lambda_{r(i,t)t} + X'_{it}\theta + \varepsilon_{it}, \end{aligned} \tag{10}$$

where $logC_{it}$ refers to log consumption measured at the household level, $UIRecipient_{it}$ is a dummy variable that equals one when the household head is unemployed and receiving benefit income, $UIExhausted_{it}$ is a dummy variable that equals one when the household head is unemployed and not receiving benefit income (being a previous recipient), $U.Rate_{r(i,t)t}$ refers to the unemployment rate in the region where individual i lives at time t, and λ_i and $\lambda_{r(i,t)t}$ are individual, and region-year fixed effects, respectively. The vector of controls X includes information on type of household.[34]

Table 1 shows the main results on consumption changes upon job loss. First, in column (1) of Panel A, we see that, on average, individuals experience a 10.1% (s.e. 1.3%) consumption drop upon unemployment with benefits. This drop is substantially larger for unemployed who exhaust benefits, reaching 22.1% (s.e. 3.1%). From columns (2) and (3) of Panel A, we see that the average consumption drop masks a clear gradient with respect to economic conditions, which is present for both groups of unemployed individuals. Focusing on the latter column, our preferred specification that includes region-year fixed effects, a 1 p.p. increase in unemployment rate results in an additional -0.6% (s.e. 0.1%) change in consumption for UI recipients relative to their pre-unemployment level. Similarly, for the group of UI exhaustees, a 1 p.p. increase in unemployment rate results in an additional -0.8% (s.e. 0.3%) change in consumption. Columns (4)-(6) of Panel A include additional covariates that intend to capture changes in household composition. Results are very similar to the ones without these covariates. Panels B and C present the estimated consumption changes upon job loss at unemployment rates corresponding to boom and recession. The estimates go from approximately 2% to 13% for benefit recipients, respectively, and are statistically different from each other. In the case of exhaustees, the difference is also statistically significant, going from 11% in boom to 25% in recession.

Panels (a) and (b) of Figure 2 depict the main result, focusing on the baseline specification and grouping individuals by their UI status and regional unemployment rate. The groups are selected so that the average unemployment rate in each group roughly approximates the national average in boom, normal times, and recession. It shows that, for both benefit recipients and exhaustees, the within individual change in consumption between employment and unemployment gets significantly and monotonically

[34]These controls are solely intended to keep household size fixed since the consumption measure is at the household level.

larger when the unemployment rate increases. This is in contrast to the findings of Kroft and Notowidigdo (2016), where the authors do not find that the individual drop gets larger when economic conditions deteriorate. The difference in findings may be explained by limited power to detect an effect in their study given the relatively small sample size and/or variation in unemployment rate, and the fact that the consumption measure only captures food components. Moreover, they are not able to focus on UI recipients and instead perform the analysis on a broader group of unemployed. In my setting, there is relatively larger variation in economic conditions, a more comprehensive consumption measure that goes beyond food consumption and information on individuals' UI status, which allows me to revisit this issue and obtain further insights.

To do so, I decompose the change in the consumption drop over the cycle by consumption category in order to understand which components of the consumption basket drive the cyclicality of the overall drop. Specifically, I estimate models of the form:

$$\begin{aligned} Y_{it}^{j} = \beta_0^j UIRecipient_{it} + \beta_1^j UIExhausted_{it} + \beta_2^j UIRecipient_{it} \times U.Rate_{r(i,t)t} \\ + \beta_3^j UIExhausted_{it} \times U.Rate_{r(i,t)t} + \lambda_i + \lambda_{r(i,t)t} + X_{it}'\theta + \varepsilon_{it}, \end{aligned} \tag{11}$$

where Y_{it}^j refers to either individual's consumption within category j relative to overall average consumption of the individual, $\tilde{c}_{it}^j = \frac{C_{it}^j}{\frac{1}{2}(C_{i1}+C_{i2})}$, or individual's expenditure shares $\tilde{s}_{it}^j = \frac{C_{it}^j}{C_{it}}$. The former measure has a clear interpretation, since adding up all the estimates across categories approximately yields the total change in the consumption drop documented before (Table 1).[35] The latter measure provides information on expenditure reallocation within total expenditure. For each category, I compute the difference between the consumption change in recession and the change in boom.[36] I focus on the preferred specification (column (3) in Table 1), now analyzing consumption patterns by categories. I start the category analysis focusing on three broad categories including food, durables and other non-durables. The results are depicted in Panels (c) and (d) of Figure 2, which show that the drop in food consumption upon unemployment presents limited cyclicality. For unemployed receiving benefits, the results show a somewhat larger, albeit still modest, drop in bad times. For unemployed without benefits, this category clearly does not vary with the unemployment rate. In contrast, the overall cyclicality in the consumption drop seems mostly explained by the behavior of the durable and other nondurable categories.

Given the richness of the data, the next part of this analysis zooms in on more granular categories. Beyond food, the larger drop in overall consumption in recession seems to arise mainly from larger drops in a few consumption categories: housing, utilities, and clothing and shoes (Panels (a) and (b) of Figure A5).[37][38] In the case of UI exhaustees, the drop is relatively larger for housing and utilities in bad times,

[35] It also allows a direct mapping across specifications by focusing on the same set of individuals, including those reporting no consumption for a given category at a given point.

[36] For instance, in the case of UI exhaustees, this statistic is computed as $\beta_3^j \times (U.Rate^R - U.Rate^B)$, where economic conditions are defined as recession corresponding to 26% and boom to 8% unemployment rate.

[37] Note that the design compares the consumption behavior of the unemployed relative to employed individuals in the same region and time period, so that the results are not explained by aggregate shocks that are common across groups such as the housing boom and bust documented in Section 3.1.

[38] The specific categories where consumption changes become most pronounced with labor market conditions may differ in

with clothing and shoes not being precisely estimated.

In relation to budget share reallocation, and given the limited cyclicality of food, the pattern is clear: a larger share of the budget is devoted to food upon unemployment, together with a share reduction in housing and utilities (Panels (c) and (d) of Figure A5). Moreover, as shown in Table A4, which presents the most detailed picture on what components of the consumption basket are cyclically adjusted by each group of unemployed, we can see that the specific components explaining the larger decline in bad times for UI exhaustees are electricity, heating and maintenance materials.

Consequently, my findings can help reconcile the empirical result in Kroft and Notowidigdo (2016). Although their theoretical model predicted a larger consumption drop in bad times, they find that it does not significantly vary with economic conditions when measured only by the food component.[39] Given my findings, it thus seems that focusing only on a component such as food consumption that is relatively inelastic may limit the understanding of the welfare costs of job loss.[40] Therefore, this result highlights the importance of empirically assessing consumption behavior using a more comprehensive measure.

After analyzing consumption changes, I turn to evaluating differences in pre-unemployment consumption levels across the unemployed in different economic conditions. This complements the focus on the insurance value and sheds light on the distributional aspect of the transfers. Specifically, with the aim of focusing on pre-unemployment consumption for this exercise, I restrict the sample to only consider the first year in which the individuals are observed in the data, when they are all employed. I consider a version of equation (10) where I compare individuals who lose their job next period with those who remain employed. Given the requirement of observing prior benefit receipt for defining UI exhaustees, this specification focuses on UI recipients. Specifically, I estimate the following regression equation:

$$logC_{it} = \beta_0 + \beta_1 UIRecipient_{it+1} + \beta_2 UIRecipient_{it+1} \times U.Rate_{r(i,t)t} + \lambda_{r(i,t)t} + X'_{it}\theta + \varepsilon_{it}, \tag{12}$$

where $UIRecipient_{it+1}$ is a dummy variable that equals one when the household head is unemployed with benefits in the next year, and zero otherwise. The results are shown in Table 2, which examines whether the pre-unemployment gap in consumption for individuals who become unemployed relative to those who do not changes over the business cycle. Columns (1)-(6) indicate that there exists a pre-unemployment consumption gap which does not change with economic conditions. This evidence implies that although the UI system redistributes between people of different pre-layoff consumption levels, it does not do so differentially in booms and recessions. Therefore, the lack of variation of this component, together

their degree of durability relative to the average category. Therefore, when evaluating consumption-smoothing benefits they would correspond to a different curvature. Although accounting for category-specific curvature is beyond the scope of this paper, the fact that consumption changes across categories are non-positive implies that the finding of larger welfare gains arising from consumption smoothing in bad times is robust to heterogeneity in preferences across categories.

[39] In this analysis there is a slightly different sample construction to differentiate between UI recipients and those who exhaust benefits. However, when constructing the sample in a similar way to them and pooling all the unemployed the results are very similar with overall consumption cyclicality that is significant and not driven by food (see Table A5, and Panels (a) and (b) of Figure A6).

[40] Although it is sufficient in theory to just use food consumption — provided that one uses the appropriate curvature of utility over food (Chetty, 2006) —, in practice obtaining estimates that are precise enough to detect cyclical variations in this component may be difficult.

with the evidence on consumption drops, suggests that the insurance component is the main driver of consumption differences during unemployment between the individuals losing their jobs under different economic conditions.

Sensitivity of consumption evidence. Further analysis presented in Table A6 explores mechanisms and sensitivity of the main results regarding consumption smoothing. While the findings above reveal that overall consumption drops are larger in bad times, here I examine whether this is solely driven by longer unemployment duration in recessions. Results are robust to controlling for either regional unemployment duration or regional unemployment duration distribution (share of unemployed longer than 6 months). This suggests that the larger drop found in bad times is not fully explained by longer unemployment spells, with larger drops at a given duration potentially being the main driver.[41] Furthermore, I assess whether the 2012 reform which slightly decreased the replacement rate after 6 months (see Section 3.1) can mechanically explain the countercyclical drop in consumption.[42] The findings indicate that this policy change does not drive the main results. Consequently, the evidence is consistent with other factors in addition to longer unemployment durations, such as lower individual asset levels in bad times, leading to larger consumption drops even conditional on unemployment duration.

Overall, the results here have shown that consumption smoothing varies substantially over the cycle, with the unemployed in bad times suffering considerably larger consumption drops upon unemployment than the unemployed in good times. Moreover, this larger drop arises from countercyclical drops in non-food categories.

4.3 Estimating Efficiency Cost

When increasing potential benefit duration, individuals may change their search behavior and thus impose some extra cost on the government budget. Equation (7) highlights the main statistics needed to quantify this component. As described in Appendix B.2, for the finite PBD extension studied here, I estimate the total changes in unemployment duration $\Delta D_b^{k_0}$ and nonemployment duration ΔD^{k_0}, and the average exhaustion rate over the extension window $\overline{S}_{P+1}^{k_0}$.

In order to gain a better understanding of the effect of PBD on unemployment and nonemployment duration, I start by providing graphical evidence of the behavior exhibited by unemployed individuals when entitled to additional benefit generosity. As outlined before, the institutional setup delivers a treatment assignment mechanism typical of a regression discontinuity design, and thus my empirical strategy exploits the discontinuities in potential benefit duration present in the UI system. Figure A7 illustrates the structure of the UI system for the main sample, where it is shown that PBD discretely increases by 2 months at specific cutoffs of recent work experience. To estimate the statistics of interest, the

[41] Kolsrud et al. (2018) show that the consumption drop gets larger over the unemployment spell in the Swedish context. Unfortunately, a measure of unemployment duration is not available at the individual level in the Spanish consumption data, so I construct it at the region-year level from the LFS, potentially introducing some measurement error that may attenuate the results.

[42] An alternative explanation for the lower ability to smooth consumption in bad times could be that effective potential benefit duration in the sample decreases in bad times, mechanically reducing benefit generosity. I explore this aspect and find it is not the case.

empirical approach pools all cutoffs together. First, I provide evidence for the validity of the design in two ways: testing whether there is any discontinuity in the density of the running variable following McCrary (2008), and assessing whether selection on observables could explain the effects on the outcomes of interest (Card et al., 2015). The reported test statistic does not reject the null hypothesis of no density discontinuity, and the covariate index of the outcomes also evolves smoothly at the threshold as depicted in Figure A8. Second, I proceed to analyze the impact of higher benefit generosity on the main labor market outcomes as shown in Figure 3. Panel (a) shows the discontinuity in PBD that arises when crossing the cutoff. Panels (b) and (c) depict the conditional expectation functions of unemployment and nonemployment duration, which also present discrete jumps exactly at the cutoff, consistent with PBD having a positive effect on both outcomes. To quantify these effects, I estimate models of the following form:

$$Y_{it} = \beta_0 + \beta_1 \mathbb{1}[Experience_i > 0] + \beta_2 \mathbb{1}[Experience_i > 0] \times U.Rate_{r(i,t)t} + \lambda_{r(i,t)} + \lambda_t + X'_{it}\theta + \varepsilon_{it}, \tag{13}$$

where Y_{it} refers to potential benefit duration, unemployment duration or nonemployment duration, $\mathbb{1}[Experience_i > 0]$ is a dummy for being on the right side of the normalized cutoff, $U.Rate_{r(i,t)t}$ refers to unemployment rate at the region level from the quarter when the individual became unemployed, $\lambda_{r(i,t)}$ and λ_t are region and year fixed effects, respectively, and X is a set of controls including the regional unemployment rate and a (linear) polynomial in normalized work experience within the last 6 years which is allowed to have a different slope at each side of the cutoff.

The results are shown in Table 3. In column (1), we see that PBD increases at the cutoff by 57.17 (s.e. 3.10) days, reflecting the discontinuous structure of the UI schedule previously described. In column (3), we see that unemployment duration increases by 30.69 (s.e. 5.33) days in response to the higher PBD enjoyed when crossing the cutoff. When looking at the effect on nonemployment duration, column (5), we see that there is an increase of 31.30 (s.e. 14.14) days. In terms of marginal effects, they correspond to an effect of PBD on unemployment duration of 0.54, and an effect of PBD on nonemployment duration of 0.55. In terms of elasticity, focusing on nonemployment duration which is the outcome that has received more attention in the literature as surveyed by Schmieder and von Wachter (2016), I find a value of 0.58. This corresponds approximately to the middle value within the top quartile of the distribution of estimates reported in the survey. All of these together imply that providing the unemployed with more generous UI benefits induces large behavioral responses.
Columns (2), (4), and (6) investigate whether these responses vary when the unemployment rate increases. I do not find that the effects are different from each other when the increase in generosity is implemented in boom relative to recession.[43]

Now I turn to assessing how the exhaustion rate, which refers to the average value of the survival function over the additional entitlement window, varies with the unemployment rate. Given the discrete nature of the PBD change, I estimate it nonparametrically using individuals below the cutoff in the

[43]The results are very similar when adding covariates as shown in Table A7.

following way:

$$\overline{S}_{P+1}^{k_0} = \frac{1}{60}\mathbb{E}[m_i|k_{i,0} = k_0], \tag{14}$$

where $m_i = \min\{D_i, P_i+60\} - \min\{D_i, P_i\}$ measures the number of days individuals remain nonemployed within the additional PBD window, and so it captures the mechanical increase in benefit coverage generated by the extension. Panel (d) of Figure 3 depicts the results, where it is shown that the share of individuals that are not able to find a job before exhausting unemployment benefits increases with the unemployment rate, going from 0.29 to 0.41 when the unemployment rate changes from 8% to 26%, respectively.

Macroeconomic effect. The distortion induced by cycle-dependent UI benefit reforms documented above should be interpreted, in principle, as a micro effect, since the design identifies labor supply responses for individuals with different benefit durations keeping labor market tightness constant. As pointed out by Landais et al. (2018a;b), when labor market tightness is not efficient, the welfare effect of UI reforms also needs to include a correction term to account for the fact that tightness may also respond to UI increases. When the correction term is countercyclical, as the authors argue, my finding can be interpreted as a lower bound, being possible that the overall cost of providing UI varies potentially even more with economic conditions than estimated here. However, when the micro and macro effects are equal, even if tightness is inefficient, UI changes do not shift tightness and the correction term vanishes. Recent meta-analysis evidence by Cohen and Ganong (2026) supports the latter scenario, in which case my finding corresponds to an exact result and no correction term is needed.

5 Welfare Implications

In relation to the value provided by the unemployment insurance transfers, the empirical analysis above has shown that there are larger social benefits of increasing the potential duration of benefits in bad times relative to good times. Table 4 presents the estimates for the marginal benefit term, combining the findings on welfare weights and consumption smoothing for individuals exhausting benefits (column (3) in Table 1). For the benchmark specification, with $\frac{\omega^R}{\omega^B} = 1.27$ and a risk aversion parameter $\gamma = 4$ (Landais and Spinnewijn, 2021), I find that the social benefits of transferring resources towards the unemployed in recession are 77% larger than for the unemployed in boom. Moreover, a main takeaway that arises from the sensitivity analysis in Table 4 is that, regardless of the implementation assumptions, the value provided by UI transfers is larger in recessions. This is true when focusing only on the consumption insurance value, but the magnitude is considerably larger when also accounting for the preferences for redistribution together with the shift in the composition of the unemployed along the effort dimension.

In relation to the cost, the estimates above imply that benefit extensions are less costly when economic conditions are bad. Using those estimates to calibrate the fiscal externality formula in equation (7), I obtain the results presented in Table 5. At the mean unemployment rate (15%), for every €1 transferred mechanically to the unemployed, the government needs to levy €0.59 on top of that because of the behavioral response. This efficiency cost is measured as euros of behavioral cost per euro of total

mechanical transfer. Moreover, the fiscal externality is larger in good times, ranging from €0.89 in boom to €0.34 in recession. Overall, these estimates on the efficiency cost are consistent with the ones obtained by Schmieder et al. (2012), where they find that the effects remain constant over the cycle, with the only relevant component that changes with economic conditions being the exhaustion rate.

Lastly, I combine social benefits and costs to quantify the net welfare impact of changing potential benefit duration according to labor market conditions. To do so, I consider the effect of the budget-balanced reform described above such that $b^R_{P+1} > b^B_{P+1}$ as shown in Appendix B.3. Normalizing by the marginal utility of unemployed exhausting benefits in boom, I find that the net welfare impact of the reform is €1.06. That is, society would gain an extra €1.06 per euro transferred mechanically from individuals unemployed in boom to individuals unemployed in recession. Moreover, I decompose the welfare gain between the different components as shown in Figure 4. I find that they are all quantitatively important. Specifically, the welfare weights account for around 30% of the total gain, while the consumption smoothing component accounts for 43% and the efficiency cost channel for 27%. This implies that there are large welfare gains from having a system where the generosity of the transfers varies with economic conditions. Importantly, this analysis highlights that the welfare gain not only increases in recession because of an efficiency cost decline as emphasized by the bulk of the empirical literature (e.g., Kroft and Notowidigdo, 2016), but also because the social benefits of the transfers increase considerably in bad times. The increase in social benefits is found when focusing only on the consumption insurance channel, but it becomes substantially more pronounced when accounting for preferences for redistribution and changes in the composition of the unemployed. Overall, these estimates strengthen the case for shifting benefit duration towards recessions, with a larger variation than implied by previous work which documented that the social benefits do not vary with economic conditions.

From local gains to optimal PBD. The baseline analysis focuses on local reforms, applying the robust framework without functional form assumptions on how social benefits and efficiency costs vary with PBD. Although finding the global optimal policy is beyond the scope of this paper, to illustrate what the estimates imply for the magnitude of PBD cyclicality I perform a simple non-local exercise that takes the expected UI budget as given, and considers reallocation of PBD across labor market conditions. For simplicity, I hold consumption moments, welfare weights, survival curves, and behavioral costs constant with respect to PBD.[44] For Spain, setting PBD in recession to the benchmark 24-month statutory maximum, while holding expected UI budget at the level of a uniform statutory median policy of 14 months, requires lowering PBD in boom to 5.8 months. At this 5.8-24 month policy allocation, equalizing marginal welfare gains per euro of fiscal cost across states requires social benefits to be only 4.5% larger in recessions relative to booms. Given the estimated social benefits in recession are 77% larger, the marginal return to recession transfers remains substantially higher even at this level of differentiation. Thus, this simple

[44]The exercise uses $\gamma = 4$ and consumption drops for UI exhaustees. Even though survival curves are constant relative to the policy, they decline with unemployment duration and so efficiency costs are allowed to vary via changes in the exhaustion rate. For Spain, entrant shares are $\alpha^B = 0.46$ and $\alpha^R = 0.54$, obtained using data on initial contributory UI payments in 2007 and 2013 from SEPE.

exercise highlights that Spanish PBD cyclicality should be large, and that benefit duration should have been considerably more generous during the Great Recession.

Applying a similar approach to the United States helps thinking about the optimality of actual policy, focusing on the extension of maximum UI duration from 26 to 99 weeks implemented during the Great Recession. To that end, I evaluate a stylized 26-99 week PBD allocation across 5% and 10% unemployment rates while holding its expected UI budget fixed.[45] Equalizing marginal welfare gains per fiscal euro requires social benefits to be 14.4% higher in recessions than in booms. Consequently, considering efficiency costs alone, which implicitly sets the social benefit variation across labor market conditions to zero, cannot rationalize the 26-99 week policy and would favor a lower duration cyclicality of 37-85 weeks. In contrast, applying the Spanish consumption drop gradient per percentage point of unemployment to the US recession yields a social benefit that is 11.9% higher in bad times, rising to 19.5% when the welfare weight gradient is also applied. Thus, in this calibration, because the increase required for optimality falls within this range, accounting for the cyclicality of social benefits helps rationalize the scale of observed US emergency extensions.

Other policies. The approach adopted in the paper is conditional on the current tax and transfer system, and specifically, on other unemployment policies. In particular, the framework assesses the welfare gains of differentiating the UI policy taking as given the distribution of unemployed types. However, some countries like Germany, and unlike Spain, enforce job search requirement policies in order to monitor search effort, which may impact the distribution of unemployed types receiving benefits. It is worth noting that the proposed framework already accounts for current unemployment policies, so that in countries where monitoring is more prevalent and low-willingness-to-work types are less likely to receive UI, we would observe smaller variation in the composition of the unemployed along this dimension over the cycle, and thus this novel channel would account for a smaller share of the welfare effect. Nevertheless, these policies are costly to implement and even if put in place, the evidence indicates that there is imperfect compliance (Arni and Schiprowski, 2019). As a result, the framework is complementary to these policies.[46]

Support for UI reform. Beyond accounting for social preferences to obtain a comprehensive assessment of the welfare effects of cycle-dependent UI policies, focusing on this dimension also provides information on the social acceptability of reforms. This is useful for policymakers considering implementation in democratic countries. I provide two pieces of suggestive evidence in this respect. First, within the online

[45] For this analysis, I combine survival curves from Kroft and Notowidigdo (2016) with the nonemployment duration response of 0.28 days per additional entitlement day summarized by Le Barbanchon et al. (2024), and $\tau/b = 0.06$ as in Spain. Survival curves beyond 50 weeks and the survival curve at 10% unemployment are extrapolated. Entrant shares are $\alpha^B = 0.42$ and $\alpha^R = 0.58$, based on number of first UI payments in 2005 and 2010, obtained from the US Department of Labor.

[46] The findings documented here on changes in the composition of the unemployed with labor market conditions are also relevant for the discussion of optimal benefit profiles over the unemployment spell. In the presence of dynamic selection in terms of search effort, the share of high-willingness-to-work individuals may also vary over the unemployment duration distribution. Evidence suggests this selection may be important (e.g., Faberman and Kudlyak, 2019; Lalive et al., 2026). Therefore, my findings have important implications for the optimal profile. Although beyond the scope of this paper, future work could apply the proposed framework for the design of UI over the spell while keeping economic conditions fixed, exploring the role of dynamic selection in terms of effort as an alternative motive for differentiating benefit generosity between the short- and long-term unemployed.

survey, I explore qualitatively whether there is hypothetical support for a UI reform of the type studied in this paper. Panel (d) of Figure 1 indicates that respondents state they would approve the proposed differentiation of the policy over the cycle. Second, in Appendix E, I also leverage an alternative public survey with data on public opinions from Spain containing views of around 38,000 individuals over the period 2005-2019. I focus on a question asking whether unemployment benefits should be same/more/less generous than in the given year. The results in Figure A9 indicate higher support for increasing benefit generosity in bad times. Although the setting does not allow to identify the drivers behind this demand for UI policy, the analysis can rule out factors such as respondent's own employment status or unemployment risk.[47]

6 Conclusion

Social benefits are the fundamental reason why unemployment insurance policies exist, and are therefore central to understanding whether benefit generosity should vary with economic conditions. Yet, the empirical literature and policy debate have focused predominantly on its efficiency costs, largely considering a narrow role for the social benefits of UI and overlooking how they change over the cycle.

This paper emphasizes the important and broad role of the social benefits, providing a comprehensive assessment of whether the duration of the benefits should be higher when economic conditions are bad. I propose a simple and robust framework where the welfare effects from allowing the UI system to vary according to the state of the economy depend on the preferences for redistribution over effort types, the consumption smoothing gains, and the efficiency cost. I exploit a newly collected online survey, rich consumption data, administrative data on labor supply, and time use data on job search to document four facts. First, society has a strong preference for transferring resources towards high-willingness-to-work individuals. Second, the pool of unemployed shifts towards high-willingness-to-work individuals in bad times. Third, individuals reduce their consumption more when hit by unemployment in recession. Fourth, the efficiency cost is large, and decreases in bad times.

Overall, I find that the welfare gains from shifting UI benefit duration towards recessions arise primarily from the greater social benefits of the transfers, rather than their lower efficiency costs. These gains are larger than previously found in the literature, highlighting the importance of accounting for the increasing social benefits of the transfers in recession. Moreover, these additional gains are not only due to the standard consumption insurance channel, but also come from the important role of society's preferences for redistribution along the willingness-to-work dimension together with shifts in the composition of the unemployed.

[47]These results should be viewed as providing suggestive evidence on the public support for UI policy generosity. However, it is generally difficult to infer social preferences from the type of data described here, as pointed out in related important work such as Almås et al. (2020). In particular, individuals may hold different beliefs about the source of unemployment and the efficiency cost of providing benefits, which cannot be measured or controlled as in the main newly-collected online survey. See Appendix E for further details on results using the public opinion survey.

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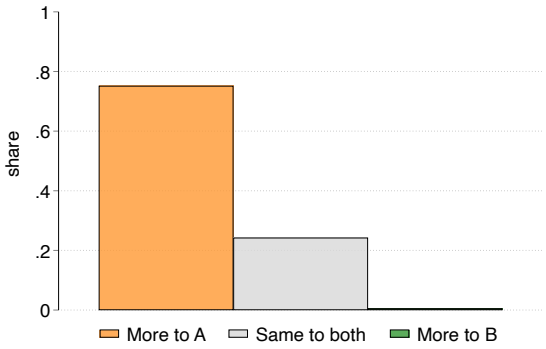
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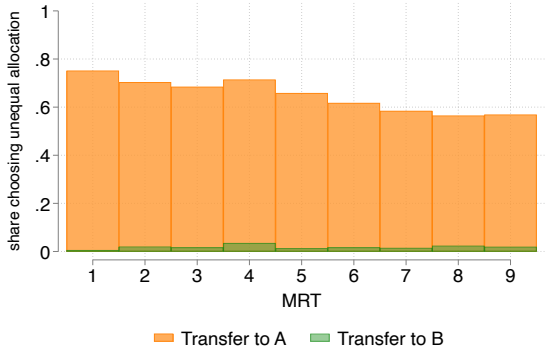
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Figure 1: Redistribution Over Search Effort Types

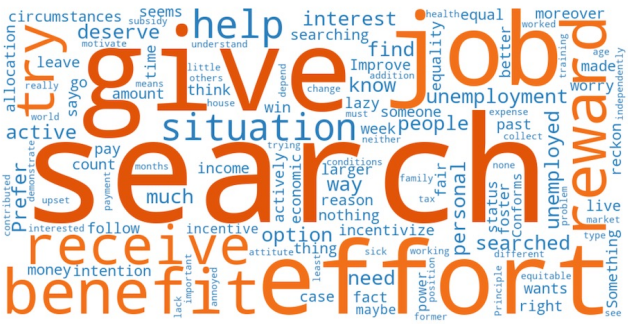
(a) Preferences for Redistribution - Qualitative



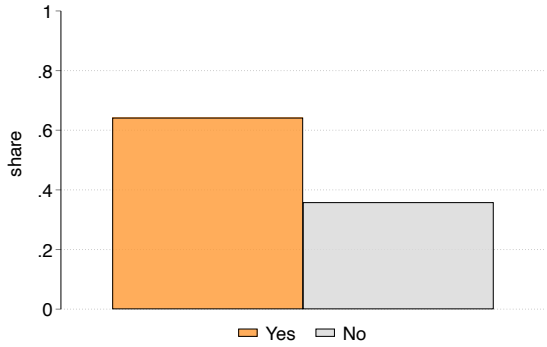
(b) Preferences for Redistribution - Quantitative



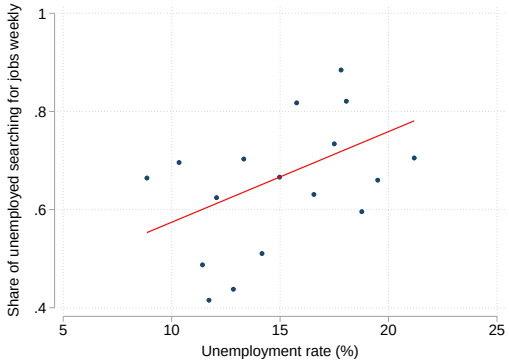
(c) Reason for Allocation



(d) Support for UI Reform



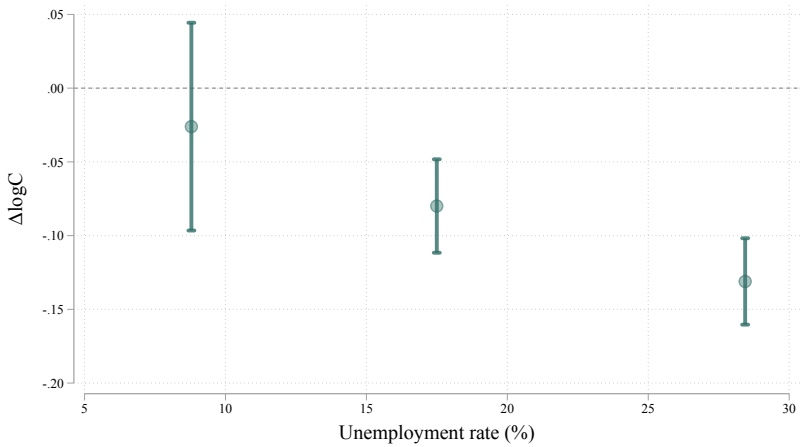
(e) Unemployed's Search Effort Over the Business Cycle



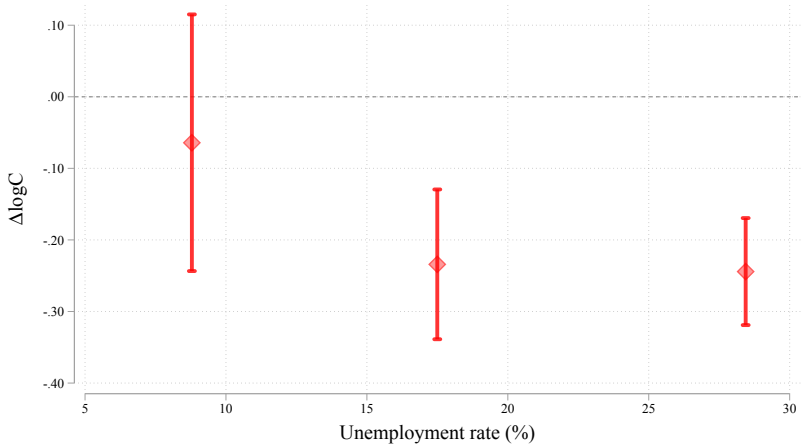
Note: The graphs in Panels (a)-(d) present results on preferences for redistribution from the online survey, while Panel (e) presents results from Time Use data on search effort changes among the unemployed. In particular, Panel (a) depicts the share of respondents choosing a given option when presented with the question on qualitative preferences for redistribution: *Imagine there are 2 unemployed individuals, A and B, who receive €950 while unemployed and are entitled to receive benefits for up to 14 months if remaining unemployed: A looked for jobs last week; B did not look for jobs last week. The government wants to transfer €1000 to individuals A and B in a one-time payment, and is thinking of how to allocate them. Please choose what allocation option you would prefer: More money to A than to B; More money to B than to A; Same amount to A and B.* Panel (b) depicts the share of respondents accepting an unequal allocation when presented with the question on quantitative preferences for redistribution: *Imagine there are 2 unemployed individuals, A and B, who receive €950 while unemployed and are entitled to receive benefits for up to 14 months if remaining unemployed: A looked for jobs last week; B did not look for jobs last week. The government wants to transfer €1000 to individuals A and B in a one-time payment, and is thinking of how to allocate them. Now imagine that an unequal allocation may be costly. That is, it can allocate €500 each, or more to one than the other, but the latter option may result in the total amount of resources that can be distributed being reduced. Please choose what allocation option you would prefer: 500 for A and 500 for B; 550 for A and 400 for B.* The offered amounts are experimentally varied as explained in Section 4.1.1, so that the implied marginal rate of transformation (MRT) or cost of the transfer varies depending on the amounts. Panel (c) depicts a wordcloud with the most common words used by respondents to describe their reasons for choosing the option in the question for Panel (a). Panel (d) shows the share of individuals stating that they would support a cycle-dependent benefit duration reform. The exact question is: *There exists the idea that benefits received by each unemployed could be more generous (higher benefit level or duration) at times when economic conditions are bad (high unemployment rate), relative to times when economic conditions are good (low unemployment rate). Some reasons for that are that when economic conditions are bad, the unemployed suffer larger consumption reductions upon job loss, and the type of unemployed at that moment are, on average, individuals who put more effort into job finding but who face more difficulties given the lack of job offers. Would you agree with having unemployment benefits changing with economic conditions in this way?*. Panel (e) shows how search effort by the unemployed varies with economic conditions. It is built using the 2002-2003 and 2009-2010 waves of the Spanish Time Use data, and unemployment rate from the LFS. First, data on the probability of daily search is collapsed at the region level. Second, it is converted to weekly terms following Faberman et al. (2022). Third, it is residualized on region FEs and weighted by population. More details on the methodology are presented in Appendix D.

Figure 2: Consumption Smoothing Gains

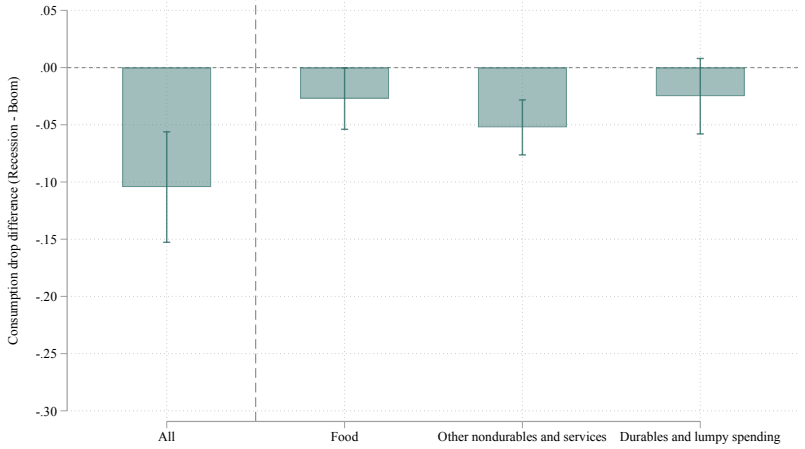
(a) Consumption Drop Upon Job Loss: UI Recipients



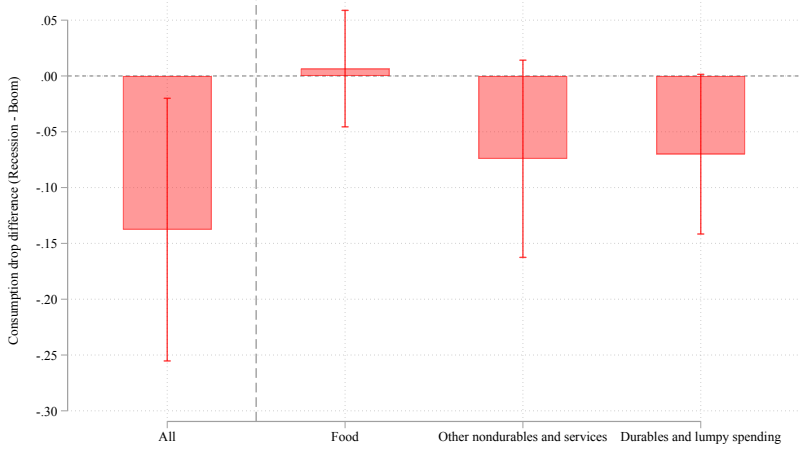
(b) Consumption Drop Upon Job Loss: UI Exhausted



(c) Cyclicality of Consumption Drop by Broad Category: UI Recipients



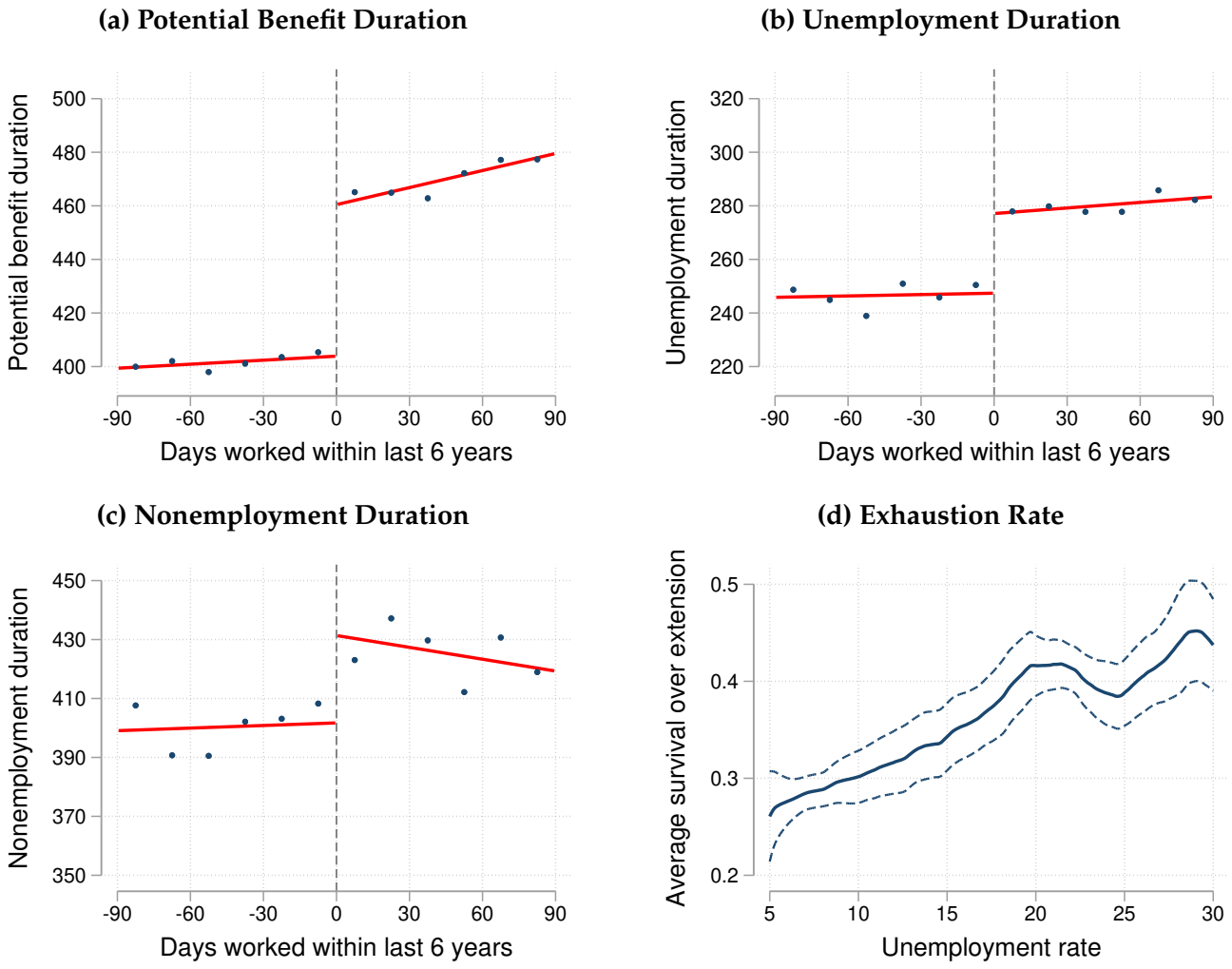
(d) Cyclicality of Consumption Drop by Broad Category: UI Exhausted



Note: Panels (a) and (b) show how the within-individual consumption drop upon job loss varies with the unemployment rate for both UI Recipients and UI exhaustees, respectively. Specifically, the graph plots the α_g^1 and α_g^2 coefficients obtained by estimating the following regression equation: $logC_{it} = \sum_g [\alpha_g^1 UIRecipient_{it} + \alpha_g^2 UIExhausted_{it}] \cdot \mathbb{1}[g(r(i,t)t) = g] + \lambda_i + \lambda_{r(i,t)t} + X'\beta + \varepsilon_{it}$, where $logC_{it}$ refers to log consumption measured at the household level, $\mathbb{1}[g(r(i,t)t) = g]$ is a dummy variable that equals one when the unemployment rate of the region where the household head lives falls within unemployment rate group g, and λ_i, $\lambda_{r(i,t)t}$, are individual, region-year fixed effects, respectively. The vector of controls X includes dummies for the regional unemployment rate groups, and information on type of household and other household composition variables, intended to control for any mechanical relationship between household composition and economic conditions or job loss. The three unemployment rate groups approximate the national average in boom, normal times and recession. 95% confidence intervals from standard errors clustered at the region-year level.

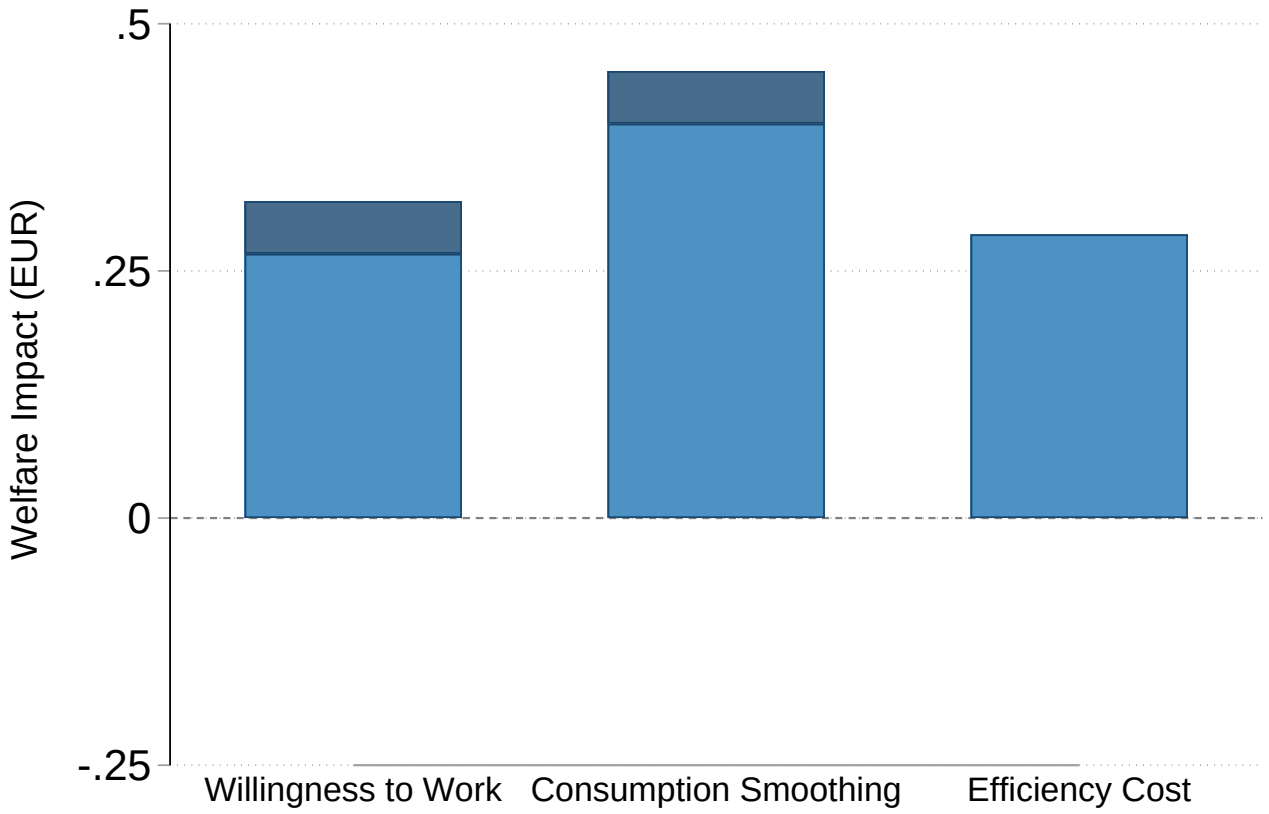
Panels (c) and (d) show how the magnitude of the broad category-specific change in consumption upon unemployment varies with economic conditions for both UI Recipients and UI exhaustees, respectively. Specifically, I first estimate models of the form in equation (11), where the dependent variable is $\tilde{c}_{i,t}^j$, the individual's consumption within category j relative to the average overall consumption of the individual. For each category, I compute the difference between the change in recession and the change in boom as $\beta_2^j \times (U.Rate^R - U.Rate^B)$ (for UI recipients) and $\beta_3^j \times (U.Rate^R - U.Rate^B)$ (for UI exhaustees). Adding up all the estimates by category yields approximately the total change in the consumption drop.

Figure 3: UI Generosity and Labor Market Outcomes



Note: Panels (a), (b), and (c) are binned scatter plots of PBD, unemployment and nonemployment duration, respectively, versus work experience within last 6 years, measured in days. They are depicted in days relative to the closest cutoff, effectively pooling together all cutoffs shown in Figure A7. Nonemployment duration is capped at 3 years. Panel (d) shows average survival over the additional 60-day PBD window, estimated nonparametrically using individuals below the cutoff as described in Section 4.3.

Figure 4: Sources of Welfare Impact



Note: The graph shows the contribution of the different components — welfare weights, consumption smoothing and efficiency cost — to the net welfare gain of implementing a budget-balanced cycle-dependent benefit duration reform. The contribution of each component is computed using a Shapley decomposition, averaging each component change over all possible orders of moving the components from no cyclical variation to the variation estimated in the main text. This assigns half of the interaction between welfare weights and consumption smoothing to each channel. The darker portions of the first two bars correspond to these interaction effects. It uses the estimates in the main text of the implementation equations (6) and (7), and follows the derivation in Appendix B.3. Welfare gains are expressed per euro mechanically transferred in recession, normalized by the average marginal utility in boom.

Table 1: Consumption Changes Upon Job Loss

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Estimates						
UI recipient	-0.101***	0.026	0.031	-0.103***	0.024	0.029
	(0.013)	(0.032)	(0.035)	(0.014)	(0.032)	(0.035)
UI exhausted	-0.221***	-0.066	-0.043	-0.222***	-0.071	-0.047
	(0.031)	(0.095)	(0.094)	(0.031)	(0.096)	(0.095)
UI recipient × unemployment rate (%)		-0.006***	-0.006***		-0.006***	-0.006***
		(0.001)	(0.001)		(0.001)	(0.001)
UI exhausted × unemployment rate (%)		-0.007*	-0.008**		-0.007*	-0.008**
		(0.003)	(0.003)		(0.003)	(0.003)
Region FE	Yes	Yes	No	Yes	Yes	No
Year FE	Yes	Yes	No	Yes	Yes	No
Region × Year FEs	No	No	Yes	No	No	Yes
Additional controls	No	No	No	Yes	Yes	Yes
N	66,450	66,450	66,450	66,450	66,450	66,450
Panel B: Post-estimation						
$\frac{c^{U,B}-c^{E,B}}{c^{E,B}}$ (UI recipient; Boom: unemp. rate 8%)		-0.020	-0.017		-0.022	-0.019
		(0.023)	(0.024)		(0.023)	(0.024)
$\frac{c^{U,R}-c^{E,R}}{c^{E,R}}$ (UI recipient; Recession: unemp. rate 26%)		-0.125***	-0.126***		-0.126***	-0.127***
		(0.012)	(0.012)		(0.012)	(0.012)
$\frac{c^{U,B}-c^{E,B}}{c^{E,B}}$ (UI exhausted; Boom: unemp. rate 8%)		-0.120*	-0.106		-0.124*	-0.109
		(0.069)	(0.068)		(0.070)	(0.069)
$\frac{c^{U,R}-c^{E,R}}{c^{E,R}}$ (UI exhausted; Recession: unemp. rate 26%)		-0.242***	-0.248***		-0.243***	-0.249***
		(0.029)	(0.030)		(0.028)	(0.029)
Panel C: Statistics						
$\frac{c^{U,R}-c^{E,R}}{c^{E,R}} - \frac{c^{U,B}-c^{E,B}}{c^{E,B}}$ (UI recipient: Recession − Boom)		-0.105***	-0.108***		-0.104***	-0.108***
		(0.023)	(0.026)		(0.023)	(0.025)
$\frac{c^{U,R}-c^{E,R}}{c^{E,R}} - \frac{c^{U,B}-c^{E,B}}{c^{E,B}}$ (UI exhausted: Recession − Boom)		-0.122*	-0.142**		-0.119*	-0.140**
		(0.062)	(0.063)		(0.062)	(0.063)

Note: The table shows how the consumption change upon unemployment evolves over the business cycle. Results are obtained from a regression with specification as in equation (10). UI recipient denotes unemployment with reported household unemployment-benefit or subsidy income. UI exhausted denotes unemployment without such income after prior observed receipt. Columns (1)-(3) include controls for regional unemployment rate and type of household. Columns (4)-(6) include controls for regional unemployment rate, type of household and the following set of covariates: age, sex, education, civil status, children, log household size, tenure, and rural status. Standard errors are clustered at the region-year level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 2: First-Year Consumption

	(1)	(2)	(3)	(4)	(5)	(6)
Future UI recipient	-0.213***	-0.254***	-0.254***	-0.129***	-0.154***	-0.153***
	(0.017)	(0.042)	(0.043)	(0.017)	(0.043)	(0.044)
Future UI recipient × unemployment rate (%)		0.002	0.002		0.001	0.001
		(0.002)	(0.002)		(0.002)	(0.002)
Region FE	Yes	Yes	No	Yes	Yes	No
Year FE	Yes	Yes	No	Yes	Yes	No
Region × year FE	No	No	Yes	No	No	Yes
Additional controls	No	No	No	Yes	Yes	Yes
N	31,399	31,399	31,399	31,399	31,399	31,399

Note: The table shows how consumption evolves over the business cycle during the first year individuals are observed in the sample. Specifically, it compares first-year (t) consumption patterns (while employed) for individuals that get unemployed in the following period (t+1) relative to those that remain employed. Future UI recipient identifies households observed unemployed with reported benefit or subsidy income in the second wave. Results are obtained from a regression specification as in equation (12). Since here I only look at individuals' first year in the sample, these specifications do not include individual FEs. Columns (1)-(3) include controls for type of household. Columns (4)-(6) include controls for type of household and the following set of covariates: age, sex, education, civil status, children, log household size, tenure, and rural status. Standard errors are clustered at the region-year level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 3: Marginal Cost

	Pot. benefit duration		Unemployment duration		Nonemp. duration	
	(1)	(2)	(3)	(4)	(5)	(6)
1[Exp. >0]	57.17***	56.13***	30.69***	31.73***	31.30**	41.75**
	(3.10)	(4.01)	(5.33)	(8.89)	(14.14)	(18.40)
1[Exp. >0] $\times$ Unemp. rate (%)		0.08		-0.07		-0.75
		(0.26)		(0.40)		(0.84)
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	13,994	13,994	13,994	13,994	13,994	13,994

Note: The table shows the effects of the PBD extension on PBD, unemployment duration and nonemployment duration. It also explores whether these relationships vary with regional unemployment rate. They are estimated following equation (13). Standard errors are clustered at the region level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 4: Marginal Benefit Calibration

	$\gamma = 1$	$\gamma = 2$	$\gamma = 3$	$\gamma = 4$	$\gamma = 5$
$\frac{\omega^R}{\omega^B}$=1	1.13	1.23	1.32	1.40	1.46
$\frac{\omega^R}{\omega^B}$=1.12	1.27	1.39	1.49	1.57	1.64
$\frac{\omega^R}{\omega^B}$=1.13	1.28	1.40	1.50	1.59	1.66
$\frac{\omega^R}{\omega^B}$=1.27	1.43	1.56	1.68	1.77	1.86
$\frac{\omega^R}{\omega^B}$=1.53	1.73	1.89	2.03	2.15	2.25

Note: The table combines the results on consumption smoothing gains for UI exhaustees and welfare weights estimated to produce estimates of the marginal benefit term $\mathbb{E}^{U,R}_{P+1}[\omega(\overline{e}_i)u'(c^U_{i,P+1})]/\mathbb{E}^{U,B}_{P+1}[\omega(\overline{e}_i)u'(c^U_{i,P+1})]$, under different implementation assumptions following the consumption-based approach in equation (6). The rows show results under different ratios of welfare weights, where I use the benchmark estimate for relative valuation across types obtained in Section 4.1.1, and vary the magnitude of search effort change over the business cycle that is due to a composition effect. Row (1) refers to the case where there is no composition change. Row (2) refers to the case where only composition changes due to observables as measured by the set of demographics in the Spanish Time Use data and LFS, as obtained from the prediction exercise using demographic-implied change for unemployed without benefits between 2007 and 2013 described in Section 4.1.2. Row (3) refers to half of the composition change found by Mukoyama et al. (2018) in the US using CPS data, while row (4) refers to the same composition change found by them. Row (5) refers to the case where all the change in effort is assumed to be due to composition effects. Columns (1)-(5) explore sensitivity of the results with respect to the parameter of relative risk aversion γ.

Table 5: Marginal Cost Calibration

	$\Delta D_b^{k_0}$ (days)	ΔD^{k_0} (days)	b (€/month)	τ (€/month)	$\overline{S}_{P+1}^{k_0}$	FE
Boom (8 % unemp. rate)	30.69	31.30	850	53	0.29	0.89
Middle (15 % unemp. rate)	30.69	31.30	850	53	0.34	0.59
Recession (26 % unemp. rate)	30.69	31.30	850	53	0.41	0.34

Note: The table quantifies how the efficiency cost varies with labor market conditions. It shows the estimates for the different components in equation (7), and how they change when the economy goes from boom to recession, with the magnitudes reflecting the national unemployment rate at different points of the business cycle. Effects of the PBD extension on unemployment duration and nonemployment duration are measured in days. Average monthly benefit level is $b = 850$, and the payroll tax is $\tau = 53$. The tax is obtained as the amount consistent with the stated benefit level and the ratio of UI recipients to employed individuals over the period 2007-2017, following Landais (2015). The number of UI recipients is obtained from OECD, and number of employed from EPA (INE). Given the discrete nature of the PBD change exploited, the fiscal externality is estimated in practice as $FE^{k_0} = [\Delta D_b^{k_0} + (\tau/b)\Delta D^{k_0}]/[60\overline{S}_{P+1}^{k_0}] - 1$.

Online Appendix

Unemployment Insurance, Inequality of Opportunity, and Labor Market Conditions

Jon Piqueras

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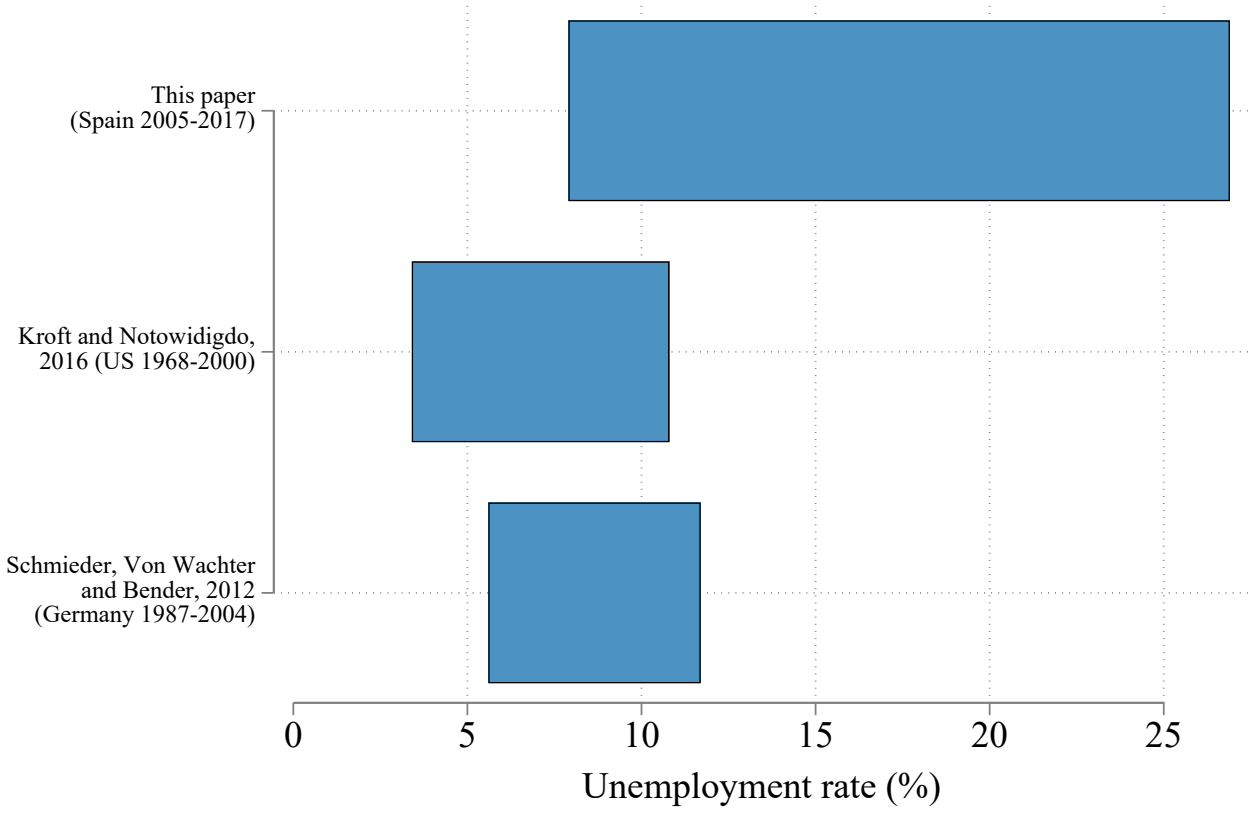
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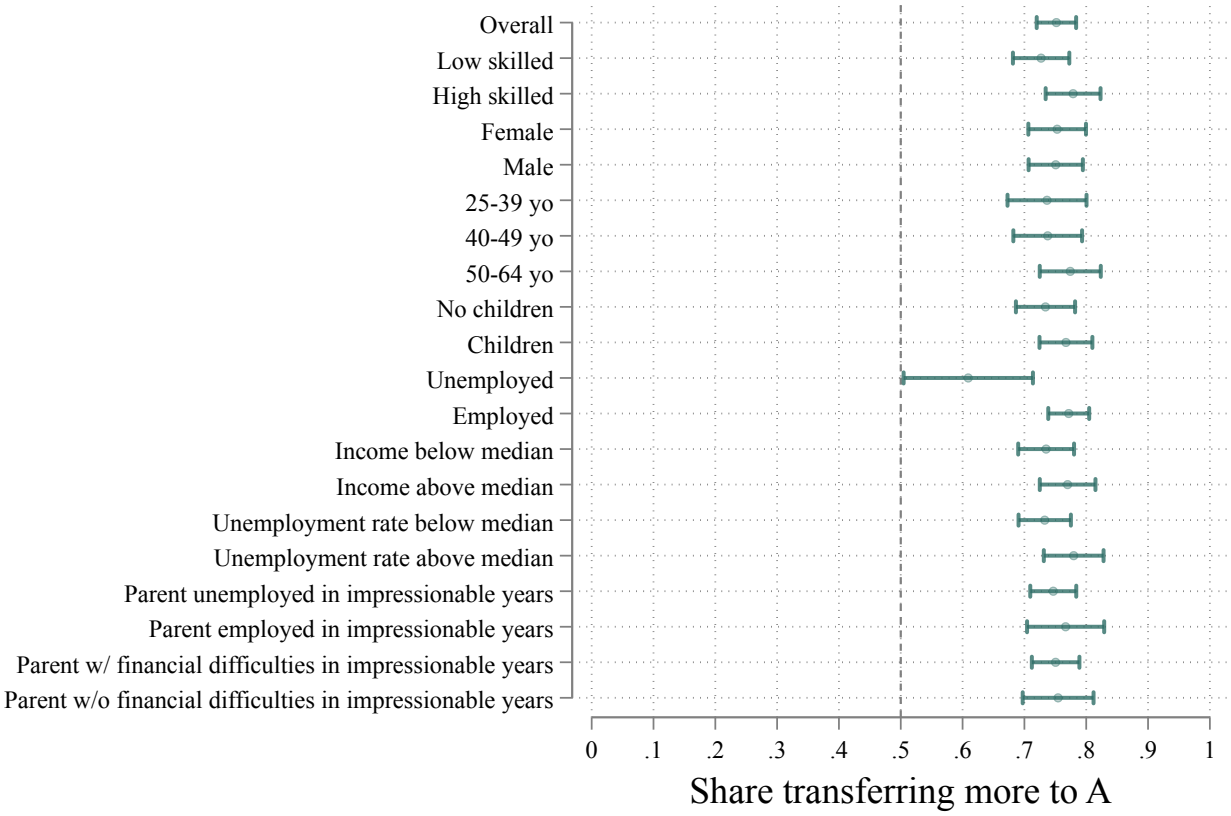
Appendix A Additional Figures and Tables

Figure A1: Spanish Unemployment Fluctuations in the Context of Recent Literature



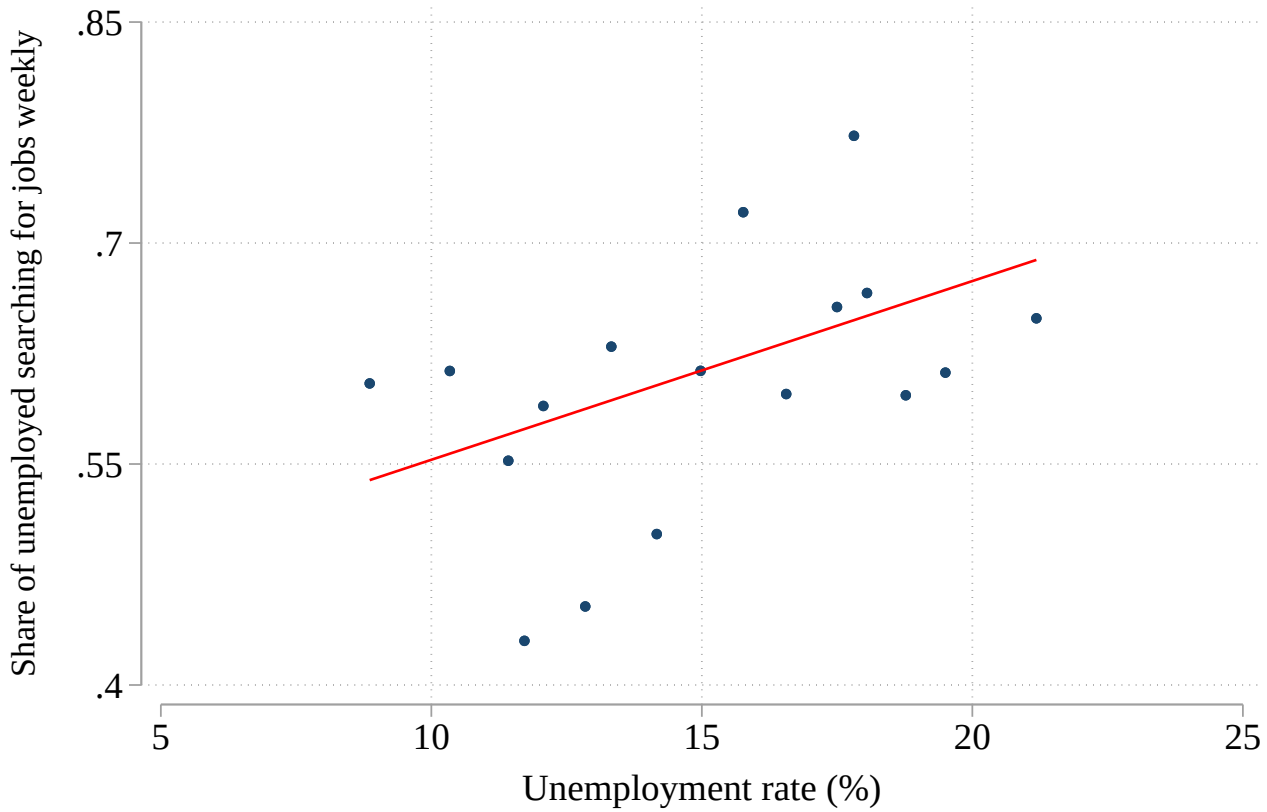
Note: This graph presents the magnitude of the Spanish unemployment fluctuations in relation to two prominent papers in the literature. Those papers use data from US (Kroft and Notowidigdo, 2016) and Germany (Schmieder et al., 2012). Blue bars correspond to the range of national unemployment rate in their country and period of study.

Figure A2: Preferences for Redistribution: Heterogeneity



Note: The graph explores heterogeneity in preferences for redistribution between the different types of unemployed. It focuses on the qualitative question: *Imagine there are 2 unemployed individuals, A and B, who receive €950 while unemployed and are entitled to receive benefits for up to 14 months if remaining unemployed: A looked for jobs last week; B did not look for jobs last week. The government wants to transfer €1000 to individuals A and B in a one-time payment, and is thinking of how to allocate them. Please choose what allocation option you would prefer: More money to A than to B; More money to B than to A; Same amount to A and B.*, as described in Section 4.1.1. I summarize the answers in a dummy variable that equals one for individuals choosing to allocate more resources to the individual who searched for jobs last week, and zero otherwise. The plot depicts the average value for that variable for the different demographic characteristics elicited in the Preliminary Questions and Demographic blocks in the online survey, as shown in Appendix C. Impressionable years refer to 18-25 (Krosnick and Alwin, 1989). Point estimates are shown along 95% confidence intervals.

Figure A3: Unemployed's Search Effort Over the Business Cycle



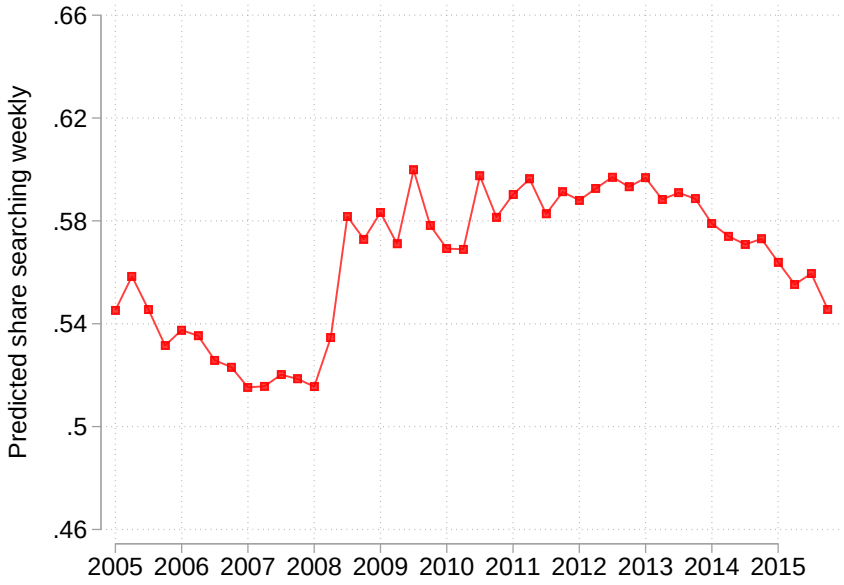
Note: The graph shows how search effort by the unemployed varies with economic conditions. It is built using the 2002-2003 and 2009-2010 waves of the Spanish Time Use data, and unemployment rate from the LFS. First, data on the probability of daily search is collapsed at the region level. Second, it is converted to weekly terms assuming search is independent across days. Third, it is residualized on region FEs, and weighted by population. More details on the methodology are presented in Appendix D.

Figure A4: Unemployed's Search Effort Over the Business Cycle: Composition Effect Based on Observables

(a) UI Recipient



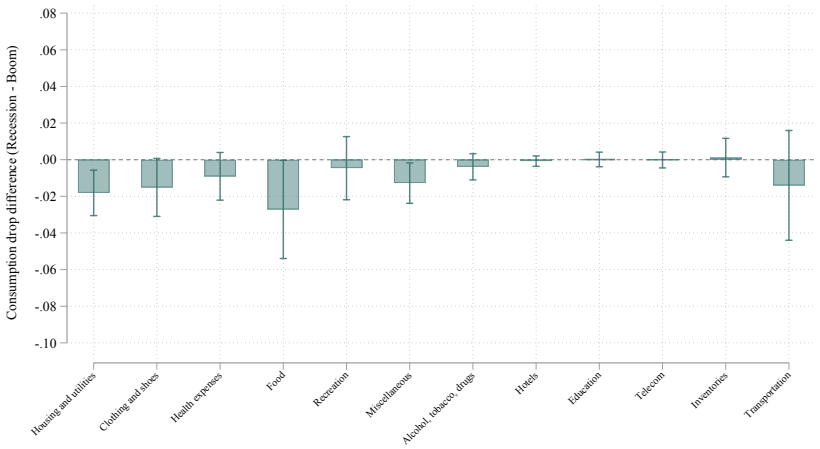
(b) UI Exhausted



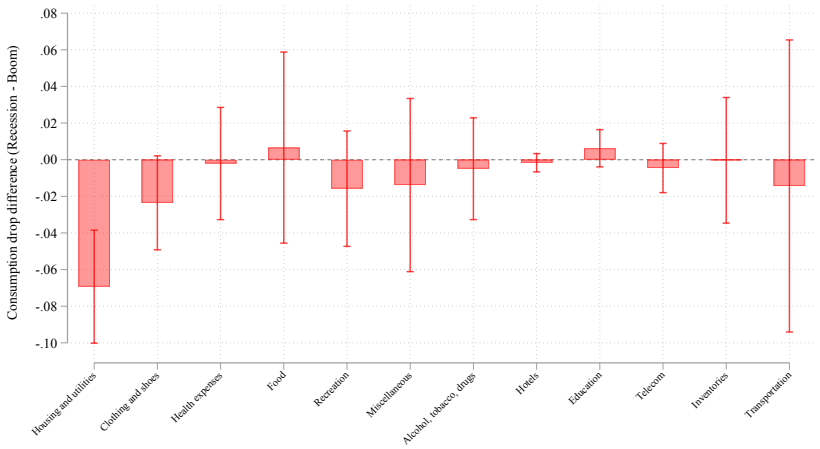
Note: The graph shows the component of search effort that varies over the business cycle due to composition shift in observable characteristics. It is built first estimating a model that relates the probability of daily search with a set of demographic characteristics in the Time Use data. These variables are: interactions between age categories and sex; interactions between education categories and sex; interaction between married status and sex; and unemployment rate, day of the week when diary was filled and region FEs. Then, I predict daily search based on that model using the same set of demographic characteristics into the Labor Force Survey. Finally, I convert to weekly terms following Faberman et al. (2022). The depicted change in search effort is therefore only due to a composition effect based on observables, and not driven by changes in search effort behavior within individual in response to economic conditions. Panel (a) refers to unemployed individuals receiving benefits, while Panel (b) refers to registered unemployed individuals without receiving benefits.

Figure A5: Consumption Smoothing Gains - Detailed Categories

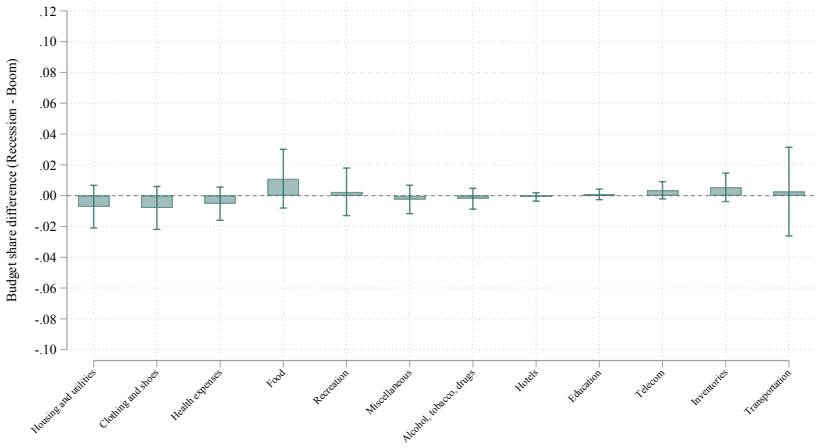
(a) Cyclicality of Consumption Drop by Category: UI Recipients



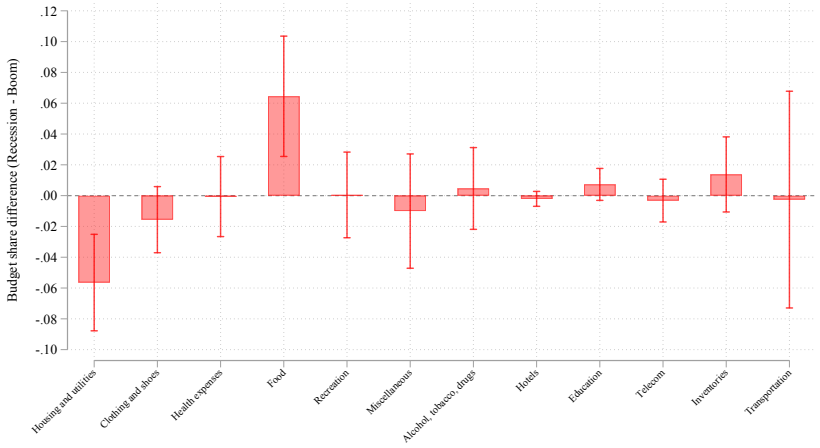
(b) Cyclicality of Consumption Drop by Category: UI Exhausted



(c) Budget-Share Cyclicality: UI Recipients



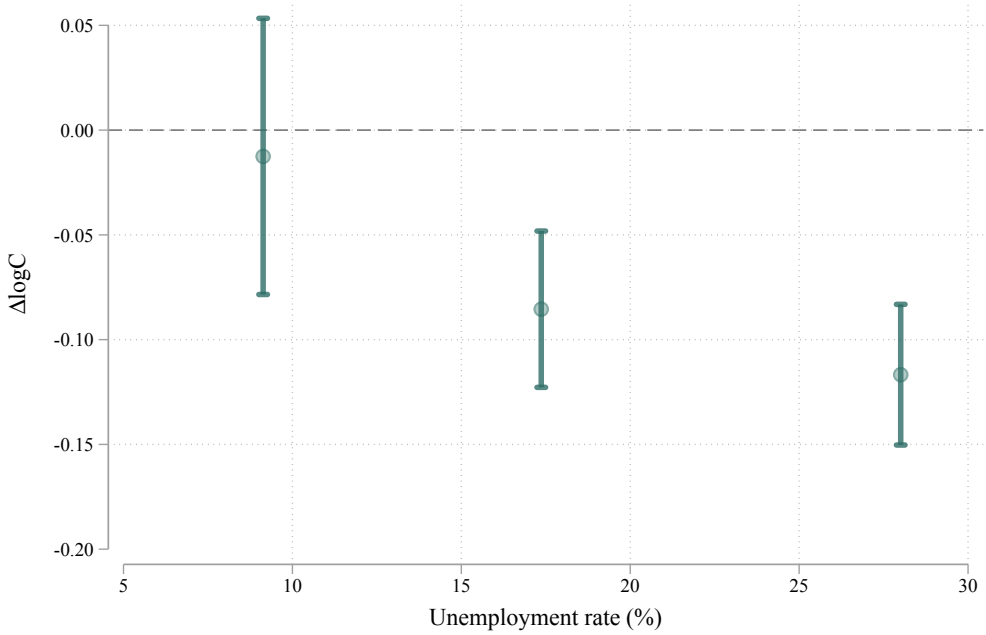
(d) Budget-Share Cyclicality: UI Exhausted



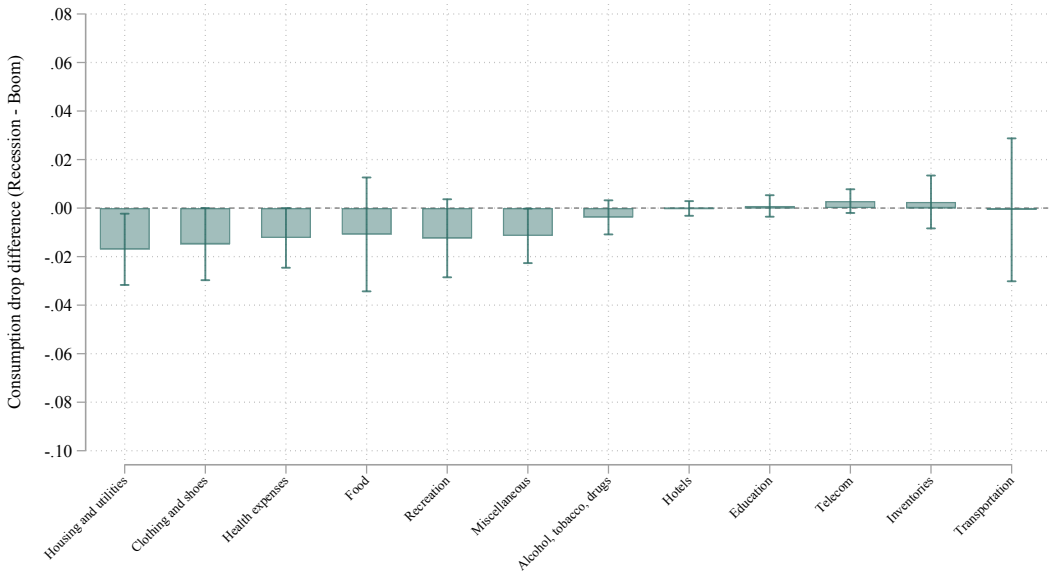
Note: Panels (a) and (b) show how the magnitude of the detailed category-specific change in consumption upon unemployment varies with economic conditions for both UI Recipients and UI exhaustees, respectively. Specifically, I first estimate models of the form in equation (11) as described in Section 4.2, where the dependent variable is $\tilde{c}^j_{i,t}$, the individual's consumption within category j relative to the average overall consumption of the individual. For each category, I compute the difference between the change in recession and the change in boom as $\beta^j_2 \times (U.Rate^R - U.Rate^B)$ (for UI recipients) and $\beta^j_3 \times (U.Rate^R - U.Rate^B)$ (for UI exhaustees). Adding up all the estimates by category yields approximately the total change in the consumption drop. Panels (c) and (d) are obtained following the same approach, using instead $\tilde{s}^j_{i,t}$ as dependent variable, which is the budget share within overall consumption and thus informs about cyclicality of consumption reallocation.

Figure A6: Consumption Smoothing Gains - All Unemployed

(a) Consumption Drop Upon Job Loss

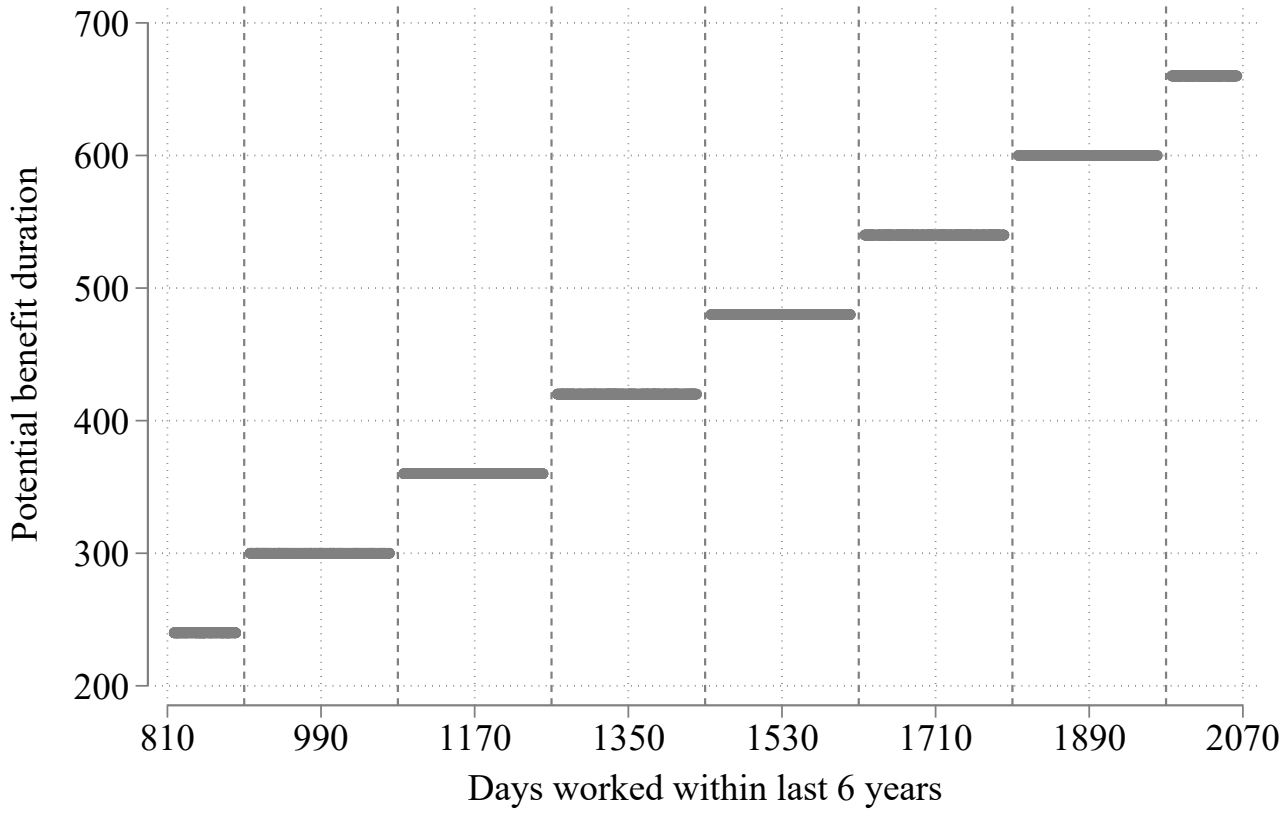


(b) Cyclicality of Consumption Drop



Note: Panel (a) shows how the within-individual consumption drop upon job loss varies with the unemployment rate for the sample of all unemployed, instead of differentiating by UI status. Specifically, the graph plots the α_j coefficients obtained by estimating the following regression equation: $logC_{it} = \sum_j \alpha_j \cdot \mathbb{1}[Unemployed_{it}] \cdot \mathbb{1}[g(r(i,t)t) = j] + \lambda_i + \lambda_{r(i,t)t} + X'\beta + \varepsilon_{it}$, where $logC_{it}$ refers to log consumption measured at the household level, $\mathbb{1}[Unemployed_{it}]$ is a dummy variable that equals one when the household head is unemployed, $\mathbb{1}[g(r(i,t)t) = j]$ is a dummy variable that equals one when the unemployment rate of the region where the household head lives falls within unemployment rate group j, and λ_i, $\lambda_{r(i,t)t}$, are individual, region-year fixed effects, respectively. The vector of controls X includes dummies for the regional unemployment rate groups, and information on type of household and other household composition variables, intended to control for any mechanical relationship between household composition and economic conditions or job loss. The three unemployment rate groups approximate the national average in boom, normal times and recession. 95% confidence intervals from standard errors clustered at the region-year level. Panel (b) shows how the magnitude of the detailed category-specific change in consumption upon unemployment varies with economic conditions for the sample of all unemployed. Specifically, I first estimate models similar to equation (11), where the dependent variable is $\tilde{c}^j_{i,t}$, the individual's consumption within category j relative to the average overall consumption of the individual. Adding up all the estimates by category yields the total change in the consumption drop.

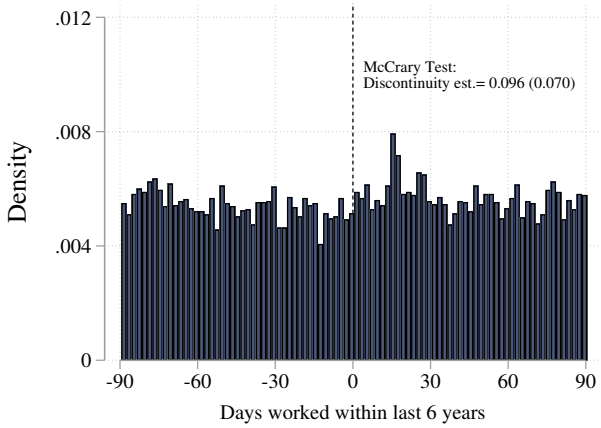
Figure A7: UI Schedule



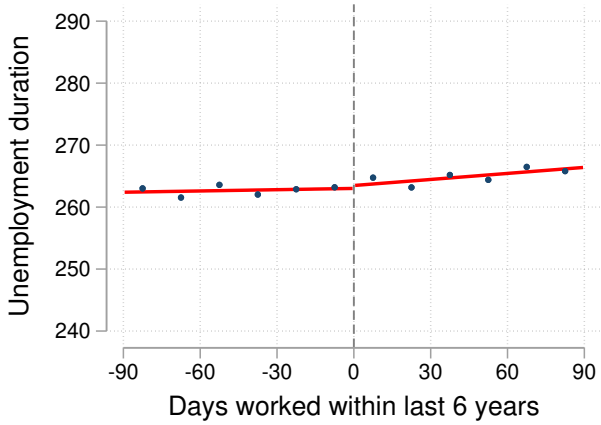
Note: The graph shows the Spanish Unemployment Insurance schedule. Specifically, it depicts the statutory relationship between recent work experience and potential benefit duration, measured in days.

Figure A8: Validity of the Design

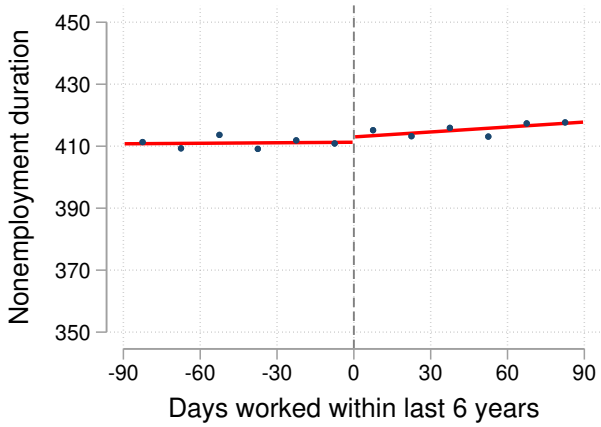
(a) Density



(b) Unemployment Duration



(c) Nonemployment Duration



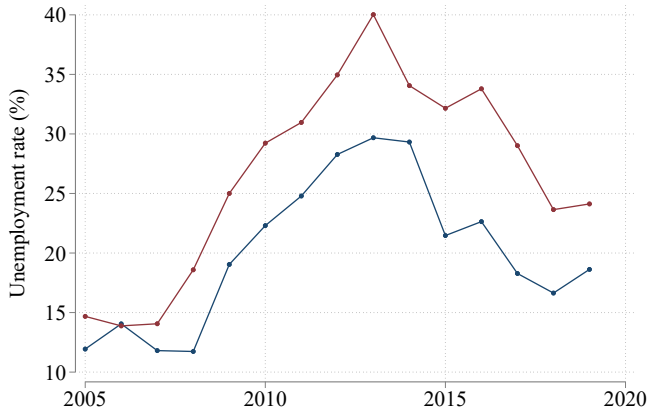
Note: Panel (a) follows McCrary (2008) to assess whether the density of the running variable presents any discontinuity at the normalized cutoff. Panels (b) and (c) plot the evolution of predicted unemployment and nonemployment duration around the normalized cutoff. This prediction is constructed using a linear model in the spirit of Card et al. (2015) including the following set of controls: regional unemployment rate, rural status, (log) previous earnings, sex, age, age squared, full-time share, three education categories, four occupation categories, and year, month and region fixed effects.

Figure A9: Support for Increase in UI Generosity - CIS Survey

(a) Support by Demographic Groups



(b) Unemployment Rate by Shock Size



(c) Support by Shock Size



Note: The plots are obtained using data from the Spanish CIS survey on public finance issues over the years 2005-2019. Panel (a) shows, on the left axis, the evolution of the share of people supporting increases in UI generosity over the business cycle for different groups of individuals. I recode the answers to the question into a dummy variable that equals one if the respondent would like a more generous UI system, and zero otherwise. National unemployment rate is shown on the right axis. Panel (b) shows the unemployment rate for two groups of regions according to the intensity of the Great Recession shock. Specifically, I compute for each region the change in unemployment rate between 2007 and 2013, and create two groups containing regions below and above the median change in the sample. Panel (c) shows the evolution of the demand for UI generosity increase split by the same groups as in Panel (b).

Table A1: Online Survey Representativeness

Characteristic	(1) Spanish Population	(2) Full Online Sample	(3) Analysis Sample
Sex			
Men	51.4	52.6	53.2
Women	48.6	47.4	46.8
Age			
25–39	26.8	26.8	26.2
40–49	33.1	36.2	34.4
50–64	40.1	37.0	39.4
Education			
Low education	55.7	53.1	52.2
High education	44.3	46.9	47.8
Employment status			
Employed	88.0	88.5	87.7
Unemployed	12.0	11.5	12.3
N		1,022	709

Note: The table compares demographic statistics between the Spanish population, obtained from the Labor Force Survey at the time of survey preparation (2022 Q1), and the respondents in the online sample. Entries are percentages of the entire sample within each demographic characteristic. Column (1) shows LFS shares for economically active individuals aged 25–64. Column (2) shows statistics for the full online sample, and Column (3) refers to the analysis sample which retains respondents after restrictions in terms of completion time and consistent qualitative and quantitative answers. High education refers to tertiary education, including higher vocational qualifications. Low education is the complement of high education.

Table A2: Unemployed's Search Effort Over the Business Cycle - Search Methods Across Countries

	All		Ask friends		Study job ads		Apply to employers		Contact public service	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Unemp. rate (%)	0.423***	0.308***	1.016***	0.833***	0.704**	0.495	1.063***	0.793***	0.949	0.795
	(0.0770)	(0.0761)	(0.227)	(0.235)	(0.344)	(0.473)	(0.246)	(0.245)	(0.599)	(0.547)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Mean	32.9	32.9	65	65	58.7	58.7	54.4	54.4	51	51
Countries	34	34	34	34	34	34	34	34	34	34
N	4,868	4,868	497	497	496	496	497	497	497	497

	Answer job ads		Contact priv. agency		Other		Look for land		Look for licenses	
	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
Unemp. rate (%)	0.737***	0.425**	0.441**	0.473	-0.615	-0.919*	-0.027	0.005	-0.042	-0.007
	(0.143)	(0.174)	(0.196)	(0.316)	(0.707)	(0.519)	(0.021)	(0.054)	(0.027)	(0.042)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Mean	41.5	41.5	20.5	20.5	14	14	1.8	1.8	1.7	1.7
Countries	34	34	34	34	34	34	34	34	34	34
N	496	496	496	496	438	438	492	492	492	492

Note: The table presents the relationship between the unemployment rate (%) and the percentage of unemployed that declared using each of the different search methods over the last 4 weeks. The data span the years 2005-2019 and 34 countries. The data are obtained from Eurostat, which collects it from national Labor Force Surveys. The annual unemployment rate and population are also from Eurostat. Columns (1)-(2) pool all search methods. Columns (3)-(20) represent different search methods used to seek work ordered in the table according to usage, which include: asking friends, relatives, and trade unions; studying job advertisements; applying directly to employers; contacting public employment services; publishing or answering job advertisements; contacting private employment agencies; other methods; looking for land, premises, or equipment; and looking for licenses, permits, or financial resources. The countries included in the sample are Austria, Belgium, Bulgaria, Switzerland, Cyprus, Czechia, Germany, Denmark, Estonia, Spain, Finland, France, Croatia, Hungary, Ireland, Iceland, Italy, Lithuania, Luxembourg, Latvia, Montenegro, North Macedonia, Malta, Netherlands, Norway, Poland, Portugal, Romania, Serbia, Sweden, Slovenia, Slovakia, Turkey and United Kingdom. For each method, regression results are presented from specifications with country FE and with/without year FE, along with the mean value of the dependent variable, the number of countries, and the number of country-year-method observations. Regressions are weighted by country population and standard errors, in parentheses, are clustered at the country level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A3: Construction of Consumption Categories

Component	COICOP/HBS code(s)	Group
Panel A: Food		
Bread and cereals	01.1.1	F
Meat and fish	01.1.2–01.1.3	F
Dairy and eggs	01.1.4	F
Fruit and vegetables	01.1.6–01.1.7	F
Oils, confectionery, and other food	01.1.5; 01.1.8–01.1.9	F
Nonalcoholic beverages	01.2.1–01.2.2	F
Restaurants and cafes	11.1.1	F
Workplace and school canteens	11.1.2	F
Panel B: Alcohol, tobacco, and drugs		
Alcoholic beverages	02.1	N
Tobacco	02.2	N
Narcotics	02.3	N
Panel C: Clothing and footwear		
Clothing	03.1	N
Footwear	03.2	N
Panel D: Housing and utilities		
Actual main-residence rent	04.1.1.1	N
Other and secondary rent	04.1.2.1; 04.1.3.1–04.1.3.3	N
Water and community charges	04.4.1.1; 04.4.2.1; 04.4.3.1; 04.4.4.1; 04.4.4.3	N
Electricity	04.5.1.1	N
Natural and bottled gas	04.5.2.1; 04.5.2.3	N
Other fuels and central heating	04.5.3.1; 04.5.4.1; 04.5.5.1	N
Maintenance materials	04.3.1.1	D
Maintenance services	04.3.2.1	D
Other housing costs	Remaining 04 codes, excluding 04.2 and the preceding housing rows	N
Panel E: Furnishings and household equipment		
Furniture, textiles, and appliances	05.1.1–05.1.3; 05.2.1; 05.3.1–05.3.3	D
Other household equipment and services	05.4.1; 05.5.1–05.5.2; 05.6.1–05.6.2; 05.7.1	D
Panel F: Health		
Medical products and equipment	06.1.1	N
Outpatient, hospital, and other health services	06.2.1–06.2.3; 06.3.1; 06.4.1	N
Panel G: Transportation		
Vehicle purchases	07.1.1–07.1.3	D
Fuel	07.2.2	N
Other private-vehicle operating costs	07.2.1; 07.2.3–07.2.4	D
Public and other transport services	07.3.1–07.3.6; 07.4.1	N
Panel H: Communications		
Postal services and communications equipment	08.1.1; 08.2.1	N
Communications services	08.3.1; 08.4.1	N
Panel I: Recreation		
Recreation goods	09.1.1–09.1.4; 09.2.1–09.2.2; 09.3.1–09.3.4; 09.5.1–09.5.4	D
Recreation and cultural services	09.1.5; 09.2.3; 09.4.1–09.4.3; 09.6.1; 09.7.1	N
Panel J: Education		
Education	10	N
Panel K: Hotels and restaurants excluding food		
Accommodation and other hospitality	11 excluding 11.1.1–11.1.2	N
Panel L: Miscellaneous		
Personal care and personal effects	12.1.1–12.1.3; 12.2.1–12.2.2; 12.3.1	N
Housing insurance	12.4.2.1	N
Other insurance and miscellaneous services	Remaining 12 codes, excluding the preceding two rows	N

Note: This table describes the mapping between consumption components and their corresponding codes included in the broad and detailed categorizations used in the main text. The broad categorization distinguishes between F: food; N: other nondurables and services; and D: durables and lumpy spending. The detailed categorization consists of 12 categories described in the different panels. Codes follow COICOP/HBS for EPF 2006–2015.

Table A4: Consumption Drop Upon Job Loss and Cyclicality by Component

Component	UI recipient		UI exhausted	
	Average	26% − 8%	Average	26% − 8%
Panel A: Food	-4.07***	-2.71**	-6.47***	0.66
	(0.52)	(1.36)	(1.23)	(2.64)
Bread and cereals	-0.24***	-0.18	-0.33	-0.13
	(0.09)	(0.18)	(0.29)	(0.90)
Meat and fish	-0.39**	-0.11	-0.75	1.70*
	(0.19)	(0.48)	(0.51)	(0.91)
Dairy and eggs	-0.11	-0.33	-0.38	0.13
	(0.07)	(0.21)	(0.23)	(0.53)
Fruit and vegetables	-0.08	-0.11	-0.69**	0.38
	(0.12)	(0.25)	(0.31)	(0.64)
Oils, confectionery, and other food	-0.15*	-0.13	-0.28	-0.40
	(0.09)	(0.21)	(0.21)	(0.54)
Nonalcoholic beverages	-0.13**	-0.43***	-0.17	0.13
	(0.06)	(0.13)	(0.14)	(0.40)
Restaurants and cafes	-2.86***	-1.50**	-3.63***	-1.24
	(0.38)	(0.73)	(0.65)	(1.34)
Workplace and school canteens	-0.12*	0.07	-0.24*	0.09
	(0.07)	(0.10)	(0.14)	(0.16)
Panel B: Alcohol, tobacco, and drugs	-0.51***	-0.39	-1.43***	-0.49
	(0.16)	(0.36)	(0.47)	(1.41)
Alcoholic beverages	-0.02	-0.23**	-0.25*	-0.74***
	(0.06)	(0.11)	(0.14)	(0.20)
Tobacco	-0.49***	-0.15	-1.18**	0.24
	(0.14)	(0.32)	(0.47)	(1.43)
Narcotics	0.00*	-0.00*	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)
Panel C: Clothing and footwear	-0.99***	-1.51*	-2.41***	-2.35*
	(0.37)	(0.80)	(0.69)	(1.30)
Clothing	-0.82***	-1.09*	-2.06***	-1.47
	(0.30)	(0.66)	(0.57)	(1.16)
Footwear	-0.16	-0.42	-0.35	-0.89
	(0.12)	(0.33)	(0.25)	(0.60)
Panel D: Housing and utilities	-0.54*	-1.81***	-1.84*	-6.93***
	(0.30)	(0.63)	(1.08)	(1.56)
Actual main-residence rent	-0.31	0.14	-0.51	-1.08
	(0.19)	(0.37)	(0.55)	(0.96)
Other and secondary rent	-0.08*	-0.14	-0.17*	-0.26*
	(0.05)	(0.09)	(0.09)	(0.14)
Water and community charges	-0.05	-0.07	-0.25**	0.09
	(0.07)	(0.15)	(0.12)	(0.31)
Electricity	-0.07	0.02	-0.53*	-1.64***
	(0.08)	(0.17)	(0.28)	(0.39)
Natural and bottled gas	-0.04	-0.29***	-0.07	-0.58
	(0.06)	(0.11)	(0.18)	(0.39)
Other fuels and central heating	-0.01	-0.08	0.01	-0.22**
	(0.03)	(0.05)	(0.05)	(0.09)
Maintenance materials	0.12	-0.43**	0.14	-1.34***
	(0.09)	(0.21)	(0.21)	(0.41)
Maintenance services	-0.08	-0.86***	-0.33	-1.65*
	(0.15)	(0.29)	(0.39)	(0.95)
Other housing costs	-0.02	-0.10*	-0.14*	-0.25
	(0.02)	(0.06)	(0.07)	(0.18)

Component	UI recipient		UI exhausted	
	Average	26% − 8%	Average	26% − 8%
Panel E: Furnishings and household equipment	-0.47	0.11	-0.12	-0.03
	(0.30)	(0.53)	(0.68)	(1.74)
Furniture, textiles, and appliances	-0.31	0.34	0.24	-0.05
	(0.24)	(0.43)	(0.56)	(1.27)
Other household equipment and services	-0.16	-0.23	-0.37	0.01
	(0.11)	(0.22)	(0.24)	(0.64)
Panel F: Health	-0.33	-0.91	-1.13	-0.21
	(0.41)	(0.66)	(0.69)	(1.55)
Medical products and equipment	-0.09	0.13	-0.21	-0.06
	(0.12)	(0.26)	(0.34)	(0.84)
Outpatient, hospital, and other health services	-0.24	-1.04*	-0.92	-0.15
	(0.39)	(0.59)	(0.59)	(1.40)
Panel G: Transportation	-1.32*	-1.40	-3.14*	-1.44
	(0.68)	(1.52)	(1.61)	(4.04)
Vehicle purchases	0.11	-0.76	0.19	-0.77
	(0.43)	(0.82)	(0.58)	(1.17)
Fuel	-0.89***	-0.21	-0.40	1.72
	(0.22)	(0.43)	(0.69)	(1.49)
Other private-vehicle operating costs	-0.21	-0.64	-2.19*	-2.85
	(0.44)	(1.11)	(1.20)	(3.07)
Public and other transport services	-0.34***	0.21	-0.75	0.47
	(0.13)	(0.25)	(0.53)	(1.30)
Panel H: Communications	-0.38**	-0.01	-0.83***	-0.46
	(0.17)	(0.22)	(0.29)	(0.68)
Postal services and communications equipment	0.04	0.06	0.01	-0.07
	(0.03)	(0.06)	(0.09)	(0.23)
Communications services	-0.41**	-0.07	-0.85***	-0.39
	(0.17)	(0.22)	(0.26)	(0.58)
Panel I: Recreation	-0.52	-0.46	-1.39**	-1.58
	(0.33)	(0.87)	(0.66)	(1.60)
Recreation goods	-0.07	0.07	0.12	-0.36
	(0.19)	(0.53)	(0.44)	(1.03)
Recreation and cultural services	-0.45**	-0.53	-1.51***	-1.22
	(0.21)	(0.50)	(0.47)	(1.23)
Panel J: Education	-0.10	0.01	-0.53**	0.62
	(0.10)	(0.20)	(0.27)	(0.52)
Education	-0.10	0.01	-0.53**	0.62
	(0.10)	(0.20)	(0.27)	(0.52)
Panel K: Hotels and restaurants excluding food	-0.18***	-0.08	-0.17	-0.17
	(0.06)	(0.14)	(0.11)	(0.25)
Accommodation and other hospitality	-0.18***	-0.08	-0.17	-0.17
	(0.06)	(0.14)	(0.11)	(0.25)
Panel L: Miscellaneous	-0.26	-1.28**	-1.93**	-1.39
	(0.32)	(0.56)	(0.85)	(2.40)
Personal care and personal effects	-0.24	-0.88***	-0.84	-2.47
	(0.19)	(0.33)	(0.67)	(1.72)
Housing insurance	0.01	-0.09	-0.17	-0.14
	(0.04)	(0.09)	(0.19)	(0.30)
Other insurance and miscellaneous services	-0.03	-0.31	-0.93	1.22
	(0.27)	(0.46)	(0.74)	(1.70)

Note: This table shows the estimated average detailed category-specific change in consumption upon unemployment for both UI recipients and exhaustees, and how this drop varies with economic conditions. Specifically, models of the form in equation (11) are estimated, where the dependent variable is $\tilde{c}^j_{i,t'}$ the individual's consumption within category j relative to the average overall consumption of the individual. Estimated changes refer to estimated coefficients multiplied by 100. Adding up all the estimates by category yields approximately the total change in the consumption drop. See Table A3 for the component definitions and COICOP/HBS codes. Standard errors are clustered at the region-year level. ***p<0.01, **p<0.05, *p<0.10.

Table A5: Consumption Changes Upon Job Loss - All Unemployed

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Estimates						
Unemployed	-0.096***	-0.006	0.003	-0.098***	-0.009	0.000
	(0.012)	(0.031)	(0.035)	(0.013)	(0.031)	(0.035)
Unemployed × unemployment rate (%)		-0.004***	-0.004***		-0.004***	-0.004***
		(0.001)	(0.001)		(0.001)	(0.001)
Region FE	Yes	Yes	No	Yes	Yes	No
Year FE	Yes	Yes	No	Yes	Yes	No
Region × Year FEs	No	No	Yes	No	No	Yes
Additional controls	No	No	No	Yes	Yes	Yes
N	63,304	63,304	63,304	63,304	63,304	63,304
Panel B: Post-estimation						
$\frac{c^{U,B}-c^{E,B}}{c^{E,B}}$ (Boom: unemp. rate 8%)		-0.039*	-0.033		-0.041*	-0.035
		(0.022)	(0.024)		(0.022)	(0.024)
$\frac{c^{U,R}-c^{E,R}}{c^{E,R}}$ (Recession: unemp. rate 26%)		-0.112***	-0.113***		-0.113***	-0.114***
		(0.011)	(0.012)		(0.012)	(0.012)
Panel C: Statistics						
$\frac{c^{U,R}-c^{E,R}}{c^{E,R}} - \frac{c^{U,B}-c^{E,B}}{c^{E,B}}$ (Recession − Boom)		-0.074***	-0.080***		-0.072***	-0.079***
		(0.023)	(0.026)		(0.022)	(0.026)

Note: The table shows how the consumption change upon unemployment evolves over the business cycle. Results are obtained from a regression with specification as in equation (10), but using the sample of all unemployed without differentiating by UI status. Columns (1)-(3) include controls for regional unemployment rate and type of household. Columns (4)-(6) include controls for regional unemployment rate, type of household and the following set of covariates: age, sex, education, civil status, children, log household size, tenure, and rural status. Standard errors are clustered at the region-year level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A6: Consumption Changes Upon Job Loss - Robustness

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
UI recipient	0.031	0.014	0.023	0.035	0.029	0.014	0.025	0.034
	(0.035)	(0.069)	(0.071)	(0.035)	(0.035)	(0.071)	(0.074)	(0.035)
UI exhausted	-0.043	0.246	0.172	-0.041	-0.047	0.210	0.141	-0.045
	(0.094)	(0.246)	(0.215)	(0.091)	(0.095)	(0.246)	(0.213)	(0.091)
UI recipient × unemployment rate (%)	-0.006***	-0.006***	-0.006***	-0.006***	-0.006***	-0.006***	-0.006***	-0.007***
	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	(0.001)	(0.002)
UI exhausted × unemployment rate (%)	-0.008**	-0.008**	-0.008**	-0.009**	-0.008**	-0.008**	-0.008**	-0.009**
	(0.003)	(0.004)	(0.004)	(0.004)	(0.003)	(0.004)	(0.004)	(0.004)
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region × Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Unemployment duration	No	Yes	No	No	No	Yes	No	No
Share unemployed 6–24 months	No	No	Yes	No	No	No	Yes	No
Post-2012 reform interaction	No	No	No	Yes	No	No	No	Yes
Additional controls	No	No	No	No	Yes	Yes	Yes	Yes
N	66,450	66,450	66,450	66,450	66,450	66,450	66,450	66,450

Note: The table explores sensitivity for the evolution of consumption changes upon unemployment over the business cycle for both UI recipients and exhaustees. Columns (1) and (5) are the same ones as columns (3) and (6) in Table 1, which report results without and with additional controls for the benchmark specification, respectively. Column (2) interacts the UI-status indicators with a first- and second-order polynomial in regional unemployment duration. Column (3) interacts unemployed's UI-status with the share of unemployed that have a duration between 6-24 months among the unemployed with duration up to 2 years. Column (4) interacts unemployed's UI-status with a dummy that equals one for the period from 2012 onward, which corresponds to the period where the benefit level after 6 months was slightly decreased. Columns (6), (7) and (8) repeat the same specifications before and also include controls for type of household and the following set of covariates: age, sex, education, civil status, children, log household size, tenure, and rural status. Standard errors are clustered at the region-year level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table A7: Marginal Cost - Robustness

	Pot. benefit duration		Unemployment duration		Nonemp. duration	
	(1)	(2)	(3)	(4)	(5)	(6)
1[Exp. > 0]	56.38***	55.64***	29.40***	30.54***	28.10*	38.96**
	(2.96)	(3.92)	(5.60)	(8.54)	(14.85)	(18.30)
1[Exp. > 0] $\times$ Unemp. rate (%)		0.05		-0.08		-0.78
		(0.26)		(0.36)		(0.73)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	13,994	13,994	13,994	13,994	13,994	13,994

Note: The table shows the effects of the PBD extension on PBD, unemployment duration and nonemployment duration. It also explores whether these relationships vary with regional unemployment rate. They are estimated following equation (13), including the following set of controls: regional unemployment rate, rural status, (log) previous earnings, sex, age, age squared, full-time share, three education categories, four occupation categories, and year, month and region fixed effects. Standard errors are clustered at the region level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Appendix B Derivations

B.1 Model

I follow closely Chetty (2006) and Kolsrud et al. (2018), but consider differentiation of the unemployment benefit policy according to economic conditions and a social planner with fairness considerations based on willingness to work. Let $\xi_{i,t}$ refer to a vector of state variables containing all relevant information up to time t, which determines individual's employment status, aggregate economic conditions $k_{i,t} \in \{B, R\}$ and behavior at time t. Conditions upon job loss are denoted by $k_{i,0}$. $F^{k_0}_{i,t}(\xi_{i,t})$ denotes the distribution of $\xi_{i,t}$ conditional on k_0. I assume $F^{k_0}_{i,t}$ is a cdf function and Ω denotes the maximal support of $F^{k_0}_{i,t}$ for $\forall i, t$.

Individuals choose each period how much to consume from income and assets, and how much effort to exert if unemployed. An employed individual earns $w_i - \tau$, and when unemployed receives $b^{k_0}_t + y^U_t$. The law of motion of assets when employed is $a_{i,t+1} = (1+r)a_{i,t} + w_i - \tau - c^E_{i,t}$; and when unemployed is $a_{i,t+1} = (1+r)a_{i,t} + b^{k_0}_t + y^U_t - c^U_{i,t}$, with $a_{i,t+1} \geq \underline{a}_i$. The Lagrange multipliers on these constraints are $\mu^{E,k}_{i,t}(\xi_{i,t})$, $\mu^{U,k}_{i,t}(\xi_{i,t})$ and $\mu^a_{i,t}(\xi_{i,t})$, respectively.

Let $\theta^{s,k}_{i,t}(\xi_{i,t})$ denote an individual's employment status s at time t in aggregate economic conditions k. When $\theta^{E,k} = 1$, the individual is employed in aggregate economic conditions k. When $\theta^{U,k} = 1$, the individual is unemployed in aggregate economic conditions k. They equal zero otherwise. Each individual is in one of four mutually exclusive labor market statuses, denoted by the combination of employment status and aggregate conditions. In each period t, an unemployed individual chooses a level of search effort $e_{i,t}$ as well. This search effort level determines the probability to leave unemployment for employment, with the mapping potentially depending on current economic conditions.

Let $\mathcal{I}_{k_0} = \{i : k_{i,0} = k_0\}$ and $\alpha^{k_0} = \Pr(k_{i,0} = k_0)$. Moreover, let H^{k_0} be the distribution of individuals within that entry cohort, with $\int_{\mathcal{I}_{k_0}} dH^{k_0}(i) = 1$. The cohort's share of the population α^{k_0} and distributions H^{k_0} are held fixed under the local policy reform. The individual probability of remaining unemployed at t, conditional on entry in k_0, and its cohort average are

$$
\begin{aligned}
S^{k_0}_{i,t} &= \int_\Omega \sum_{k \in \{B,R\}} \theta^{U,k}_{i,t}(\xi_{i,t})\, dF^{k_0}_{i,t}(\xi_{i,t}), \\
S^{k_0}_t &= \int_{\mathcal{I}_{k_0}} S^{k_0}_{i,t}\, dH^{k_0}(i) = \Pr(s_{i,t} = U \mid k_{i,0} = k_0; \mathbf{P}).
\end{aligned}
$$

Each individual i chooses a program (e_i, c^U_i, c^E_i) where

$$
\begin{aligned}
e_i &= \{e_{i,t}(\xi_{i,t})\}_{t \in \{1,2..T\}, \xi_{i,t} \in \Omega, \theta^{U,k}(\xi_{i,t})=1}, \\
c^U_i &= \{c^U_{i,t}(\xi_{i,t})\}_{t \in \{1,2..T\}, \xi_{i,t} \in \Omega, \theta^{U,k}(\xi_{i,t})=1}, \\
c^E_i &= \{c^E_{i,t}(\xi_{i,t})\}_{t \in \{1,2..T\}, \xi_{i,t} \in \Omega, \theta^{E,k}(\xi_{i,t})=1},
\end{aligned}
$$

to solve

$$
\begin{aligned}
V_{i,0}(\mathbf{P}) = \max \Sigma_{t=1}^{T} \beta^{t-1} & \int \{\Sigma_k [u\left(c_{i,t}^{U}(\xi_{i,t}), e_{i,t}(\xi_{i,t})\right) \theta_{i,t}^{U,k}(\xi_{i,t}) + u\left(c_{i,t}^{E}\right) \theta_{i,t}^{E,k}(\xi_{i,t})]\} dF_{i,t}^{k_0}(\xi_{i,t}) \\
& + \Sigma_{t=1}^{T} \beta^{t-1} \int \{\Sigma_k \mu_{i,t}^{U,k}(\xi_{i,t}) [(1+r) a_{i,t}(\tilde{\xi}_{i,t-1}) + b_t^{k_0} + y_t^{U} - c_{i,t}^{U}(\xi_{i,t}) - a_{i,t+1}] \theta_{i,t}^{U,k}(\xi_{i,t})\} dF_{i,t}^{k_0}(\xi_{i,t}) \\
& + \Sigma_{t=1}^{T} \beta^{t-1} \int \{\Sigma_k \mu_{i,t}^{E,k}(\xi_{i,t}) [(1+r) a_{i,t}(\tilde{\xi}_{i,t-1}) + w_i - \tau - c_{i,t}^{E}(\xi_{i,t}) - a_{i,t+1}] \theta_{i,t}^{E,k}(\xi_{i,t})\} dF_{i,t}^{k_0}(\xi_{i,t}) \\
& + \Sigma_{t=1}^{T} \beta^{t-1} \int \mu_{i,t}^{a}(\xi_{i,t}) [a_{i,t+1} - \underline{a}_i] dF_{i,t}^{k_0}(\xi_{i,t}),
\end{aligned}
$$

where the distribution $F_{i,t}^{k_0}$ reflects the individual's choices and the exogenous process for current conditions. While in the empirical implementation in the main text I consider separability of utility between consumption and effort, here I allow for non-separability in the characterization. In the framework, the individual starts unemployed and remains active until T, with employment being an absorbing state. The individual's exit rate out of unemployment in aggregate economic conditions k at time t depends on her search effort at time t. The present value of the government's budget is

$$
G(\mathbf{P}) = \Sigma_{t=1}^{T} \frac{1}{(1+r)^{t-1}} \int \int \{\Sigma_k [-b_t^{k_0} \theta_{i,t}^{U,k}(\xi_{i,t}) + \tau \theta_{i,t}^{E,k}(\xi_{i,t})]\} dF_{i,t}^{k_0}(\xi_{i,t}) di.
$$

The government solves

$$
\max_{\mathbf{P}} \int \omega(\bar{e}_i) V_{i,0}(\mathbf{P}) di + \lambda [G(\mathbf{P}) - \overline{G}],
$$

where $\omega(.)$ is a welfare weight, λ is the Lagrange multiplier on the government's budget constraint and $\overline{G}$ is an exogenous revenue constraint. The welfare weight is assumed to depend on a willingness-to-work type $\bar{e}_i$, which is defined as the average effort level exerted by the individual per period spent unemployed when faced with fixed economic conditions $\overline{k}$ and a fixed baseline benefit policy $\overline{\mathbf{P}}$:

$$
\bar{e}_i = \mathbb{E}[e_{i,t} \mid \overline{k}, \overline{\mathbf{P}}] = \frac{\sum_{t=1}^{T} \int e_{i,t}(\xi_{i,t}) \theta_{i,t}^{U,\overline{k}}(\xi_{i,t}) \, dF_{i,t}^{\overline{k}}(\xi_{i,t}; \overline{\mathbf{P}})}{\sum_{t=1}^{T} S_{i,t}^{\overline{k}}(\overline{\mathbf{P}})},
$$

where $F_{i,t}^{\overline{k}}(\xi_{i,t}; \overline{\mathbf{P}})$ and $S_{i,t}^{\overline{k}}(\overline{\mathbf{P}})$ denote the state distribution and survival probability when economic conditions are held fixed at $\overline{k}$ and the benefit policy is fixed at $\overline{\mathbf{P}}$. The characterization is based on local policy changes and thus only allows for local tests and recommendations. For the ability to translate globally, the program would need to be strictly concave in $\mathbf{P}$. To provide tractable expressions of local welfare implications I assume that the social welfare function is differentiable.

Unemployment Policy. Following Chetty (2006), I assume that lifetime utility is smooth, increasing and strictly quasi-concave in (c_i^U, c_i^E, e_i) and that the value function $V_{i,0}(\mathbf{P})$ is differentiable such that the envelope theorem applies. Consider a baseline policy paying a fixed benefit b up to duration P and zero afterwards. Define

$$
D_b^{k_0} = \sum_{t=1}^{T} \frac{b_t^{k_0}}{b} S_t^{k_0}, \qquad D^{k_0} = \sum_{t=1}^{T} S_t^{k_0}.
$$

where $D_b^{k_0}$ measures expected benefit duration in full-benefit days, and D^{k_0} measures expected nonemployment duration. Under a flat-benefit policy, $D_b^{k_0} = \sum_{t=1}^{P^{k_0}} S_t^{k_0}$ equals expected benefit duration.
The welfare impact of a change in the unemployment benefit $b_{P+1}^{k_0}$ equals

$$\frac{\partial W(\mathbf{P})}{\partial b_{P+1}^{k_0}} = \int \omega(\overline{e}_i) \frac{\partial V_{i,0}(\mathbf{P})}{\partial b_{P+1}^{k_0}} di + \lambda \frac{\partial G(\mathbf{P})}{\partial b_{P+1}^{k_0}},$$

where, using $S_t^{r,k_0} = S_t^{k_0}/[1+r]^{(t-1)}$,

$$\begin{aligned}
\frac{\partial G(\mathbf{P})}{\partial b_{P+1}^{k_0}} &= -\alpha^{k_0} S_{P+1}^{r,k_0} - \alpha^{k_0} \sum_{t=1}^{T} (b_t^{k_0} + \tau) \frac{\partial S_t^{r,k_0}}{\partial b_{P+1}^{k_0}} - \alpha^{k_0'} \sum_{t=1}^{T} (b_t^{k_0'} + \tau) \frac{\partial S_t^{r,k_0'}}{\partial b_{P+1}^{k_0}} \\
&\approx -\alpha^{k_0} \left[S_{P+1}^{r,k_0} + \sum_{t=1}^{T} (b_t^{k_0} + \tau) \frac{\partial S_t^{r,k_0}}{\partial b_{P+1}^{k_0}} \right] \\
&= -\alpha^{k_0} \left[\frac{S_{P+1}^{k_0}}{(1+r)^P} + \sum_{t=1}^{T} \frac{b_t^{k_0} + \tau}{(1+r)^{t-1}} \frac{\partial S_t^{k_0}}{\partial b_{P+1}^{k_0}} \right] \\
&= -\alpha^{k_0} \left[b \frac{\partial D_b^{k_0}}{\partial b_{P+1}^{k_0}} + \tau \frac{\partial D^{k_0}}{\partial b_{P+1}^{k_0}} \right] \\
&= -\alpha^{k_0} S_{P+1}^{k_0} \frac{b \frac{\partial D_b^{k_0}}{\partial b_{P+1}^{k_0}} + \tau \frac{\partial D^{k_0}}{\partial b_{P+1}^{k_0}}}{S_{P+1}^{k_0}} \\
&= -\alpha^{k_0} S_{P+1}^{k_0} (1 + FE^{k_0}).
\end{aligned}$$

where the second line approximation relies on the effect of extending benefit duration in a given economic condition having small effect on individuals' behavior who became unemployed in other economic conditions. The fourth line sets $r = 0$, as in the main text. For a general subjective discount factor β, we have

$$\begin{aligned}
\int \omega(\overline{e}_i) \frac{\partial V_{i,0}(\mathbf{P})}{\partial b_{P+1}^{k_0}} di &= \beta^P \int_{\mathcal{I}_{k_0}} \int \omega(\overline{e}_i) \frac{\partial u\left(c_{i,P+1}^U(\xi_{i,P+1}), e_{i,P+1}(\xi_{i,P+1})\right)}{\partial c_{i,P+1}^U} \sum_k \theta_{i,P+1}^{U,k}(\xi_{i,P+1})\, dF_{i,P+1}^{k_0}(\xi_{i,P+1})\, di \\
&= \alpha^{k_0} \beta^P S_{P+1}^{k_0} \mathbb{E}_{P+1}^{U,k_0} \left[\omega(\overline{e}_i) \frac{\partial u(c_{i,P+1}^U(\xi_{i,P+1}), e_{i,P+1}(\xi_{i,P+1}))}{\partial c_{i,P+1}^U} \right].
\end{aligned}$$

The expectation operator is $\mathbb{E}_{P+1}^{U,k_0}[\,\cdot\,] = \mathbb{E}\,[\,\cdot \mid s_{i,P+1} = U,\ k_{i,0} = k_0; \mathbf{P}]$. It thus averages over all potential states in which individuals are unemployed at time $P+1$ after entering in economic condition k_0, integrating over current economic conditions. The weight of individual i's marginal utility in calculating the average marginal utility among the unemployed at time t is scaled by $S_{i,t}^{k_0}/S_t^{k_0}$ relative to the entry-cohort distribution H^{k_0}.

Combining the two expressions and imposing $\beta = 1$, I find

$$\frac{\partial W(\mathbf{P})}{\partial b_{P+1}^{k_0}} = 0 \Leftrightarrow \mathbb{E}_{P+1}^{U,k_0} \left[\omega(\overline{e}_i) \frac{\partial u(c_{i,P+1}^U(\xi_{i,P+1}), e_{i,P+1}(\xi_{i,P+1}))}{\partial c_{i,P+1}^U} \right] = \lambda \left[\frac{1}{S_{P+1}^{k_0}} \left[\frac{\partial D_b^{k_0}}{\partial b_{P+1}^{k_0}} b + \frac{\partial D^{k_0}}{\partial b_{P+1}^{k_0}} \tau \right] \right].$$

Thus, changing the unemployment benefit duration in both boom and recession, and combining expressions, I obtain:

$$\frac{\mathbb{E}^{U,R}_{P+1}\left[\omega(\overline{e}_i)\dfrac{\partial u(c^U_{i,P+1}(\xi_{i,P+1}),e_{i,P+1}(\xi_{i,P+1}))}{\partial c^U_{i,P+1}}\right]}{\mathbb{E}^{U,B}_{P+1}\left[\omega(\overline{e}_i)\dfrac{\partial u(c^U_{i,P+1}(\xi_{i,P+1}),e_{i,P+1}(\xi_{i,P+1}))}{\partial c^U_{i,P+1}}\right]} = \frac{\dfrac{1}{S^R_{P+1}}\left[\dfrac{\partial D^R_b}{\partial b^R_{P+1}}b+\dfrac{\partial D^R}{\partial b^R_{P+1}}\tau\right]}{\dfrac{1}{S^B_{P+1}}\left[\dfrac{\partial D^B_b}{\partial b^B_{P+1}}b+\dfrac{\partial D^B}{\partial b^B_{P+1}}\tau\right]}.$$

which, under additive separability of consumption and effort, corresponds to equation (5) in the main text.

B.2 Implementation

Marginal Benefit. We make the following assumptions:

- *Assumption 1* - Relevant consumption heterogeneity within unemployed in the same aggregate economic condition is negligible: $c^{U,k_0}_{i,t} \approx c^{U,k_0}_t$ [48]
- *Assumption 2* - Homogeneous relative risk aversion: $\gamma^{k_0} \approx \gamma$ (where $\gamma^{k_0} = \frac{-u''(c^{E,k_0})c^{E,k_0}}{u'(c^{E,k_0})}$)
- *Assumption 3* - Taylor approximation for $u'(c^{U,k_0})$ around c^{E,k_0}:

$$\begin{aligned} u'(c^{U,k_0}) &\approx u'(c^{E,k_0}) + u''(c^{E,k_0})(c^{U,k_0}-c^{E,k_0}) \\ &\approx u'(c^{E,k_0})\left[1+\gamma\frac{c^{E,k_0}-c^{U,k_0}}{c^{E,k_0}}\right] \end{aligned}$$

Thus, we arrive at the expression in the main text:

$$\begin{aligned} \frac{\mathbb{E}^{U,R}_{P+1}[\omega(\overline{e}_i)u'(c^U_{i,P+1})]}{\mathbb{E}^{U,B}_{P+1}[\omega(\overline{e}_i)u'(c^U_{i,P+1})]} &\approx \frac{\omega^R}{\omega^B}\frac{u'(c^{U,R})}{u'(c^{U,B})} \\ &\approx \frac{\omega^R}{\omega^B}\frac{1+\gamma\frac{c^{E,R}-c^{U,R}}{c^{E,R}}}{1+\gamma\frac{c^{E,B}-c^{U,B}}{c^{E,B}}}\frac{u'(c^{E,R})}{u'(c^{E,B})} \\ &\approx \frac{\omega^R}{\omega^B}\frac{1+\gamma\frac{c^{E,R}-c^{U,R}}{c^{E,R}}}{1+\gamma\frac{c^{E,B}-c^{U,B}}{c^{E,B}}}\left[1+\gamma\frac{c^{E,B}-c^{E,R}}{c^{E,B}}\right] \end{aligned}$$

where the third line follows from another Taylor approximation of $u'(c^{E,R})$ around $c^{E,B}$, and $\omega^{k_0} = \mathbb{E}^{U,k_0}_{P+1}[\omega(\overline{e}_i)]$ denotes the average welfare weight among individuals remaining unemployed at benefit exhaustion in economic condition $k_0 \in \{B,R\}$.

In the empirical implementation, I proxy willingness to work by classifying individuals into binary types $\overline{e}_i \in \{\overline{e}_A, \overline{e}_B\}$, corresponding to high and low willingness to work. Let $s^{k_0}_g = \Pr(\overline{e}_i = \overline{e}_g \mid s_{i,P+1} = U,\ k_{i,0} = k_0; \mathbf{P})$ for $g \in \{A,B\}$ denote the share of type-g individuals among benefit exhaustees in economic condition k_0, with $s^{k_0}_A + s^{k_0}_B = 1$. The average welfare weight in state k_0 is then the weighted

[48] This is similar to the cohort assumption in the context of pension design (Kolsrud et al., 2024). When there is heterogeneity within the same economic conditions, one needs to account for the covariance between marginal utility and welfare weights (Andrews and Miller, 2013). In this case, a normalized covariance correction that is equal across economic conditions also suffices.

average $\omega^{k_0} = s_A^{k_0}\omega(\overline{e}_A) + s_B^{k_0}\omega(\overline{e}_B)$, and so the ratio of welfare weights is:

$$\begin{aligned}\frac{\omega^R}{\omega^B} &= \frac{\mathbb{E}_{P+1}^{U,R}[\omega(\overline{e}_i)]}{\mathbb{E}_{P+1}^{U,B}[\omega(\overline{e}_i)]} \\ &= \frac{s_A^{Recess.}\omega(\overline{e}_A) + s_B^{Recess.}\omega(\overline{e}_B)}{s_A^{Boom}\omega(\overline{e}_A) + s_B^{Boom}\omega(\overline{e}_B)} \\ &= \frac{s_A^{Recess.}\frac{\omega(\overline{e}_A)}{\omega(\overline{e}_B)} + s_B^{Recess.}}{s_A^{Boom}\frac{\omega(\overline{e}_A)}{\omega(\overline{e}_B)} + s_B^{Boom}}.\end{aligned}$$

Marginal Cost. Let $dM^{k_0} = S_{P+1}^{k_0} db_{P+1}^{k_0}/b$ denote the mechanical increase in full-benefit days. Let $dD_b^{k_0}$ and dD^{k_0} denote the total duration changes induced by the same benefit change $db_{P+1}^{k_0}$. Multiplying by the same payment change delivers:

$$1 + FE^{k_0} = \frac{dD_b^{k_0} + (\tau/b)dD^{k_0}}{dM^{k_0}}.$$

In the empirical implementation, I exploit a finite PBD extension of 60 days. The estimates thus provide the discrete counterpart of the marginal changes: the total changes in unemployment and nonemployment duration, which are denoted by $\Delta D_b^{k_0}$ and ΔD^{k_0}, respectively. Therefore, $\Delta D_b^{k_0}$ includes both mechanical and behavioral changes. The mechanical effect is estimated using the survival rate of individuals below the cutoff over the additional entitlement window as $\Delta M^{k_0} = 60\overline{S}_{P+1}^{k_0}$. As a result, the empirical exercise implements $FE^{k_0} = [\Delta D_b^{k_0} + (\tau/b)\Delta D^{k_0}]/\Delta M^{k_0} - 1$.

B.3 Welfare Impact

Consider a reform where $\frac{db_{P+1}^R}{db_{P+1}^B} < 0$ such that the budget remains balanced:
$\alpha^R S_{P+1}^R(1+FE^R)db_{P+1}^R = -\alpha^B S_{P+1}^B(1+FE^B)db_{P+1}^B$

$$\begin{aligned}dW =& \alpha^R S_{P+1}^R \mathbb{E}_{P+1}^{U,R}[\omega(\overline{e}_i)u'(c_{i,P+1}^U)]db_{P+1}^R + \alpha^B S_{P+1}^B \mathbb{E}_{P+1}^{U,B}[\omega(\overline{e}_i)u'(c_{i,P+1}^U)]db_{P+1}^B \\ &- \lambda\left[\alpha^R S_{P+1}^R(1+FE^R)db_{P+1}^R + \alpha^B S_{P+1}^B(1+FE^B)db_{P+1}^B\right] \\ =& \alpha^R S_{P+1}^R \mathbb{E}_{P+1}^{U,R}[\omega(\overline{e}_i)u'(c_{i,P+1}^U)]db_{P+1}^R - \alpha^B S_{P+1}^B \mathbb{E}_{P+1}^{U,B}[\omega(\overline{e}_i)u'(c_{i,P+1}^U)]\frac{\alpha^R S_{P+1}^R(1+FE^R)}{\alpha^B S_{P+1}^B(1+FE^B)}db_{P+1}^R\end{aligned}$$

where the second equality follows from the budget neutrality of the reform.

$$\begin{aligned}\frac{dW/(\alpha^R S_{P+1}^R db_{P+1}^R)}{\mathbb{E}_{P+1}^{U,B}[\omega(\overline{e}_i)u'(c_{i,P+1}^U)]} &= \underbrace{\frac{\mathbb{E}_{P+1}^{U,R}[\omega(\overline{e}_i)u'(c_{i,P+1}^U)]}{\mathbb{E}_{P+1}^{U,B}[\omega(\overline{e}_i)u'(c_{i,P+1}^U)]}}_{=1.77} - \underbrace{\frac{1+FE^R}{1+FE^B}}_{=0.71} \\ &= \mathbf{1.06}\end{aligned}$$

where welfare is measured per euro mechanically transferred in recession and expressed in monetary units relative to $\mathbb{E}_{P+1}^{U,B}[\omega(\overline{e}_i)u'(c_{i,P+1}^U)]$, which denotes the average welfare-weighted marginal utility for the individuals unemployed in boom (with boom and recession referring to 8% and 26% unemployment rate, respectively).

Appendix C Online Survey



Instructions (Information Sheet)

You are invited to participate in this survey, which investigates the preferences of Spanish people for public policies. It takes approximately 8 minutes. When you complete it, you will receive panel points worth €1.15 as a reward.
We are a non-partisan group of researchers affiliated to University College London. Our goal is to understand preferences about various social policies. **It is very important that you respond honestly. No matter what your political views are, this is an important question and by completing this survey, you are contributing to our knowledge as a society.**
Anytime you don't know an answer, just give your best guess.

All the information collected about you is treated confidentially. The information is only used for scientific purposes and the survey is registered at University College London. This is intended to be used for academic publication, a chapter of a PhD thesis and potentially for subsequent research. You have been chosen to participate because you are aged 25-64 and are representative of the Spanish population. Participation is voluntary and you can withdraw at any point if you wish. **The data are anonymized and you will not be individually identified from responses.** There are no direct risks or benefits for you from taking part in the survey, beyond the monetary reward. This project has received approval by the UCL Research Ethics Committee (#21191/001). In case you have doubts or complaints, you can contact the corresponding member of the Research Team (jon.piqueras.17@ucl.ac.uk) or the Ethics Committee (ethics@ucl.ac.uk). The title of the study is "Redistribution and Social Insurance Design". This project is funded by the Institute for Fiscal Studies.

Local Data Protection Privacy Notice:

The controller for this project will be University College London (UCL). The UCL Data Protection Officer provides oversight of UCL activities involving the processing of personal data. This "local" privacy notice sets out the information that applies to this particular study. Further information on how UCL uses participant information can be found in our "general" privacy notice (here). The categories of personal data used will be as follows: year of birth, gender, province of residence, employment status, income, socio-economic status, education, postcode. The lawful basis that will be used to process your personal data are: "Public task" for personal data and "Research purposes" for special category data. Your data will be anonymized at any point and therefore it cannot be withdrawn. If you are concerned about how your personal data is being processed, or if you would like to contact us about your rights, please contact UCL in the first instance at data-protection@ucl.ac.uk.

Thank you for reading this information sheet and for considering to take part in this research study. If you agree with these terms, please continue reading the consent form below before you start.

Consent Form

I confirm that I understand that by clicking "Next" below I am consenting to ALL these elements of the study. I understand that by not giving consent to these elements I may be deemed ineligible for the study.

- I confirm that I have read and understood the Information Sheet for the above study. I have had an opportunity to consider the information and what will be expected of me.
- I consent to participate in the study. I understand that my personal information will be used for the purposes explained to me. I understand that according to data protection legislation, "public task" will be the lawful basis for processing.
- I understand that my data gathered in this study will be stored anonymously and securely. It will not be possible to identify me in any publications.
- I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.
- I understand the direct/indirect benefits and risks of participating.
- I understand that I will be compensated if I complete the survey.
- I agree that my anonymized research data may be used by others for future research.
- I hereby confirm that I understand the inclusion criteria as detailed in the Information Sheet.
- I am aware of who I should contact if I wish to lodge a complaint.
- I voluntarily agree to take part in this study.

Next

Preliminary Questions

What is your year of birth?

What is your sex?

- ○ Male
- ○ Female

What is your current employment status?

- ○ Full-time employment
- ○ Part-time employment
- ○ Unemployed looking for jobs
- ○ Other (for example, student or retired)
- ○ Disabled/unable to work

What is your highest educational level?

- ○ Primary education
- ○ Secondary education
- ○ Baccalaureate/professional training
- ○ Bachelors degree
- ○ Masters degree
- ○ PhD

Next

Demographics

What is your province of residence?

What is the duration of your usual workday? (for example, 100% corresponds to full time and 50% to part time)

0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%

What is your gross monthly labor income if employed, or benefits if unemployed? (that is, before taxes, without decimals, in €)

What is your current type of job?

- Permanent
- Temporary
- Self employed
- Not working

How many children do you have (younger than 26 years old)?

How much do you have in savings (savings in current account + value of other assets you own - debt, in €)?

While you were 18-25 years old, was any of your parents more than one year unemployed?

- Yes
- No

While you were 18-25 years old, did your parents have difficulties to make ends meet?

- Yes
- No

Next

20%

Opinions on Public Policy

Imagine there are 2 unemployed individuals, A and B, who receive €950 while unemployed and are entitled to receive benefits for up to 14 months if remaining unemployed:

- *Individual A looked for jobs last week*
- *Individuo B did not look for jobs last week*

The government wants to transfer €1000 to individuals A and B in a one-time payment, and is thinking of how to allocate them. Please choose what allocation option you would prefer:

- ○ More money to A than to B
- ○ More money to B than to A
- ○ Same amount to A and B

Next

40%

Opinions on Public Policy

Imagine there are 2 unemployed individuals, A and B, who receive 950€ while unemployed and are entitled to receive benefits for up to 14 months if remaining unemployed:

- *Individual A looked for jobs last week*
- *Individual B did not look for jobs last week*

The government wants to transfer €1000 to individuals A and B in a one-time payment, and is thinking of how to allocate them. Now imagine that an unequal allocation may be costly. That is, it can allocate €500 each, or more to one than the other, but the latter option may result in the total amount of resources that can be distributed being reduced.

Please choose, for each pair of options below, what allocation option you would prefer:

Individual A looked for jobs last week
Individual B did not look for jobs last week

What amounts (€) would you prefer for individuals A and B?

- ○ 550 for A, 450 for B
- ○ 500 for A, 500 for B

Individual A looked for jobs last week
Individual B did not look for jobs last week

What amounts (€) would you prefer for individuals A and B?

- ○ 350 for A, 550 for B
- ○ 500 for A, 500 for B

Individual A looked for jobs last week
Individual B did not look for jobs last week

What amounts (€) would you prefer for individuals A and B?

- ○ 150 for A, 550 for B
- ○ 500 for A, 500 for B

Individual A looked for jobs last week
Individual B did not look for jobs last week

What amounts (€) would you prefer for individuals A and B?

- ○ 500 for A, 500 for B
- ○ 450 for A, 550 for B

Individual A looked for jobs last week
Individual B did not look for jobs last week

What amounts (€) would you prefer for individuals A and B?

- ○ 500 for A, 500 for B
- ○ 250 for A, 550 for B

Next

60%

Opinions on Public Policy

Individual A looked for jobs last week
Individual B did not look for jobs last week

What amounts (€) would you prefer for individuals A and B?

- 50 for A, 550 for B
- 500 for A, 500 for B

Individual A looked for jobs last week
Individual B did not look for jobs last week

What amounts (€) would you prefer for individuals A and B?

- 500 for A, 500 for B
- 550 for A, 300 for B

Individual A looked for jobs last week
Individual B did not look for jobs last week

What amounts (€) would you prefer for individuals A and B?

- 550 for A, 350 for B
- 500 for A, 500 for B

Individual A looked for jobs last week
Individual B did not look for jobs last week

What amounts (€) would you prefer for individuals A and B?

- 500 for A, 500 for B
- 300 for A, 550 for B

Individual A looked for jobs last week
Individual B did not look for jobs last week

What amounts (€) would you prefer for individuals A and B?

- 500 for A, 500 for B
- 550 for A, 400 for B

Next

80%

General Opinions (last questions)

When you were asked before about your preferred allocation of the €1000, you answered you would prefer to give "More money to A than to B".
(Note: Remember that A looked for jobs last week, and B did not looked for jobs last week).

Please, explain why you chose that option:

There exists the idea that the benefits received by each unemployed could be more generous (higher benefit level or duration) at times when economic conditions are bad (high unemployment rate), relative to times where economic conditions are good (low unemployment rate).
Some reasons for that are that when the economic conditions are bad, the unemployed suffer larger consumption reductions upon job loss and the type of unemployed at that moment are, on average, people who put more effort into job finding but who face more difficulties given the lack of job offers.

Would you agree with having unemployment benefits changing with economic conditions in this way?

- ○ Yes
- ○ No

We want to improve the survey and your opinion matters. Please, write below if you have any comment about the questions, the survey structure or whether the questions are clear/intuitive (in case you do not have comments, write NO).

Next



Many thanks for answering the questions!

Please click on "End survey" to end your participation.

End survey

Appendix D Time Use Data

A key input of the framework in the main text is the share of unemployed individuals who searched for jobs over the last week, and how it varies with economic conditions. Here I provide a more detailed explanation of how I estimate these statistics. The data used are the two available waves (2002-2003 and 2009-2010) of the Spanish Time Use survey, which is similar to the American Time Use Survey (ATUS) in the US. Since this survey oversamples individuals during Fridays, Saturdays and Sundays, I estimate averages of the variable of interest for each day, which is the share of individuals with positive search time, and then aggregate by adding up all the days of the week. I compute these averages at the region (CCAA)-year level, which is the lowest geographical level available. I also employ data on unemployment rate at this same level from the National Statistics Institute (INE).

While time use surveys provide very detailed information on the activities of individuals throughout the day, they have the caveat that individuals only fill the diary once, and so we cannot see how the same individual behaves over the whole week. Given that my statistic of interest is the share of individuals who searched last week, I estimate this from the daily probability of search. To do so, I follow Faberman et al. (2022) who compare their estimates from the Survey of Consumer Expectations (SCE) with the ones obtained with ATUS, and provide a framework to map daily probability of job search into weekly share of searchers. The advantage of SCE is that individuals are asked specifically about whether they searched for jobs last week, and not just one day as in ATUS. For this exercise, they leverage the fact that in the UK time use survey, individuals fill diaries for two consecutive days, which provides information about what share of individuals search frequently. They find that 35% of the unemployed who search one day, repeat again the next day. They assume that (i) there are two types of searchers: steady or intermittent job seekers; (ii) individuals observed searching two days in a row are steady job seekers who search every day; (iii) individuals who do not search two days in a row are intermittent job seekers who search only once per week and could therefore be randomly observed looking for work on any day of the week. Based on this statistic and their two-type model, they set $\mu_j^{uk} = 0.35$ for the unemployed, where μ_j^{uk} is the share of steady searchers among individuals observed searching on a given day. This yields the following probability of weekly search:

$$\pi_{\text{week}_j} = \mu_j^{uk}\pi_j^{day} + 7(1-\mu_j^{uk})\pi_j^{day}$$

Assuming that the same conditional share applies in Spain, I apply this mapping to construct the weekly search measure shown in Panel (e) of Figure 1. The boom and recession comparisons and the composition adjustment are described in Section 4.1.2. The analysis residualizes the variables on region FEs to not compare regions that are structurally different and focus on the variation over the business cycle.

Moreover, I also provide results estimating the share of weekly searchers using a different approach to prove the robustness of the findings. To do so I instead assume that search is independent across days, and just compute the probability of searching at least once in a week as $\pi_{\text{week}_j} = 1-(1-\pi_j^{day})^7$. A related

conversion based on independence across days is used by Braun (2024). Results are shown in Figure A3. Reassuringly, even in this case under a strong independence assumption, the share ranges between 53% and 75%, which implies a relatively lower increase but still a substantial shift which is in line with the baseline finding.

Appendix E Public Opinion Polls

As described in the main text, the empirical implementation of the framework suggests the UI system should be more generous when economic conditions worsen. Here I explore public opinions about this type of reform, presenting two pieces of suggestive evidence on the political economy of a cycle-dependent benefit reform. First, in the online survey I designed, respondents are also asked whether they would support a UI reform of the type described in this paper. The objective of this question is to assess whether they agree with the interpretation of their answers through the lens of my framework. Panel (d) of Figure 1 shows a majority of people state they would be likely to support a UI reform of the type described above.

Second, I draw on annual public opinion surveys over the years 2005-2019. This information on public attitudes towards certain public policies comes from the individual-level files of the Public Opinion and Fiscal Policy Survey (*Encuesta de Opinión Pública y Política Fiscal*) carried out by the Spanish CIS. This is a representative sample, eliciting opinions on several public policy issues over time. I combine the waves for the years 2005-2019.[49] My analysis focuses on a question asked every year about whether they think that the government should devote more/less/same resources to UI than are provided at each point in time. I classify individuals as demanding more generous UI when they state that more resources should be devoted. The evolution of the answers to this question over the cycle is shown in Figure A9. Panel (a) shows that people demand more generous UI when economic conditions are bad, with a remarkable increase in this demand that parallels the increase in the unemployment rate. Although I am not able to isolate mechanisms in this analysis, a potential explanation is that the probability of receiving benefits goes up in recession (probability of paying taxes goes down), so people just respond in that way because they will be more likely to benefit directly from the reform, and not because they consider that there is a higher value of transferring resources towards the type of individuals unemployed in recessions. However, the same plot reveals that the increase in UI demand is common across groups (employed, unemployed, retired)[50], and thus does not depend on the group-specific probability of receiving benefits which varies across them. To gain further insights, in Panels (b) and (c) I divide regions in two groups by the size of the region unemployment shock experienced in order to see whether regions that are hit relatively harder by the Great Recession increase their support for higher UI generosity more. I find again that the increase is common across groups and not related to the change in the unemployment risk. Although more suggestive, this complementary evidence is consistent with people taking into account other dimensions beyond their own unemployment risk, and supporting more generous UI when economic conditions deteriorate.

[49] The annual samples consist of around 2,500 individuals, so the total sample contains information on approximately 38,000 individuals.

[50] The increase in support for a more generous UI system when economic conditions worsen could be just driven by a group composition effect: if the unemployed support more UI increases than the employed on average, the fact that more people become unemployed in recession could explain the increase in the support for UI. A shift-share decomposition of the change in UI support from trough to peak (2005–2014) shows that this increase is primarily driven by changes within groups: $UI_t - UI_{t-1} = \sum_g [s_{g,t-1}(UI_{g,t} - UI_{g,t-1}) + (s_{g,t} - s_{g,t-1})UI_{g,t}]$, where g indexes labor-market groups, $s_{g,t}$ denotes group g's share of the analysis population, and $UI_{g,t}$ denotes the share within group g supporting more generous UI at time t. The within-group component, given by the first term, accounts for 95% of the overall increase.